14.0 INITIAL TEST PROGRAM

14.1 SPECIFIC INFORMATION TO BE INCLUDED IN PRELIMINARY SAFETY ANALYSIS REPORTS

This information is applicable only to a Preliminary Safety Analysis Report.

14.2 INITIAL TEST PROGRAM

14.2.1 SUMMARY OF TEST PROGRAM AND OBJECTIVES

This chapter provides information covering the initial test program of the VEGP. It encompasses the period following installation and construction-related inspections and tests of components and systems, through the completion of power ascension testing.

The initial test program established for the VEGP demonstrates that components and systems operate in accordance with design requirements. This test program is based on the requirements of 10 CFR 50, Appendix B, Criterion XI. The initial test program is conducted in two phases: the preoperational test program and the startup test program. At the conclusion of these phases, the plant is ready for commercial operations.

Prior to preoperational testing of a particular system, certain prerequisite and construction tests will be conducted in order to verify the integrity, proper installation, cleanliness, and functional operability of the system components.

Based on the experience gained on Unit 1, the preoperational test phase for Unit 2 has been enhanced to incorporate a new subphase called release/release for test. This allows for a more interactive interface with construction than did Unit 1. The strategy for Unit 2 is to allow the commencement of construction acceptance test, and preventive maintenance on components and systems that installation/construction is complete but has not been turned over to the nuclear operation department. The controls for this interface are described in the VEGP startup manual. Preoperational and acceptance tests will commence only after the component or system has been turned over and certain prerequisite and construction acceptance tests have been conducted to verify the integrity, proper installation, cleanliness, and functional operability of the system components.

Conformance with the recommendations of Regulatory Guide 1.68, Initial Test Programs for Water-Cooled Reactor Power Plants, is discussed in subsection 14.2.3.

14.2.1.1 Preoperational Test Program

The preoperational test program is that part of the initial test program that commences with the completion of construction and construction-related inspections and tests and terminates with the commencement of nuclear fuel loading.

The preoperational test program (table 14.2.1-1) includes tests on both safety-related and nonsafety-related systems. The tests conducted on safety-related systems will demonstrate the capability of the structures, systems, and components to meet performance requirements and design criteria. The nonsafety-related tests will verify the system's or component's operability.

As a minimum, testing will be conducted on those systems that:

A. Are relied upon for safe shutdown and cooldown of the reactor under normal plant conditions and for maintaining the reactor in a safe condition for an extended shutdown period.

B. Are relied upon for safe shutdown and cooldown of the reactor under transient and postulated accident conditions and for maintaining the reactor in a safe condition for an extended shutdown period following such conditions.

C. Are relied upon for establishing conformance with safety limits or limiting conditions for operations.

D. Are classified as engineered safety features actuation systems (ESFAS) or are relied upon to support or ensure operation of ESFAS within design limits.

E. Are assumed to function during an accident or for which credit is taken in the accident analysis.

F. Are used to process, store, control, or limit the release of radioactive material.

The objectives of the preoperational test program are to:

A. Verify that plant components and systems, as constructed, fulfill their design intent.

B. Demonstrate, to the extent practicable, proper system/component response to postulated accidents.

C. Familiarize plant operating, technical, and maintenance personnel with plant operation.

Due to the proven design resulting from the Unit 1 test program, the Unit 2 testing methodology differs from Unit 1 in that, acceptance tests instead of preoperational tests will be performed on the nonsafety-related systems which have tests listed in table 14.2.1-1 and are not required to have preoperational tests performed.

The objective of acceptance tests is to demonstrate the operability of the plant components and systems which are nonsafety-related.

14.2.1.2 Startup Test Program

The startup test program is that part of the initial test program that commences with the start of nuclear fuel loading and terminates with the completion of power ascension testing. The startup tests (table 14.2.1-2):

A. Ensure that fuel loading is accomplished in a safe manner.

B. Confirm the design basis.

C. Demonstrate, where practical, that the plant operates and responds to anticipated transients and postulated accidents.

D. Ensure that the plant can be safely brought to rated capacity and can sustain power operation.

The objectives of the startup test program are to:

A. Accomplish a controlled, orderly, and safe initial core loading.

B. Accomplish a controlled, orderly, and safe initial criticality.

C. Conduct low-power testing sufficient to ensure that design parameters are satisfied and safety analysis assumptions are conservative.

D. Perform a controlled, orderly, and safe power ascension with testing terminating at plant-rated conditions.

E. Provide sufficient testing of transient and accident conditions to verify safe operation during transient or accident conditions.

The completion of startup testing constitutes the completion of the initial test program.

14.2.2 TEST PROCEDURES

The initial test program is conducted in accordance with detailed preoperational and startup test procedures. The general-manager Vogtle nuclear operations maintains the overall responsibility for initial test program procedure preparation, review, and approval. These activities shall be completed so that preoperational test and acceptance test (Unit 2 only) procedures listed in this chapter will be available 60 days prior to implementation and so that startup test procedures will be available 60 days prior to fuel loading for review by the Nuclear Regulatory Commission.

The following paragraphs describe the general methods employed to control procedure development and review. The detailed controls and methods are described in the Startup Manual and Startup Test Administrative Procedures.

14.2.2.1 Procedure Preparation

The general manager-Vogtle nuclear operations is responsible for preparation of initial test program procedures. Technical assistance is provided by Georgia Power Company (GPC), Bechtel Western Power Corporation, Westinghouse Electric Corporation, and Southern Company Services to develop detailed procedures as described below.

The cognizant design organization will review the objectives and acceptance criteria of the initial test program procedures. Comments resulting from the review will be provided for any technical comments not incorporated. This review will occur prior to procedure implementation.

The test procedures are prepared using the latest design information available. This information is utilized in developing the detailed test methods which verify the ability of systems and components to function within their design specifications.

The format and content of the test procedures developed for the initial test program for VEGP reflect the guidance provided in Regulatory Guide 1.68. The format consists of the following sections:

A. Objectives

This section identifies the objectives of the test. They shall be clear, concise, and not quantitative. Examples of some of the possible contents of this section are listed below:

1. Normal starting and stopping, both manual and automatic.
2. Normal operation.
3. Control from local and remote stations.
4. Loss of motive or control power alarm verification.
5. System parameter verification.
6. System capacity verification.

B. References

Reference documents used in the preparation of the test procedure shall be listed in this section. Drawings, both domestic and vendor, will be listed with the revision of the document used.

C. Test Equipment

This section shall list the test equipment, other than installed plant equipment, used to perform the test and collect data. The name and model number of each piece of equipment shall be listed along with ranges where applicable. The test equipment to be used is determined by the test requirements and vendor information.

D. Notes and Precautions

This section contains any notes and/or precautions applicable during the performance of the test. Items generally pertinent to the entire test are included in this section. Specific precautions are to be included in the text of the test just prior to the step to which they apply.

E. Prerequisites and Initial Conditions

Prerequisites are those items which must be completed prior to performance of the test or subsection of the test.

Conditions that must exist prior to starting the test (i.e., alignment, temporary modifications, support systems, etc.) are initial conditions and shall be listed in this section.

F. Test Procedure

The test procedure section provides the detailed step- by-step instructions required to demonstrate that test acceptance criteria are satisfied and to obtain baseline operating data. The instructions must include activities that demonstrate that system and component performances meet acceptance criteria.

This section is subdivided by headings for the purpose of procedure organization and clarity. Each subdivision consists of a continuous series of operations prepared under the assumption that performance of a subsection, once started, will be carried to completion without interruption.

Test data are normally recorded in the body of the procedure immediately following the steps that set up the required test conditions and adjacent to acceptance criteria where practicable. However, where extensive or repetitive lists of data are required, data sheets may be used.

G. Test Data Sheets

The test results (of more than a half-page accumulation) shall be entered onto data sheets contained in this section. Each data sheet shall be numbered sequentially beginning with data sheet 7.1.

H. System Restoration

This section will provide, if necessary, instructions beyond those in the test procedure section for restoring the systems. Jumper removal and disconnection of temporary instruments are examples of section content.

I. Acceptance Criteria

The acceptance criteria section identifies the criteria which shall be met to verify that the system performance is acceptable.

Values and tolerances shall be specified for quantitative criteria. However, some acceptance criteria may be qualitative in nature.

J. Attachments

This section shall contain any attachments necessary to fully document the test results. Recorder traces, computer printouts, pump curves, etc., may be included as attachments.

14.2.2.2 Procedure Review and Approval

14.2.2.2.1 Preoperational/Startup Test Procedures

The general manager-Vogtle nuclear operations (GMVNO), or his designee is responsible for the review and approval of preoperational test procedures and startup test procedures. The general manager's designee for preoperational test procedures is the assistant plant startup manager; for startup tests his designee is the manager operations. In general preoperational test procedures and startup test procedures go through a minimum of two review cycles prior to submittal to the test review board for preoperational test procedures and plant review board for startup test procedures. Each cycle includes reviews by designated Georgia Power Company personnel, Bechtel Western Power Corporation representative, and/or the Westinghouse Electric Corporation representative as appropriate.

When the Unit 1 preoperational test procedure forms the basis for the Unit 2 preoperational test procedure, only one review cycle will be used prior to submittal to the test review board.

As a minimum, the preoperational and startup test procedures listed in this chapter are submitted for review to the test review board or the plant review board. Test procedures that are not satisfactory are resolved under the direction of the board prior to recommendation for approval. The test review board or the plant review board recommends approval to the GMVNO or his designee of procedures found acceptable. The GMVNO or his designee approves the procedure.

Subsequent revisions to the preoperational test procedure and startup test procedure are handled in a manner similar to a new procedure.

Qualifications of the above plant personnel who review and approve preoperational and startup test procedures are addressed in section 13.1.2.

The above processes are further described in the VEGP startup manual and in the startup test administrative procedures.

14.2.2.2.2 Acceptance Test Procedures (Unit 2 only)

The general manager-Vogtle nuclear operations, or his designee is responsible for the review and approval of acceptance test procedures for nonsafety related systems/components identified in table 14.2.1-1. The general manager's designee for acceptance test procedures is the assistant plant startup manager (APSM). In general, acceptance tests go through a

minimum of two review cycles prior to submittal to the assistant plant startup manager for approval. Each cycle includes reviews by the designated Georgia Power Company test supervisor, and either the Bechtel Western Power Corporation, or the Westinghouse Electric Corporation representative, whichever is designated responsible for the procedure.

When the Unit 1 preoperational test procedure forms the basis for the Unit 2 acceptance test procedure, one review cycle will be used prior to submittal to the assistant plant startup manager for approval.

Subsequent revisions to the acceptance test procedure is handled in a manner similar to a new procedure.

Qualification of the above referenced plant personnel who review and approve test procedures is addressed in section 13.1.2.

The above process is further described in the VEGP startup manual.

14.2.2.3 Conduct of the Test Program

Georgia Power Company administers and conducts the initial test program using approved procedures. System turnover is made by the construction department to the nuclear operations department.

Upon acceptance of the turnover, administrative controls (detailed in the startup manual) become effective.

All phases of testing (with the exception of certain construction-performed tests) are under the control of the test supervisor.

Construction acceptance testing generally begins at the time of equipment/system turnover, release, or release for test. This testing must be completed or excepted (as provided for in the startup manual) prior to beginning the preoperational test. The procedure for development of construction acceptance testing procedures is provided in the startup manual.

Whenever practical, cleanliness verification testing will be performed, using approved procedures, prior to the initiation of fluid flow in the system during preoperational testing. Upon completion of construction acceptance testing, subject to the priorities of the startup schedule(s), the preoperational tests are performed.

The preoperational test phase will be governed by procedures in the startup manual. The startup test phase will be governed by plant procedures as supplemented by the Startup Test Administrative Procedures.

14.2.2.4 Review, Evaluation, and Approval of Test Results

14.2.2.4.1 Preoperational/Startup Test Procedure

Upon completion of a preoperational test and/or startup test, the test supervisor prepares a test results package and submits it for review.

The general manager-Vogtle nuclear operations (GMVNO) or his designee is responsible for the review and approval of preoperational test and startup test results. The general manager's designee for preoperational test results review and approval is the plant support manager; for startup tests results review and approval, his designee is the operations manager. In general,

preoperational and startup test results are reviewed and approved by the same organizations and the same or higher level of authority that reviewed and approved the preoperational test procedure and startup test procedure. This includes a review of comments by the Georgia Power Company test supervisor, Bechtel Western Power Corporation representative, and/or Westinghouse Electric Corporation representative, as appropriate, prior to submittal of a preoperational and startup test results package to the appropriate review board. As a minimum, results of preoperational and startup tests listed in this chapter are submitted to the appropriate review board.

The test review board reviews the preoperational test results package and the plant review board reviews the startup test results package. Test results packages that are not satisfactory are resolved under direction of the board prior to recommendation of the package for approval. The test review board and plant review board recommend approval to the GMVNO or his designee of test results packages found acceptable. The GMVNO or his designee approves the test results package as complete. Completed test result packages become part of the permanent plant records.

Qualifications of the above referenced plant personnel who review and approve preoperational and startup test results is addressed in section 13.1.2.

The above process is further described in the VEGP Startup Manual and the Startup Test Administrative Procedures.

14.2.2.4.2 Acceptance Test (Unit 2 only)

Upon completion of an acceptance test procedure, the test supervisor prepares a test results package and submits it for review.

The general manager-Vogtle nuclear operations, or his designee is responsible for the review and approval of acceptance test results. The general manager designates the plant support manager or his designee for acceptance test results review and approval. In general, acceptance test results are reviewed and approved by the same organizations and same or higher level of authority that reviewed the procedure. Test results that are not satisfactory are resolved by the test engineering superintendent prior to submittal to the plant support manager for approval.

Qualification of the above referenced plant personnel who review and approve test results is addressed in section 13.1.2.

The above process is further described in the VEGP startup manual.

14.2.3 CONFORMANCE OF TEST PROGRAM WITH REGULATORY GUIDES (HISTORICAL)

[The initial test program for VEGP Units 1 and 2 is, in general, consistent with the requirements of Regulatory Guide 1.68. The following is a list of regulatory guides that are specifically associated with VEGP's initial test program and will be conformed with except as noted in FSAR section 1.9:

A. Regulatory Guide 1.18, Structural Acceptance Test for Concrete Primary Reactor Containments.

B. Regulatory Guide 1.20, Comprehensive Vibration Assessment Program for Reactor Internals During Preoperational and Initial Startup Testing.

C. Regulatory Guide 1.30, Quality Assurance Requirements for the Installation, Inspection, and Testing of Instrumentation and Electric Equipment (Safety Guide 30).

D. Regulatory Guide 1.37, Quality Assurance Requirements for Cleaning of Fluid Systems and Associated Components of Water-Cooled Nuclear Power Plants.

E. Regulatory Guide 1.41, Preoperational Testing of Redundant Onsite Electric Power Systems to Verify Proper Load Group Assignments.

F. Regulatory Guide 1.52, Design, Testing, and Maintenance Criteria for Engineered Safety Features Atmosphere Cleanup System Air Filtration and Absorption Units of Light-Water-Cooled Nuclear Power Plants.

G. Regulatory Guide 1.68, Initial Test Programs for Water-Cooled Nuclear Power Plants.

H. Regulatory Guide 1.68.2, Initial Startup Test Program to Demonstrate Remote Shutdown Capability for Water-Cooled Nuclear Power Plant.

I. Regulatory Guide 1.68.3, Preoperational Testing of Instrument and Control Air Systems.

J. Regulatory Guide 1.79, Preoperational Testing of Emergency Core Cooling Systems for Pressurized Water Reactors.

K. Regulatory Guide 1.95, Protection of Nuclear Power Plant Control Room Operators Against an Accidental Chlorine Release.

L. Regulatory Guide 1.108, Periodic Testing of Diesel Generators Used as Onsite Electric Power Systems at Nuclear Power Plants.

M. Regulatory Guide 1.116, Quality Assurance Requirements for Installation, Inspection, and Testing of Mechanical Equipment and Systems.

N. Regulatory Guide 1.129, Maintenance Testing and Replacement of Large Lead Storage Batteries for Nuclear Power Plants.

O. Regulatory Guide 1.139, Guidance for Residual Heat Removal (For Comment).

P. Regulatory Guide 1.140, Design, Testing, and Maintenance Criteria for Normal Ventilation Exhaust System Air Filtration and Absorption Units of Light-Water-Cooled Nuclear Power Plant.]

14.2.4 UTILIZATION OF REACTOR OPERATING AND TESTING EXPERIENCE IN THE DEVELOPMENT OF THE TEST PROGRAM

The general manager-Vogtle nuclear operations (GMVNO) is responsible for ensuring that the VEGP initial test program will be developed utilizing information gained from operating and testing experiences in other similar nuclear plants.

Various documents (listed below) will be reviewed on a continuing basis to determine whether there is any information that should be included in VEGP's initial test program. Any pertinent information will be routed to the GMVNO or his designee for review by the appropriate startup personnel and, if necessary, for inclusion in the initial test program. Specific procedures are included in the plant procedures manual:

- Nuclear Regulatory Commission inspection and enforcement bulletins, notices, and circulars.

- Significant event report.
- Significant operating event report.
- Westinghouse data letters and technical bulletins.

The GMVNO also receives operating and testing information through the Westinghouse Owners Group, from the Institute of Nuclear Power Operations, and from Georgia Power Company employees who have been temporarily stationed at other nuclear plants that are conducting the initial test program.

14.2.5 TRIAL USE OF PLANT OPERATING AND EMERGENCY PROCEDURES

The plant operating procedures are utilized, where applicable, during the initial test program to support testing, maintain plant conditions, and facilitate training. The trial use of operating procedures serves to familiarize operating personnel with systems and plant operation during the testing phase and also serves to ensure the adequacy of the procedures under actual or simulated operating conditions. Emergency procedures are verified to the extent practical during the plant startup as plant conditions, testing, and training warrant; surveillance tests are performed as the test program permits to demonstrate the adequacy of the procedure.

VEGP will conform with the requirements of TMI action item I.G.1 of NUREG-0737. During low-power natural circulation testing, certain operators will be designated to conduct the test to gain hands-on experience in this mode of operation. Other operators will observe the conduct of the test as the startup and shift schedules permit.

14.2.6 INITIAL FUEL LOADING AND INITIAL CRITICALITY

Prior to the commencement of fuel loading, required preoperational test procedures are evaluated, and appropriate remedial action is taken if the acceptance criteria are not satisfied.

14.2.6.1 Initial Fuel Loading

The reactor containment structure will be completed, and the containment integrity established and maintained during fuel loading.

Fuel handling tools and equipment will have been checked out and dry runs conducted in the use and operation of the equipment.

The reactor vessel and associated components will be in a state of readiness to receive fuel. Water level will be maintained above the bottom of the nozzles and recirculation maintained to ensure a uniform boron concentration.

Plant management exercises the overall responsibility and direction for initial core loading. The overall process of initial core loading is, in general, directed from the operating floor of the containment structure. Standard procedures for the control of personnel and the maintenance of containment security are established prior to fuel loading.

The core is assembled in the reactor vessel and is submerged in reactor grade water containing enough dissolved boric acid to maintain a calculated core effective multiplication factor of 0.95 or lower. The refueling cavity may be dry during initial core loading. Core moderator chemistry conditions (particularly, boron concentration) are prescribed in the core loading procedure

document and are verified periodically by chemical analysis of moderator samples taken prior to and during core loading operations.

Core loading instrumentation consists of two permanently installed source range (pulse type) nuclear channels and temporary incore pulse-type channels. The permanent channels, when responding, are monitored in the main control room by licensed plant operators; the temporary channels are installed in the containment structure and are monitored by test personnel. At least one permanent channel is equipped with an audible count rate indicator. Both plant channels have the capability of displaying the neutron flux level on a strip chart recorder. The temporary channels indicate on scalers, with a minimum of one channel recorded on a strip chart recorder.

Minimum count rates of at least 1/2 counts/sec, attributable to core neutrons, are required on at least two of the available pulse-type nuclear channels at all times following installation of the initial nucleus of fuel assemblies. A response check of nuclear instruments to a neutron source shall be performed within 8 h prior to loading of the core or upon resumption of loading if delay is for more than 8 h.

At least two artificial neutron sources are introduced into the core at appropriate specified points in the core loading program to ensure a detector response of at least 1/2 counts/sec attributable to neutrons.

Fuel assemblies together with inserted components (control rod assemblies, burnable poison inserts, source spider, or thimble plugging devices) are placed in the reactor vessel one at a time according to a previously established and approved sequence which was developed to provide reliable core monitoring with minimum possibility of core mechanical damage. The core loading procedure documents include detailed tabular check sheets which prescribe and verify the successive movements of each fuel assembly and its specified inserts from its initial position in the storage racks to its final position in the core. Multiple checks are made of component serial numbers and types at successive transfer points to guard against possible inadvertent exchanges or substitutions of components, and fuel assembly status boards are maintained throughout the core loading operation.

An initial nucleus of approximately eight fuel assemblies, one of which contains a neutron source, is the minimum source-fuel nucleus which permits subsequent meaningful inverse count-rate monitoring. This initial nucleus is determined by calculation to be markedly subcritical ($k_{eff} \leq 0.95$) under the required conditions of loading.

Each subsequent fuel addition is accompanied by detailed neutron count-rate monitoring to determine that the just-loaded fuel assembly does not excessively increase the count rate and that the extrapolated inverse count-rate ratio is not decreasing for unexplained reasons. The results of each step are evaluated before the next prescribed step is started. The final as-loaded core configuration is subcritical ($k_{eff} \leq 0.95$) under the required loading conditions.

Criteria for safe loading require that loading operations stop immediately if:

A. An unanticipated increase in the neutron count rate by a factor of two occurs on all responding nuclear channels during any single loading step after the initial nucleus of fuel assemblies is loaded.

B. An unanticipated increase in the count rate by a factor of five occurs on any individual responding nuclear channel during any single loading step after the initial nucleus of fuel assemblies is loaded.

C. A decrease in boron concentration greater than 20 ppm is determined from two successive samples of reactor coolant system water until the decrease is explained.

An alarm in the containment and main control room is coupled to the source range channels with a setpoint equal to or less than five times the current count rate; this alarm automatically alerts the loading operation personnel of the high count rate and requires an immediate stop of operations until the situation is evaluated. Normally, the alarm used for this purpose is the containment evacuation alarm. In the event that the evacuation alarm is actuated during core loading and the alarm is validated by a second responding detector, the containment is evacuated. After it has been determined that no hazards to personnel exist, preselected personnel are permitted to reenter the containment vessel to evaluate the cause and determine future action.

Core loading procedures specify the condition of fluid systems to prevent inadvertent dilution of the reactor coolant, specify the movement of fuel to preclude the possibility of mechanical damage, prescribe the conditions under which loading can proceed, identify responsibility and authority, and provide continuous and complete fuel and core component accountability.

14.2.6.2 Postloading Tests

Upon completion of core loading, the reactor upper internals and the pressure vessel head are installed, and additional mechanical and electrical tests are performed prior to initial criticality. The final pressure test is conducted after filling and venting is completed to check the integrity of the vessel head installation.

Mechanical and electrical tests are performed on the control rod drive mechanisms (CRDMs). These tests include a complete operational checkout of the mechanisms and calibration of the rod position indication system.

Tests are performed on the reactor trip circuits to test manual trip operation. The actual control rod assembly drop times are measured for each control rod assembly. The reactor control and protection system is checked with simulated signals to produce a trip signal for the various conditions that require plant trip.

At times that the CRDMs are being tested, the boron concentration in the coolant moderator is maintained such that the shutdown margin requirements specified in the Technical Specifications are met. During individual rod cluster control assembly or rod cluster control bank motion, source range instrumentation is monitored for unexpected changes in core reactivity.

A complete functional electrical and mechanical check is made of the incore nuclear flux mapping system at the operating temperature and pressure.

After final precritical tests, nuclear operation of the reactor begins. This final phase of startup and testing includes initial criticality, low-power testing, and power level escalation. The purpose of these tests is to establish the operational characteristics of the unit and core, to acquire data for the proper calibration of setpoints, and to ensure that operation is within license requirements. A brief description of the testing is presented in the following sections. Table 14.2.1-2 summarizes the tests which are performed from initial core loading to rated power.

Precritical tests shall be completed prior to initial criticality and the results evaluated. Prerequisites for performing a test are specified in the individual test procedure. The sequence of testing is outlined in a startup test sequence, such that required prerequisite testing is completed prior to performing subsequent testing. Any special test instruments required are specified to be installed, calibrated, and checked in the test procedure that specifies the test equipment. Where these test instruments are not left installed for future use, they are removed from the systems and removal is verified.

Initial criticality is achieved by a combination of shutdown and control bank withdrawal and reactor coolant system (RCS) boron concentration reduction. A predicted boron concentration for criticality will be determined for the precritical shutdown and control bank configuration specified in the initial criticality procedure. This configuration will require all banks to be fully withdrawn, with the exception of the last control bank, which will remain partially inserted to provide effective control when criticality is achieved.

Prior to beginning the approach to initial criticality, a neutron count rate of at least 1/2 count/s shall register on the startup channels, and the signal-to-noise ratio should be known to be greater than two. Core response during rod bank withdrawal and subsequent RCS boric acid concentration reduction will be monitored in the main control room by observing the change in neutron count rate as indicated by the permanent startup nuclear instrumentation.

Inverse count-rate ratio monitoring, using data from the normal plant source range instrumentation, is used as an indication of the proximity and rate of approach to criticality. Inverse count-rate ratio data are plotted as a function of rod bank position during rod motion and as a function of primary water addition during RCS boron concentration reduction.

Initially, the shutdown and control banks for control rods are withdrawn incrementally in the normal withdrawal sequence, leaving the last withdrawn control bank inserted as specified in the procedure.

The boron concentration in the RCS is then reduced. Criticality is achieved during boron dilution or by subsequent rod withdrawal following boron dilution. The rate of boron dilution, and hence the rate of approach to criticality, may be reduced as the reactor approaches criticality to ensure that effective control is maintained. At no time may a sustained startup rate of 1 decade/min be exceeded. Throughout this period, samples of the primary coolant are obtained and analyzed for boron concentration.

14.2.6.3 Low-Power Testing

Following initial criticality, a program of reactor physics measurements is undertaken to verify that the basic static and kinetic characteristics of the core are as expected, and that the values of the kinetic coefficients assumed in the safeguards analysis are conservative.

Procedures specify the sequence of tests and measurements to be conducted, and the conditions under which each is to be performed to ensure both safety of operation and the validity and consistency of the results obtained. If test results deviate significantly from design predictions, if unacceptable behavior is revealed, or if unexplained anomalies develop, the plant is brought to a safe, stable condition and the situation reviewed to determine the course of subsequent plant operation.

These measurements are made at low power and primarily at or near normal operating temperature and pressure. Measurements are made to verify the calculated values of control rod bank reactivity worths, the isothermal temperature coefficient under various core conditions, differential boron concentration reactivity worth, and critical boron concentration as functions of control rod configuration. In addition, measurements of the relative power distributions are made, and concurrent tests are conducted on the instrumentation, including source and intermediate range nuclear channels.

Gamma and neutron radiation surveys are performed at selected points throughout the plant. Periodic sampling is performed to verify chemical and radiochemical analysis of the reactor coolant.

14.2.6.4 Power Level Ascension

After the operating characteristics of the reactor have been verified by low-power testing, a program of power level ascension brings the unit to its full-rated power level in successive stages. At each successive stage, hold points are provided to evaluate and approve test results prior to proceeding to the next stage. The minimum test requirements for each successive stage of power ascension are specified in the applicable startup test procedures.

Measurements are made to determine the relative power distribution in the core as functions of power level and control assembly bank position.

Secondary system heat balance measurements ensure that the indications of power level are consistent and provide for calibration of the power range nuclear channels. The ability of the RCS to respond effectively to signals from primary and secondary instrumentation under a variety of conditions encountered in normal operations is verified.

At prescribed power levels, the dynamic response characteristics of the primary and secondary systems are evaluated. System response characteristics are measured for design step-load changes, rapid load reduction, and plant trips.

Adequacy of radiation shielding is verified by gamma and neutron radiation surveys at selected points throughout the plant at various power levels. Periodic sampling is performed to verify the chemical and radiochemical analysis of the reactor coolant.

14.2.7 TEST PROGRAM SCHEDULE

The sequential schedule for the preoperational and startup test program is provided in figure 14.2.7-1.

The general manager-Vogtle nuclear operations or his designee is responsible for the preparation, review, and maintenance of detailed schedules for testing.

Preoperational testing is scheduled to commence approximately 18 months prior to fuel loading. Included in the Unit 1 preoperational test schedule is the time needed to energize and test the plant's switchyard and associated equipment. The preoperational and construction acceptance tests are performed and sequenced during this period as a function of system turnover, system interrelationships, and acceptance for testing.

Startup testing is scheduled to be conducted over a period of approximately 4 to 6 months, commencing with fuel loading. The sequential schedule for startup testing ensures, insofar as practicable, that test requirements are completed prior to exceeding 25-percent power for all plant structures, systems, and components that are relied upon to prevent, limit, or mitigate the consequences of postulated accidents.

As shown in figure 14.2.7-1, there are 18 months between the fuel loading dates of Unit 1 and Unit 2. This results in Unit 2 test phases starting at least 18 months after the equivalent Unit 1 test phase. Also, Unit 1 is scheduled to complete fuel loading prior to construction acceptance testing and preoperational tests starting on Unit 2. This prevents any significant division of responsibility or dilution in the required personnel to conduct either unit's test program.

14.2.8 INDIVIDUAL TEST DESCRIPTIONS

The test abstracts provided are intended to show what systems/components will be tested and what minimum information will be verified. Information to be verified has been listed under the test abstract most likely to cover the required testing.

14.2.8.1 Preoperational Test Procedures

The following paragraphs give the test abstracts for each preoperational test listed in table 14.2.1-1.

14.2.8.1.1 Main Steam System Preoperational Test

A. Objective

To demonstrate the functional performance of the main steam system, including the following:

1. Acceptable closing times of the main steam isolation valves (MSIVs) and the MSIV bypass valves with steam lines at normal operating temperature and pressure.
2. Operability of the power-operated atmospheric relief valves with steam lines at normal operating temperature and pressure.
3. Operability of the main steam safety valves in a bench test at temperature and pressure with steam as the pressurizing fluid.

B. Prerequisites

The required portions of the following prerequisites are completed as necessary to support the preoperational test:

1. Construction acceptance testing completed.
2. Component testing and instrument calibration completed.
3. Test instrumentation available and calibrated.
4. Support systems available.

C. Test Method

1. Verify manual and automatic system controls.
2. The MSIV system's manual and automatic operation and MSIV and MSIV bypass valve closure times will be demonstrated during hot functional testing.
3. Demonstrate operability of main steam safety valves and seal leaktightness in a bench test at temperature and pressure with steam as the pressurizing fluid.
4. Demonstrate operability of the power-operated atmospheric relief valves during hot functional testing.
5. Verify alarms and status lights are functional.

D. Acceptance Criterion

The main steam system operates as described in section 10.3.

14.2.8.1.2 Steam Dump Preoperational Test

A. Objective

To demonstrate operation of the steam dump system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing completed.
2. Component testing and instrument calibration completed.
3. Test instrumentation available and calibrated.
4. Support systems available.

C. Test Method

1. Verify manual and automatic system controls.
2. Verify alarms, indicating instruments, and status lights are functional.
3. Demonstrate dynamic operation of steam dump to condenser during hot functional testing.

D. Acceptance Criterion

The steam dump system operates as described in paragraph 7.7.1.8.

14.2.8.1.3 Condensate System Preoperational Test

A. Objectives

1. To demonstrate that the condensate pumps provide rated flow and pressure with operation of controls, alarms, and interlocks.
2. To demonstrate the automatic operation of system valves along with status lights and alarms.
3. To demonstrate the operation of the hotwell level control system.

B. Prerequisites

1. Required construction acceptance testing has been completed.
2. Required electrical power supplies and control circuits are operational.
3. The turbine plant closed cooling water system, the demineralized water system, and the condensate storage tank are available for use.
4. The feedwater system is available to receive flow from the condensate pump discharge header.

C. Test Method

1. Simulate pump overloads and discharge pressure signals to verify alarms and pump operation.
2. Operate pumps and verify pump performance characteristics.
3. Verify operation, status lights, and alarms of system valves and switches.

D. Acceptance Criteria

1. Condensate pumps, control valves, and isolation valves perform as described in subsection 10.4.7.
2. The hotwell level control system operates as described in subsection 10.4.1.

14.2.8.1.4 Main Feedwater System Preoperational Test

A. Objective

To demonstrate the proper operation of the feedwater control system valves, the steam generator feedwater pumps (SGFPs), manual and automatic function indications, and alarms.

B. Prerequisites

1. Required construction activities including component testing, instrument calibration, and system flushing/cleaning are complete.
2. Required test equipment has been calibrated and is operable.
3. Required ac and dc power is available.
4. Instrument air is available to air-operated valves.
5. The main turbine steam seal system is available to provide seal steam and packing exhaust for the SGFP turbines.
6. The main condenser is available to receive steam flow from the SGFP turbine.
7. The condensate system is available to supply suction and seal injection for the SGFPs.
8. Steam is available to the SGFP turbines.
9. The closed cooling water system is available to provide cooling water to the SGFP lube oil coolers.

C. Test Method

1. Manual and automatic operation of the feedwater system control valves (MFCVs and CBVs) to demonstrate the modes of operation, interlocks, response, failure mode indications, and alarms.
2. The SGFPs are operated; pump performance and flow and status indications are recorded. Simulated signals are used to verify low feedwater header discharge pressure alarm annunciation and low suction pressure alarm and trip functions.

3. Simulated signals are used to verify the operation of the condensate miscellaneous drain tank level alarm annunciation.

D. Acceptance Criterion

The main feedwater system operates as discussed in subsection 10.4.7.

14.2.8.1.5 Motor-Driven Auxiliary Feedwater System Preoperational Test

A. Objectives

1. To demonstrate the ability of the motor-driven auxiliary feedwater system to supply feedwater to the steam generators.
2. To verify operation of system valves and components.
3. To verify the operation of the condensate storage degasifiers.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing completed.
2. Component testing and instrument calibration completed.
3. Test instrumentation available and calibrated.
4. Support systems available.

C. Test Method

1. Verify manual and automatic system controls.
2. Verify system flowpaths, flowrates, and operating parameters within limits.
3. Verify pump and valve operating parameters.
4. Verify instruments, status lights, indications, and alarms.
5. Verify 48-h continuous operation and operation of the HVAC system for the auxiliary feedwater pumphouse (paragraph 14.2.8.1.35).
6. Verify 1-h operation on restart after cooldown.

D. Acceptance Criterion

The motor-driven auxiliary feedwater system operates within design limits, as described in subsections 10.4.7 and 10.4.9.

14.2.8.1.6 Turbine-Driven Auxiliary Feedwater System Preoperational Test

A. Objective

To demonstrate that the turbine-driven auxiliary feedwater system supplies feedwater to the steam generators.

B. Prerequisites

The required portions of the following prerequisites are completed as necessary to support the preoperational test:

1. Construction acceptance testing completed.
2. Component testing and instrument calibration completed.
3. Test instrumentation available and calibrated.
4. Support systems available.

C. Test Method

1. Verify manual and automatic system controls.
2. Verify system flowpaths, flowrates, and operating parameters within limits.
3. Verify turbine, pump, and valve operating parameters.
4. Verify that instruments, status lights, indications, and alarms are functional.
5. Verify 48-h continuous operation and operation of the HVAC system (paragraph 14.2.8.1.35).
6. Verify 1-h operation on restart after cooldown.
7. Verify cold quick start.
8. Verify start and 2-h operation with loss of ac power. This will include defeat of ventilation fan.

D. Acceptance Criterion

The turbine-driven auxiliary feedwater system operates within design limits, as described in subsection 10.4.9.

14.2.8.1.7 Reactor Coolant System (RCS) Preoperational Test

A. Objectives

1. To verify the control circuitry and operation of the system valves.
2. To demonstrate the operation of the head vent system.
3. To verify the operation of the boron analyzer.
4. To verify the operability of the pressurizer safety valves in a bench test at temperature and pressure with steam as the pressurizing fluid.

B. Prerequisite

Required component testing and instrument calibration are complete.

C. Test Method

1. The control circuitry and operation of system valves will be verified.
2. During a fill and vent of the RCS, the head vent will be operated.

3. The boron analyzer will be functionally tested using known concentrations of boric acid solutions.

4. The operability of the pressurizer safety valves will be demonstrated in a bench test at temperature and pressure with steam as the pressurizing fluid.

D. Acceptance Criteria

1. The valves respond to control signals. (See drawings 1X6AA02-225, 1X6AA02-226, 1X6AA02-227, 1X6AA02-228, 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, 1X6AA02-233, 1X6AA02-234, 1X6AA02-235, 1X6AA02-236, 1X6AA02-237, 1X6AA02-238, 1X6AA02-239, 1X6AA02-240, 1X6AA02-494, 1X6AA02-495, 1X6AA02-496, and 1X6AA02-519.)

2. The readings of the boron analyzer are in accordance with the known boron concentration. (See table 7.7.1-2.)

3. The pressurizer safety valves actuate at a pressure of 2485 $\pm$ 25 psig.

14.2.8.1.8 Pressurizer Relief Tank (PRT) Preoperational Test

A. Objectives

1. To demonstrate that design PRT spray flow can be obtained against design backpressure in the tank.

2. To verify the operation of the PRT nitrogen pressurization system.

3. To demonstrate the filling and draining operation of the PRT.

4. To demonstrate a cooldown of the PRT using the reactor coolant drain tank (RCDT) heat exchanger.

5. To demonstrate the operation of system valve control circuitry.

B. Prerequisites

1. Required component testing and instrument calibration is complete.

2. The RCDT part of the liquid waste processing system is available to recirculate the PRT.

3. Reactor makeup is available to supply water to the PRT.

4. The nitrogen gas system is available to supply the PRT.

5. The plant is in hot functional testing for the PRT cooldown demonstration.

C. Test Method

1. With design backpressure in the PRT, the required spray flow is pumped to the PRT.

2. While draining and filling the PRT, the nitrogen pressurization system operation is observed.

3. During hot functional testing, the PRT is cooled down through the RCDT heat exchanger.

4. The control circuitry for system valves will be verified.

D. Acceptance Criteria

1. The required spray flow is obtained during design conditions. (See NSSS Startup Manual, SU-2.1.4.)
2. The pressurization system properly pressurizes and vents the PRT. (See NSSS Startup Manual, SU-2.1.4.)
3. The PRT can be cooled down by the RCDT heat exchanger. (See table 5.4.11-1.)
4. The system valves operate as designed. (See NSSS Startup Manual, SU-2.1.4.)

14.2.8.1.9 RCS Hydrostatic Preoperational Test

A. Objective

To verify the integrity and leaktightness of the RCS and the high-pressure portions of associated systems by performing a hydrostatic pressure test in conformance with Section III of the American Society of Mechanical Engineers (ASME) Code.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. The reactor coolant pumps are available to support this test.
3. The positive displacement charging pump or a test pump is available.
4. The reactor vessel's lower and upper internals and the closure head are installed. The studs are tensioned to the design value for the associated hydrostatic test pressure.
5. Temporary temperature instrumentation is installed for measuring the temperature of the steam generator tube sheets, the bottom of the pressurizer, and the closure flange of the reactor vessel.
6. System relief valves and instrumentation that are within the test boundaries are either recalibrated or are verified to be able to withstand the hydrostatic test pressure.
7. Welds that are within the test boundaries have been verified as ready for hydrostatic testing.

C. Test Method

1. The system will be filled using the chemical and volume control system (CVCS) and residual heat removal system (RHRS).
2. The reactor coolant pumps will be operated, as required, to vent the system and to establish the required hydro temperature.
3. The system will then be pressurized to test pressure in stages; system welds, piping, and components will be monitored for leaks at each stage.

D. Acceptance Criterion

There shall be no leaks at welds or piping within the test boundaries during the final inspection. However, evidence of leaks at valves, flanges, or mechanical

fittings does not invalidate the hydro. If there is evidence of leakage at a weld or piping, the leak must be repaired prior to the final inspection, or the leak may be isolated, repaired, and retested at a later date.

14.2.8.1.10 RCS Leak Rate Preoperational Test

A. Objectives

1. To determine during hot functional testing the amount of identified and unidentified leakage from the RCS and verify that the leakage is within allowable limits.
2. The operation of the containment particulate and radioactive gas monitoring portions of the leak detection system are verified in the process radiation monitoring system preoperational test.

B. Prerequisites

Hot functional testing is in progress for the determination of the amount of leakage.

C. Test Method

The identified and unidentified RCS leakage rates are determined by monitoring the system water inventory over a specified period of time.

D. Acceptance Criteria

1. The identified leakage rate is less than 10 gal/min.
2. The unidentified leakage rate is less than 1 gal/min.

14.2.8.1.11 Pressurizer Pressure and Level Control Preoperational Test

A. Objectives

1. To demonstrate the stability and response of the pressurizer pressure control system, including the verification of alarm and control functions.
2. To demonstrate the stability and response of the pressurizer level control system, including the verification of alarm and control functions.
3. To perform preliminary adjustment of the pressurizer continuous spray flow valves. Final adjustment of the continuous spray flow valves is performed during startup testing.
4. To demonstrate operation and valve response times for the pressurizer power-operated relief valves.
5. To demonstrate operation of the automatic block valves.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. Required electrical power supplies and control circuits are energized and operational.

3. The plant is at normal operating temperature and pressure with reactor coolant pumps running, and hot functional testing is in progress.
4. The pressurizer continuous spray flow valves are full open.
5. The letdown and charging portions of the CVCS are available to vary pressurizer level.
6. The PRT is at a normal operating level and is aligned for normal operation.
7. The liquid waste processing system is available for cooldown of the PRT via the RCDT heat exchanger.

C. Test Method

1. Pressurizer pressure and level are varied separately, and the ability of the control system to automatically control and stabilize pressurizer pressure and level is verified.
2. Pressurizer pressure and level are varied separately, and the pressure and level alarm and control setpoints are verified.
3. Pressurizer continuous spray flow valve initial setting determination is made, and proper spray line temperatures are verified.
4. Pressurizer power-operated relief valves are operated, and response times are determined.
5. Pressurizer block valves are operated and their interlocks demonstrated.

D. Acceptance Criteria

1. The pressurizer pressure and level control systems respond to an increase or decrease in pressurizer pressure and level. (See drawings 1X6AA02-225, 1X6AA02-226, 1X6AA02-227, 1X6AA02-228, 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, 1X6AA02-233, 1X6AA02-234, 1X6AA02-235, 1X6AA02-236, 1X6AA02-237, 1X6AA02-238, 1X6AA02-239, 1X6AA02-240, 1X6AA02-494, 1X6AA02-495, 1X6AA02-496, and 1X6AA02-519.)
2. Pressurizer pressure and level control system alarm and control functions have been operationally verified. (See drawings 1X6AA02-225, 1X6AA02-226, 1X6AA02-227, 1X6AA02-228, 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, 1X6AA02-233, 1X6AA02-234, 1X6AA02-235, 1X6AA02-236, 1X6AA02-237, 1X6AA02-238, 1X6AA02-239, 1X6AA02-240, 1X6AA02-494, 1X6AA02-495, 1X6AA02-496, and 1X6AA02-519.)
3. Pressurizer continuous spray flow valve initial settings are determined, and spray line temperatures are maintained within design specifications (precautions, limitations, and setpoints).
4. Pressurizer power-operated relief valve operating times are within design specifications. (See NSSS Startup Manual, SU-2.1.2.)
5. Pressurizer power-operated relief block valve low pressure isolation and manual override are operable.

14.2.8.1.12 Reactor Coolant Pump Initial Operation Preoperational Test

A. Objectives

1. To measure and record system and operating parameters of the reactor coolant pumps during cold hydrostatic testing.
2. To measure and record pump operating parameters during hot functional testing.
3. To verify the operation of the associated oil lift pumps.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. The CVCS is available to provide seal water to the reactor coolant pumps.
3. The auxiliary component cooling water (ACCW) system is available for cooling.
4. The RCS is filled and pressurized.

C. Test Method

1. The reactor coolant pumps and associated oil lift pumps will be operated during cold hydrostatic testing, and operating data will be recorded.
2. The reactor coolant pumps and associated oil lift pumps will be operated during hot functional testing, and operating data will be recorded at various temperature plateaus.

D. Acceptance Criterion

The reactor coolant pump and oil lift pump operating characteristics are within design specifications as described in subsection 5.4.1.

14.2.8.1.13 RCS Hot Functional Preoperational Test[(1)]

A. Objectives

1. To operate the RCS at full-flow conditions for a minimum of 240 h to provide the necessary vibration cycles on the reactor vessel's internal components prior to their final inspection before core loading. At least one-half of this operating time must occur with the reactor coolant temperature at or above 515°F.
2. To provide coordination and establishment of initial conditions necessary for the conduct of those preoperational tests to be performed during heatup, normal operating temperature and pressure, and cooldown of the RCS.
3. To provide the plant operators an opportunity to actually control the plant from an operations viewpoint and put into practice the plant operation

[1] See footnote 10 of table 14.2.1-1 for Unit 2 specific scope.

procedures for the RCS, auxiliary systems, and principal secondary systems prior to nuclear operation.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. The reactor vessel's internals and head are installed.
3. All systems and components required to support heatup, operation at normal temperature and pressure, and cooldown of the RCS are available.
4. The diesel generators are operable and are ready for emergency power requirements.

C. Test Method

1. The plant is taken from cold shutdown conditions to hot standby for the first time prior to placing nuclear fuel in the reactor core.
2. The required energy input for this operation will be from operating reactor coolant pumps and the pressurizer heaters. The preoperational tests that require these hot and/or dynamic conditions will be conducted during hot functional testing as coordinated by this procedure.

D. Acceptance Criterion

1. The RCS has operated at full-flow conditions for a minimum of 240 h with at least half of it above 515°F.
2. The acceptance criteria for individual systems are a part of the individual test procedures sequenced by this procedure.

14.2.8.1.14 Reactor Internals and RCS Vibration Test

A. Objectives

1. To demonstrate that dynamic effects experienced during both steady flow and flow-induced transients will not cause structural damage or malfunction of the RCS, pressurizer surge line, and pressurizer relief line piping and components.
2. To perform a post-hot functional inspection of the reactor vessel and internals for vibration-induced damage and to document inspection results.

B. Prerequisites

1. Special test instrumentation for monitoring system and component vibration is installed at the required locations.
2. RCS hot functional testing is in progress.
3. The inspection of the reactor vessel and internals will be performed following cooldown from hot functional testing at full flow for greater than 240 h.

C. Test Method

1. The dynamic response of the RCS, pressurizer surge line, and pressurizer

pressurizer relief line piping and components is verified during both steady flow and flow-induced transients.

2. Following cooldown from hot functional testing, the reactor vessel and internals are inspected and the results documented.

D. Acceptance Criteria

1. The dynamic response of the RCS, pressurizer surge line, and pressurizer relief line piping and components is acceptable when compared with analytical results.

2. An inspection of the reactor vessel and internals, following 240 h of full-flow operation, has been performed, and inspection results are in accordance with drawing AX6DD401. (See paragraph 3.9.B.2.1.)

14.2.8.1.15 Resistance Temperature Detectors (RTDs)/Thermocouple Cross Calibration Preoperational Test

A. Objectives

1. To collect data to verify expected resistance versus temperature characteristics of installed RTDs and millivolt versus temperature characteristics of thermocouples.

2. To determine RTD deviation from RCS average temperature for individual RTDs and isothermal corrections for individual thermocouples.

3. To demonstrate that temperature instrumentation is reading correctly after disconnecting special test instrumentation and reconnecting RTDs to normal plant terminations.

B. Prerequisites

1. The reactor vessel upper internals are installed and thermocouple wiring is complete.

2. The temperature instruments have been aligned in accordance with manufacturer's calibration data.

3. Hot functional testing is in progress.

4. The special test instrumentation required for test performance is available and has been calibrated.

5. The plant computer is operational to obtain thermocouple data in the trend mode while taking RTD readings.

C. Test Method

1. RTD resistance data is taken at specified temperatures during hot functional testing, and the RTD resistance versus temperature characteristics is verified.

2. Thermocouple data is taken at the same temperature points used for recording RTD data, and the thermocouple millivolt versus temperature characteristics is verified.

3. RTD deviation from RCS average temperature and isothermal corrections for individual thermocouples are determined from the recorded data.

4. After disconnecting special test instrumentation and reconnecting RTDs to normal plant terminations, temperature data are taken to verify that all temperature instruments are reading correctly.

D. Acceptance Criteria

1. Test data are recorded for future alignment purposes only, and specific acceptance criteria are not provided. However, any individual RTD reading that differs from the calculated average temperature by a specified amount will not be used for average temperature calculations.

2. All temperature instruments disconnected for measurement indicate approximately reactor coolant system average temperature when their RTDs are reconnected to normal plant terminations.

14.2.8.1.16 RTD Bypass Flow Measurements

A. Objectives

1. To determine the flowrate necessary to achieve the design reactor coolant transport time in each RTD bypass loop.

2. To measure the flowrate in each RTD bypass loop and ensure that the transport times are acceptable.

B. Prerequisites

1. Required component testing and instrument calibration are complete.

2. The installed pipe length measurements are made with the plant cold before insulation is installed.

3. The bypass loop flow measurement will be performed during hot functional testing.

C. Test Method

1. The flowrate necessary to achieve the design reactor coolant transport time for each hot and cold leg bypass loop is calculated utilizing the piping length of each leg.

2. The hot and cold leg RTD bypass loop flow data are SU-2.1.9.)

D. Acceptance Criterion

The flowrate in each hot and cold leg RTD bypass loop is within design specifications. (See NSSS Startup Manual, SU-2.1.9.)

14.2.8.1.17 RHRS Preoperational Test

A. Objectives

1. To demonstrate operation of the residual heat removal (RHR) pumps and system valves and their associated control and interlock circuitry, including response to simulated safety signals.

2. To demonstrate RHR pump and system performance during discharge to the reactor coolant hot and cold loops and during RCS recirculation and to verify the head/flow characteristics of each installed pump.
3. To demonstrate RHR pump cold leg injection branch line orifice sizing and flowrates and pump runout flowrates.
4. To demonstrate RHR pump runout flowrates during reactor coolant hot and cold leg recirculation.
5. To demonstrate the time required for the RHR pumps to reach their rated flow.
6. To demonstrate RHRS operation during RCS heatup with letdown through the RHRS (to be performed during hot functional test).
7. To demonstrate RHRS operation during RCS cooldown (to be performed during hot functional test).
8. To demonstrate that each RHR pumproom cooler fan starts and stops when the associated RHR pump is started and stopped.
9. To demonstrate operation of the RHRS during filling and draining of the refueling cavity (to be performed during integrated control logic (safety injection) testing).

B. Prerequisites

1. Required component testing, instrument calibration, and system flushing/cleaning are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. The refueling water storage tank (RWST) contains an adequate supply of demineralized water for performance of the test.
4. The reactor vessel head is removed and provisions have been made to remove water from the vessel.
5. For the RCS recirculation portion of the test, RCS cleaning is complete, and vessel water level is above the nozzles.
6. Temporary suction strainers are installed for both RHR pumps.
7. The component cooling water (CCW) system is available to supply water to the RHR heat exchangers and the pump seal coolers.
8. The nuclear service cooling water (NSCW) system is available to supply cooling water to the RHR pump motors.

C. Test Method

1. System component control and interlock circuits are verified, including the operation of the RHR pumps and systems valves on receipt of simulated safety signals.
2. RHR pump and system performance characteristics are verified during discharge to the reactor coolant hot and cold loops and during RCS recirculation.

3. RHR pump cold leg injection branch line orifice sizing, branch line flowrates, and pump runout flowrates are verified during simulated cold leg injection.
4. RHR pump runout flowrates are verified during reactor coolant hot and cold leg recirculation.
5. RHRS operation is verified during RCS heatup and cooldown in conjunction with the RCS hot functional test.
6. Operation of the RHRS during filling and draining of refueling cavity is verified in conjunction with integrated control logic (safety injection) testing.
7. RHR pumproom cooler fan and RHR pump interlocking is verified during pump operation.

D. Acceptance Criteria

1. RHRS components respond properly to normal control and interlock signals and to simulated safety signals. (See NSSS Startup Manual, SU-2.4.1.)
2. RHR pump and system performance characteristics are within design specifications. (See NSSS Startup Manual, SU-2.3.3.)
3. RHR pump cold leg injection branch line orifice sizing, branch line flowrates, and RHR pump runout flowrates are within design specifications. (See NSSS Startup Manual, SU-2.3.3.)
4. RHR pump runout flowrates during reactor coolant hot and cold leg recirculation are within design specifications. (See NSSS Startup Manual, SU-2.3.3.)
5. The time required for each RHR pump to reach rated flow conditions is within design specifications. (See NSSS Startup Manual, SU-2.3.3.)
6. RHR pump and room cooler fan interlock operates properly. (See drawings 1X5DN030-1, 1X5DN030-3, 1X5DN030-4, 1X5DN030-5, and 1X5DN065-1.)
7. The RHR system functions as designed during RCS heatup and cooldown. (See NSSS Startup Manual, SU-2.4.1.)
8. The RHR functions as designed during refueling cavity filling and draining. (See paragraph 5.4.7.2.3.6.)

14.2.8.1.18 Chemical and Volume Control System

A. Objectives

1. To demonstrate the operation and verify the operating characteristics of the positive displacement charging pump.
2. To demonstrate the ability of the CVCS to supply adequate seal water injection flow to the reactor coolant pumps.
3. To verify operation of the control circuitry of the charging pumps and system valves including their response to safety signals.

4. To demonstrate that each room cooler fan starts and stops with its respective charging pump.
5. To demonstrate that the auxiliary spray will cool the pressurizer during cooldown of the RCS.
6. To demonstrate that the volume control tank level control system operates as designed.

B. Prerequisites

1. The required component testing and instrumentation calibration are complete.
2. The NSCW system is available to provide cooling water to the centrifugal charging pumps.
3. The ACCW system is available to provide cooling water to the positive displacement charging pump and the seal water heat exchanger.
4. Initial pump testing will be performed with the reactor head removed. Auxiliary spray cooldown of the pressurizer will be made during hot functional testing.

C. Test Methods

1. The centrifugal and positive displacement charging pumps will be operated, the appropriate room cooler interlocks will be verified, and performance characteristics of the positive displacement charging pump will be verified.

   Performance characteristics of the centrifugal charging pumps will be verified in the safety injection preoperational test.
2. Seal injection throttle valves will be adjusted to establish flow within the required tolerances for each reactor coolant pump.
3. During hot functional testing, a cooldown of the pressurizer using auxiliary spray will be demonstrated.
4. The control circuitry of the charging pumps and system valves, including their response to safety signals, will be verified.
5. Levels and pressure in the volume control tank will be varied and system response will be recorded.

D. Acceptance Criteria

1. The positive displacement charging pump operating characteristics are within design specifications. (See table 9.3.4-2.)
2. Seal water injection flow to each reactor coolant pump has been established in accordance with the limits imposed in NSSS Startup Manual 2.3.3.
3. System valve and pump control circuits and interlocks operate as designed. (See drawings 1X6AA02-225, 1X6AA02-226, 1X6AA02-227, 1X6AA02-228, 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, 1X6AA02-233, 1X6AA02-234, 1X6AA02-235, 1X6AA02-236, 1X6AA02-237, 1X6AA02-238, 1X6AA02-239, 1X6AA02-240, 1X6AA02-494, 1X6AA02-495, 1X6AA02-496, and 1X6AA02-519.)

14.2.8.1.19 Letdown System Preoperational Test

A. Objectives

1. To verify operation of the normal and excess letdown systems.
2. To verify operation of the system valves and control circuitry including the response to safety signals.
3. To demonstrate the ability of the charging and letdown system to maintain solid RCS pressure control.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. The system tests will be performed during hot functional testing.

C. Test Method

1. Control circuitry for system valves including the response to safety signals will be verified.
2. The system will be operated during hot functional testing, and system performance characteristics will be recorded.
3. During hot functional test heatup, with the system solid, the charging and letdown system pressure control will be demonstrated by varying the charging and letdown flows and observing the system response.

D. Acceptance Criteria

1. System valves and interlocks operate as designed. (See NSSS Startup Manual SU-2.2.1; DC-1208; and FSAR table 6.2.4-1.)
2. System operating characteristics are within the design limitations (tables 9.3.4-1 and 9.3.4-2).

14.2.8.1.20 Boron Thermal Regeneration System Preoperational Test (See table 14.2.1-1 for Unit 2.)

A. Objectives

1. To verify the operation of the boron thermal regeneration system, including the chiller units, pumps, and other system components.
2. To demonstrate that the system will heat and cool the letdown as designed.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. Portions of this test involving the heating and cooling cycles will be performed during hot functional testing.

C. Test Method

1. Control circuitry and operation of system valves and motors will be verified.

2. The system will be operated during hot functional testing, and system performance characteristics will be recorded.

D. Acceptance Criteria

1. System valves, motors, and interlocks operate as designed. (See NSSS Startup Manual, SU-2.2.3.)

2. The system functions to heat and cool the letdown to within design limitations. (See NSSS Startup Manual, SU-2.2.3.)

14.2.8.1.21 Boric Acid Blending Operation

A. Objectives

1. To verify the operation of the boric acid transfer pumps and the ability of the boric acid blending system to make up at design flowrates and boric acid concentrations to the CVCS in each mode of operation.

2. To verify the operation of system valves and control circuitry, including the response to safety signals.

3. To demonstrate the heating system for the boric acid batch tank.

4. To demonstrate the ability of the boric acid system to supply an emergency boration flow to the charging pump suction.

B. Prerequisites

1. The required component testing and instrument calibration are complete.

2. The reactor makeup water system is available to supply water to the boric acid blending tee and batch tank.

3. Auxiliary steam is available to heat the batch tank as required.

4. A charging pump must be available for operation.

5. Parts of the system that contain boric acid at 4-percent concentration will have to be kept at least at 65°F.

C. Test Method

1. The boric acid pumps will be operated, and performance characteristics will be verified.

2. The control circuitry for system valves and motors will be verified.

3. The makeup and blending system controls will be demonstrated in all modes of operation: auto makeup, dilute, alternate dilute, borate, and manual.

4. Emergency boration to charging pump suction operation will be demonstrated.

5. Steam heating of the boric acid batch tank will be demonstrated.

D. Acceptance Criteria

1. The boric acid transfer pump operating characteristics are within the design specifications. (See table 9.3.4-2.)

2. System valves and interlocks operate as designed. (See NSSS Startup Manual, SU-2.2.2.)
3. The flowrates and boric acid concentrations associated with the blender are within the design specifications for each operating mode. (See NSSS Startup Manual, SU-2.2.1.)
4. The emergency boration flowrate to the charging pumps is within the design specifications. (See NSSS Startup Manual, SU-2.2.2.)

14.2.8.1.22 Safety Injection System (SIS) Preoperational Test

A. Objectives

1. To demonstrate centrifugal charging pump performance and to verify that the head/flow characteristics of each installed pump are approximately the same.
2. To demonstrate operation of the safety injection pumps and system valves and their associated control circuitry including response to safety signals.
3. To demonstrate safety injection pump performance and to verify that the head/flow characteristics of each installed pump are approximately the same.
4. To demonstrate miniflowrate for the centrifugal charging pumps and safety injection pumps.
5. To position the centrifugal charging pump cold leg injection line throttling valves and to demonstrate injection branch line flow balancing and pump runout flowrate.
6. To position the safety injection pump cold and hot leg injection line throttling valves and to demonstrate injection branch line flow balancing and pump runout flowrates.
7. To demonstrate the time required for the centrifugal charging and safety injection pumps to reach their rated flow.
8. To demonstrate operation of the boron injection tank outlet motor-operated valves under maximum differential pressure conditions.
9. To demonstrate that each safety injection pump room cooler fan starts and stops when the associated safety injection pump is started and stopped.

B. Prerequisites

1. Required component testing, instrument calibration, and system flushing/cleaning are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. The RWST contains an adequate supply of demineralized water for performance of the test.

4. The reactor vessel head is removed and provisions have been made to remove water from the vessel.
5. Temporary suction strainers are installed for both safety injection pumps.
6. The NSCW system is available to supply cooling water to the centrifugal charging pump and safety injection pump motors and oil coolers.
7. The ACCW system is available to supply cooling water to the CVCS seal water heat exchanger.

C. Test Method

1. System component control and interlock circuits, including the operation of the safety injection pumps and system valves on receipt of simulated safety signals, are verified.
2. Centrifugal charging pump and safety injection pump performance characteristics are verified during simulated reactor coolant cold leg injection.
3. The centrifugal charging pump cold leg injection line throttling valves are properly positioned, and injection branch line flows and pump runout flowrates are verified.
4. The safety injection pump cold and hot leg injection line throttling valves are properly positioned, and injection branch line flows and pump runout flowrates are verified.
5. The boron injection tank outlet valves operated under conditions of maximum differential pressure, and operation is verified.
6. Safety injection pump room cooler fan and safety injection pump interlocking is verified during pump operation.

D. Acceptance Criteria

1. SIS components respond to normal control and interlock signals and to simulated safety signals. (See figure 6.3.2-1.)
2. The centrifugal charging pumps, safety injection pumps, and their associated system performance characteristics are within design specifications. (See NSSS Startup Manual, SU-2.3.3.)
3. Centrifugal charging pump and safety injection miniflowrates are within design specifications. (See NSSS Startup Manual, SU-2.3.3.)
4. The centrifugal charging pump cold leg injection line throttling valves are positioned to provide branch line flows and pump runout flowrates within design specifications. (See NSSS Startup Manual, SU-2.3.3.)
5. The safety injection pump cold and hot leg injection line throttling valves are positioned to provide branch line flows and pump runout flowrates within design specifications. (See NSSS Startup Manual, SU-2.3.3.)
6. The time required for the centrifugal charging pumps and safety injection pumps to reach rated flow conditions is within design specifications. (See NSSS Startup Manual, SU-2.3.3.)
7. Boron injection tank outlet valve response times are within design specifications. (See table 6.3.2-1.)

8. Safety injection pump and room cooler fan interlock operates. (See drawings 1X5DN030-1, 1X5DN030-3, 1X5DN030-4, 1X5DN030-5 and 1X5DN065-1.)

14.2.8.1.23 Safety Injection Check Valve Preoperational Test

A. Objectives

1. To demonstrate the operability of the various safeguard systems injection line check valves which are expected to be at higher than ambient temperature when the RCS is at normal operating temperature.
2. To demonstrate the integrity of the accumulator and safeguard systems injection line check valves by performing a backleakage check.
3. To demonstrate that each accumulator motor-operated isolation valve opens on reset of the pressurizer pressure (P-11) manual block.

B. Prerequisites

1. Required component testing, preoperational testing, and instrument calibrations are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. RCS pressure and temperature are established near the normal operating values, and hot functional testing is in progress.

C. Test Method

1. Flow from a safety injection pump is directed through the SIS test line to the RCS, and the accumulator and injection line check valve operability is verified.
2. SIS test line valves are aligned to provide a leakoff path from each of the accumulator and injection line primary and backup check valves, and valve backleakage is verified.
3. Accumulator motor-operated isolation valve opening is verified on reset of the pressurizer pressure (P-11) manual block.

D. Acceptance Criteria

1. The accumulator and injection line check valves operate as demonstrated by verification of flow through the check valves.
2. Check valve integrity test results (measured backleakage flowrate) conform to technical specification requirements (table 16.3.4-1).
3. Accumulator motor-operated isolation valves open automatically on pressurizer pressure (P-11) manual block.

14.2.8.1.24 Safety Injection Accumulator Testing

A. Objectives

1. To demonstrate operation of the safety injection accumulators and system valves and their associated control and interlock circuitry.
2. To demonstrate accumulator injection and obtain blowdown rate data during a low-pressure blowdown of each accumulator.

B. Prerequisites

1. Required component testing, instrument calibration, and system flushing/cleaning are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. The reactor vessel head and reactor internals have been removed, and provisions have been made to remove water from the vessel during the test.
4. The nitrogen system is available to supply nitrogen to the accumulators.
5. The RWST contains an adequate supply of demineralized water for the performance of this test.
6. The SIS is available for filling the accumulators.

C. Test Method

1. Accumulator system component control and interlock circuits, including their response to simulated safety injection and containment isolation signals, are verified.
2. Each accumulator is filled and partially pressurized with the discharge valves closed. The discharge valves are opened, discharging the accumulators to the reactor vessel, and accumulator blowdown rate data are determined.
3. Each accumulator discharge line isolation valve is operated under maximum differential pressure conditions of normal accumulator precharge pressure and zero reactor coolant pressure.

D. Acceptance Criteria

1. Safety injection accumulator system components respond to normal control and interlock signals and to simulated safety injection and containment isolation signals. (See drawings 1X4DB119, 1X4DB120, and 1X4DB121.)
2. Safety injection accumulator system valve response times are within design specifications. (See table 6.3.2-1.)

14.2.8.1.25 Boron Injection Tank and Recirculation Pump Test

This test has been deleted.

14.2.8.1.26 Containment Spray System Preoperational Test

A. Objectives

1. To demonstrate that the spray nozzles in the containment spray headers are clear of obstructions.
2. To demonstrate operation of the containment spray pumps and system valves and their associated control circuitry.
3. To demonstrate containment spray pump and system performance during operation in the test and simulated accident modes.
4. To demonstrate proper pump performance with simultaneous operation of two containment spray pumps, two centrifugal charging pumps, two safety injection pumps, and two RHR pumps.
5. To demonstrate that each containment spray pump room cooler fan starts and stops when the associated containment spray pump is started and stopped.
6. To demonstrate eductor performances by measuring spray additive flowrates.

B. Prerequisites

1. Required component testing, instrument calibration, and system flushing/cleaning are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. The RWST contains an adequate supply of demineralized water for the performance of this test.
4. Temporary suction strainers are installed for both containment spray pumps.
5. Temporary piping is installed from the spray header test connections.
6. A source of compressed air is available to pressurize the spray headers.
7. The NSCW system is available to supply cooling water to the containment spray pump motors.

C. Test Method

1. System component control circuits, including the operation of the containment spray pumps and system valves on receipt of simulated safety signals, are verified.
2. Containment spray pumps and system performance characteristics are verified during operation in the test and simulated accident modes.
3. Spray additive eductor operating characteristics are verified during system operation.
4. Airflow is initiated through the containment spray headers, and unobstructed flow is verified through each nozzle.
5. Containment spray pumproom cooler fan and containment spray pump interlocking is verified during system operation.

6. The containment spray pumps, centrifugal charging pumps, safety injection pumps, and RHR pumps are operated simultaneously, and adequate suction flow is verified.

D. Acceptance Criteria

1. Containment spray system components respond to normal control signals and to simulated containment spray actuation load, load shed, and load sequencing signals. (See drawings 1X3D-BD-J01A, 1X3D-BD-J01B, 1X3D-BD-J02A, 1X3D-BD-J02B, 1X3D-BD-J02C, and 1X3D-BD-J02D.)
2. Containment spray pump and system performance characteristics are within design specifications. (See NSSS Startup Manual, SU-4.1.8.)
3. Spray additive eductor operating characteristics are within design specifications. (See FSAR 6.2.2-4.)
4. All containment spray nozzles are unobstructed, as evidenced by air passing through each nozzle. (See NSSS Startup Manual, SU-4.1.8.)
5. Containment spray pump and room cooler fan interlocking operates properly. (See drawings 1X5DN030-1, 1X5DN030-3, 1X5DN030-4, 1X5DN030-5, and 1X5DN065-1.)
6. Pump pressures, flow, and available net positive suction head requirements are met for the containment spray pumps, centrifugal charging pumps, safety injection pumps, and RHR pumps when all eight pumps are run simultaneously. (See NSSS Startup Manual 2.3.3.)

14.2.8.1.27 Reactor Makeup Water Storage Tank and Degasifier System Preoperational Test

A. Objective

To demonstrate operation of the reactor makeup water storage tank and degasifier system.

B. Prerequisites

The required portions of the following prerequisites are completed as necessary to support the preoperational test:

1. Construction acceptance testing completed.
2. Component testing and instrument calibration completed.
3. Test instrumentation available and calibrated.
4. Support systems available.

C. Test Method

1. Verify manual and automatic system controls.
2. Verify system flowrates.

D. Acceptance Criterion

The reactor makeup water storage tank and degasifier system operates as described in subsection 9.2.7.

14.2.8.1.28 Refueling Water Storage Tank

A. Objectives

1. To demonstrate operation of refueling water storage and sludge mixing/heating system components and their associated control and interlock circuitry.
2. To demonstrate sludge mixing pump performance during circulation operation.
3. To demonstrate operation of the electric circulation heater.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. The demineralized water system is available for filling the RWST.

C. Test Method

1. System component control circuits, including response to normal control, interlock, and alarm signals, are verified.
2. The sludge mixing/heating system is operated to circulate the tank contents, and system performance characteristics are verified.
3. The electric circulation heater capacity is verified by measuring heater voltage and amperage during operation.

D. Acceptance Criteria

1. The RWST and sludge mixing/heating system components respond properly to normal control, interlock, and alarm signals.
2. The sludge mixing system performance characteristics are within system design specifications.
3. The electric circulation heater capacity is within design specifications.

14.2.8.1.29 Condenser Air Ejection System Preoperational Test

A. Objectives

To demonstrate operation of each steam jet air ejector (SJAE), vacuum pump, motor-actuated valve, and the condenser air ejection portion of the turbine building heating, ventilation, and air-conditioning (HVAC) system.

B. Prerequisites

1. Required construction acceptance testing is complete.
2. Permanently installed instrumentation necessary to support the test has been properly calibrated and is operable.
3. The turbine building HVAC system is available.
4. Appropriate ac and dc power sources are available.

5. The condensate and feedwater system, auxiliary steam system, and turbine plant cooling water system are available.
6. Turbine is on turning gears.

C. Test Method

1. Achieve and maintain a condenser vacuum to verify that:
   a. Mechanical vacuum pumps operate in accordance with design.
   b. The SJAEs operate in accordance with design.
2. Operate mechanical equipment in the system to verify operation.
3. Simulate a high radiation level and verify operation of the condenser air ejection filtration dampers.

D. Acceptance Criterion

Equipment listed in the objectives performs as described in subsections 9.4.4 and 10.4.2.

14.2.8.1.30 Circulating Water System Preoperational Test

A. Objectives

To demonstrate operation of the circulating water pumps, the system's associated motor-actuated valves, and the turbine plant cooling tower blowdown valve.

B. Prerequisites

1. Required construction acceptance testing has been completed.
2. Required permanently installed instrumentation has been properly calibrated and is operable.
3. Appropriate ac and dc power sources are available.
4. The following support systems are available:
   a. Instrument and service air systems.
   b. River makeup water pump system.
   c. Traveling water screen and wash system.
   d. Utility water system.
   e. Turbine plant cooling water system.

C. Test Method

1. Check circulating water pump controls.
2. Check motor-, control-, and air-operated valves.
3. Simulate high circulating water conductivity, and verify operation of the turbine plant cooling tower blowdown valves.
4. Test circulating water pumps and verify conformance to their design specifications.

D. Acceptance Criterion

Equipment associated with the circulating water system performs as described in subsection 10.4.5.

14.2.8.1.31 Spent Fuel Pool Cooling and Cleanup System (SFPCCS) Preoperational Test

A. Objectives

1. To demonstrate operation of the SFPCCS components and their associated control circuitry.
2. To demonstrate spent fuel pool cooling pump, purification pump, skimmer pump, and system performance during operation in their various modes.
3. To demonstrate that the spent fuel pool can be filled from the demineralized water system, RWST, or the reactor makeup water system.
4. To demonstrate that the antisiphon design prevents gravity drainage of the spent fuel pool.
5. To demonstrate the operability of the low level water alarm.
6. To verify the operability of the gates and drains and to perform leak tests of the gates, drains, and gaskets in the refueling canal and fuel storage pool.

B. Prerequisites

1. Required component testing, instrument calibration, and system flushing/cleaning are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. The CCW system is available to provide cooling water to the spent fuel pool cooling heat exchangers.
4. The RWST contains an adequate supply of demineralized water for the performance of this test.
5. The demineralized water system and reactor makeup water system are both available to provide fill water to the spent fuel pool.
6. The spent fuel pool gate has been installed to keep the fuel transfer canal dry, and the fuel transfer tube valve is closed.
7. The spent fuel pool is filled to the normal level with demineralized water.

C. Test Method

1. System component control circuits, including response of the spent fuel pool cooling pumps to simulated safety signals, are verified.
2. The spent fuel pool cooling, spent fuel pool purification, and spent fuel pool skimmer pumps are operated in their various modes, and the pump and system performance characteristics are verified.

3. The ability to partially fill the spent fuel pool from the RWST, demineralized water system, and reactor makeup water system is verified.

4. The spent fuel pool cooling system piping is altered to allow gravity draining of the spent fuel pool by siphon effect. The minimum level to which the spent fuel pool can be drained is verified.

5. The gates, drains, and gaskets in the refueling canal and fuel storage pool are checked for unacceptable leakage.

D. Acceptance Criteria

1. The SFPCCS components respond properly to normal control signals and to simulated load shed signals. (See drawing 1X4DB130.)

2. The spent fuel pool cooling pumps, purification pumps, skimmer pumps, and system performance characteristics are within design specifications. (See table 9.1.3-1.)

3. The spent fuel pool can be filled from the RWST, demineralized water system, or the reactor makeup water system. (See Design Basis DC-1213.)

4. The spent fuel cooling system antisiphon design prevents draining the spent fuel pool below an acceptable minimum level. (See Design Basis DC-1213.)

14.2.8.1.32 ACCW Preoperational Test

A. Objectives

1. To demonstrate the capability of the ACCW system to provide cooling water flow during shutdown with loss of offsite power and containment isolation modes of operation.

2. To verify operation of system valves and control circuitry and pump control circuitry, including the response to safety signals.

3. To demonstrate the operation and verify the operating characteristics of the ACCW pumps.

B. Prerequisites

1. Required component testing and instrument calibration are complete.

2. Demineralized water or reactor makeup water is available for system makeup.

3. NSCW is available.

C. Test Method

1. The control circuits of the ACCW pumps and valves will be verified.

2. The ACCW pumps will be operated, and performance characteristics will be verified.

3. The system flows will be verified in the shutdown with loss of power and containment isolation modes of operation.

D. Acceptance Criteria

1. The performance characteristics of the ACCW pumps are as designed. (See table 9.2.8.2.)
2. The pump and valve controls and interlocks operate as designed. (See drawings 1X4DB138-1, 1X4DB138-2 and 1X4DB139.)
3. Components that are supplied with ACCW receive flows that are within the design specifications in each operating mode. (See table 9.2.8-1.)

14.2.8.1.33 NSCW System Preoperational Test

A. Objective

To demonstrate operation of the NSCW system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing completed.
2. Component testing and instrument calibration completed.
3. Test instrumentation available and calibrated.
4. Support systems available.

C. Test Method

1. Verify manual and automatic system controls.
2. Verify system flowrates.
3. Verify alarms and status lights are functional.
4. Verify adequate NPSH for pump operation at tower basin minimum level and maximum design temperature.

D. Acceptance Criterion

The NSCW system operates within design limits, as described in subsection 9.2.1.

14.2.8.1.34 Component Cooling Water Preoperational Test

A. Objectives

1. To demonstrate the capability of the CCW system to provide cooling water during normal operation, normal cooldown, and postulated loss-of-coolant accident (LOCA) modes of operation.
2. To verify operation of system valves and control circuitry.
3. To demonstrate the operation and verify the operating characteristics of the CCW pumps, including their response to safety signals.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. Demineralized water or reactor makeup water is available for system makeup.
3. NSCW is available to cool the CCW motors.

C. Test Method

1. The control circuitry of the CCW pumps and valves will be verified.
2. The CCW system pumps will be operated, and performance characteristics will be verified.
3. System flows will be verified in each mode of operation.

D. Acceptance Criteria

1. The performance characteristics of the CCW pumps are within design specifications. (See table 9.2.2-1.)
2. Components that are supplied with CCW receive flows that are within the design specifications in each of the operating modes. (See table 9.2.2-2.)
3. The pump control and interlocks operate as designed. (See drawings 1X5DN091-1, 1X5DN091-2, 1X5DN091-3, 1X5DN092-1, and 1X5DN092-2.)

14.2.8.1.35 Auxiliary Feedwater Pumphouse HVAC System Preoperational Test

A. Objective

To demonstrate operation of the auxiliary feedwater pumphouse HVAC system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.
5. The turbine- and motor-driven auxiliary feedwater pumps are available for operation.

C. Test Method

1. Verify manual and automatic system controls.
2. Verify alarms, and status lights are functional.
3. Verify design airflow.
4. Each engineered safety feature (ESF) fan will be shown to meet design requirements with the motor-driven pump running and without assistance from any non ESF equipment.

5. The non-ESF fan will be tested with the turbine-driven pump operating.

D. Acceptance Criterion

1. The auxiliary feedwater pumphouse HVAC system operates as described in subsection 9.4.8.
2. The ESF fan maintains the motor-driven pump room at or below 120°F (9.4.8.2.3).
3. The non-ESF fan maintains the turbine-driven pump room at or below 120°F (9.4.8.2.3).

14.2.8.1.36 Fuel Handling Building HVAC System Preoperational Test

A. Objective

To demonstrate operation of the fuel handling building HVAC system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

1. Verify manual and automatic system controls.
2. Verify alarms, indicating instruments, and status lights are functional.
3. Verify design airflow.
4. Demonstrate ability of fuel handling building exhaust system to maintain a negative pressure in the fuel handling building.
5. Verify normal exhaust system isolation and post-accident exhaust system initiation on simulated high radiation signal from the exhaust ductwork radiation instrumentation.

D. Acceptance Criterion

The fuel handling building HVAC system operates as described in subsection 9.4.2.

14.2.8.1.37 (Deleted)

14.2.8.1.38 Essential Chilled Water System Preoperational Test

A. Objective

To demonstrate operation of the essential chilled water system.

B. Prerequisites

The required portions of the following prerequisites are completed as necessary to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

1. Verify manual and automatic system controls.
2. Verify alarms and status lights are functional.
3. Verify system flowrates.

D. Acceptance Criterion

The essential chilled water system operates as described in subsection 9.2.9.

14.2.8.1.39 Control Building HVAC Preoperational Test

A. Objective

To demonstrate operation of the control building HVAC system, including electrical tunnel ventilation and steam tunnel ventilation systems.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

1. Verify manual and automatic controls in the normal and emergency operating modes.
2. Verify alarms and status lights are functional.
3. Verify design airflow.
4. Verify control building isolation on simulated signal from the intake ductwork radiation instrumentation.

D. Acceptance Criterion

The control building HVAC system operates as described in section 6.4, subsections 9.4.1 and 9.4.5, and paragraph 9.4.9.2.

14.2.8.1.40 Auxiliary Building HVAC System Preoperational Test

A. Objective

To demonstrate operation of the auxiliary building HVAC system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

1. Verify manual and automatic controls in the normal and emergency operating modes.
2. Verify alarms and status lights are functional.
3. Verify design airflow.
4. Verify auxiliary building isolation on simulated containment ventilation isolation signal.

D. Acceptance Criterion

The auxiliary building HVAC system operates as described in subsection 9.4.3.

14.2.8.1.41 Diesel Generator Building HVAC Preoperational Test

A. Objective

To demonstrate operation of the engineered safety features (ESF) fans, non-ESF fans, non-ESF unit heaters, automatic dampers, fuel oil day tank exhaust fans, and all alarms and status indication.

B. Prerequisites

1. Required construction acceptance testing is complete.
2. Required system instrumentation and test equipment is available and calibrated.
3. Required diesel generator building HVAC system air handling ductwork has been tested and balanced as necessary.
4. Required ac and dc power is available.

C. Test Method

1. Cycle the thermostat for each non-ESF unit heater and fan and verify operation.
2. Simulate start and interlock signals for each ESF fan (with its breaker in the test position) and verify operation and annunciation.

3. Simulate high and low room air temperature signals and verify alarm annunciation.
4. Operate diesel fuel oil day tank exhaust fans and verify operation and indication.
5. Verify design airflow.
6. Operate dampers under simulated normal and emergency conditions and verify operation and indication.

D. Acceptance Criteria

1. Each of the diesel generator building train A and train B unit heaters automatically start with a signal from their respective temperature controller.
2. Diesel generator building train A and train B ESF supply fans maintain respective room temperature at 120°F or below with the diesel generator running.
3. All high and low air temperature alarms annunciate properly.
4. Diesel generator building train A and train B ESF supply fans operate on the proper signals.
5. The diesel generator building train A and train B non-ESF exhaust fans operate on the proper signals.
6. The train A and train B automatic dampers operate properly in normal and emergency operation.

14.2.8.1.42 Containment Heat Removal System Preoperational Test

A. Objective

To demonstrate operation of the containment heat removal system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

1. Verify manual and automatic system controls.
2. Verify alarms, indicating instruments, and status lights are functional.
3. Verify design airflow.
4. Verify heat removal capability during hot functional testing with RCS at operating temperature.

D. Acceptance Criterion

The containment heat removal system operates as described in section 6.2 and subsection 9.4.6.

14.2.8.1.43 Containment Integrated Leak Rate Test (ILRT) Preoperational Test

A. Objective

To demonstrate that the total leakage from the containment does not exceed the maximum allowable leakage rate at the calculated peak containment internal pressure.

B. Prerequisites

1. The containment local leak rate tests are complete.
2. Containment isolation valves are closed by normal actuation methods.
3. Portions of fluid systems, which are part of the containment boundary that may be opened directly to the containment or outside atmosphere under post-accident conditions, are opened or vented to the appropriate atmosphere to place the containment in conditions as close to post-accident conditions as possible.
4. Containment penetration, including equipment and personnel airlocks, is closed.

C. Test Method

1. The ILRT is conducted, using the absolute method described in the American National Standard (ANSI) N45.4-1972 and ANSI/ANS-56.8-1981 which describe containment system leakage testing requirements. Measurements will be taken to calculate the leakage rate for the containment.
2. A verification test will be conducted to confirm the capability of the data acquisition and reduction system to satisfactorily determine the calculated integrated leakage rate. The verification test is accomplished by imposing a known leakage rate on the containment, or by pumping back a known quantity of air into the containment through a calibrated flow measurement device.

D. Acceptance Criterion

The containment integrated leakage does not exceed the maximum allowable leakage rate as defined in 10 CFR 50, Appendix J.

14.2.8.1.44 CRDM Cavity and Vessel Support Cooling System Preoperational Test

A. Objective

To demonstrate operation of CRDM cavity and vessel support cooling system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

1. Verify manual and automatic system controls.
2. Verify design airflow.
3. Verify alarms, indicating instruments, and status lights are functional.

D. Acceptance Criterion

The CRDM cavity and vessel support cooling system operates as described in subsection 9.4.6.

14.2.8.1.45 Post-LOCA Purge Exhaust System Preoperational Test

A. Objective

To demonstrate operation of the post-LOCA purge exhaust system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

1. Verify manual and automatic controls.
2. Verify design airflow.
3. Verify alarms and status lights are functional.

D. Acceptance Criterion

The post-LOCA purge exhaust system operates as described in subsection 6.2.5.

14.2.8.1.46 Hydrogen Monitor and Removal System Preoperational Test

A. Objective

To demonstrate operation of the hydrogen monitor (electric recombiner) and removal system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

1. Verify manual and automatic controls.
2. Verify alarms and status lights are functional.
3. Verify design airflow.

D. Acceptance Criterion

The hydrogen monitor and removal system operates as described in subsection 6.2.5.

14.2.8.1.47 Containment Air Purification Cleanup System Preoperational Test

A. Objective

To demonstrate operation of the containment air purification cleanup system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

1. Verify manual and automatic controls.
2. Verify alarms and status lights are functional.
3. Verify design airflow.
4. Verify particulate removal of HEPA filter.

D. Acceptance Criterion

The containment air purification cleanup system operates as described in subsection 9.4.6.

14.2.8.1.48 Gaseous Waste Processing System Preoperational Test

A. Objectives

1. To demonstrate operation of the waste gas compressors, catalytic hydrogen recombiners, and waste gas system components and their associated control and interlock circuitry.
2. To demonstrate waste gas compressor, catalytic hydrogen recombiner, and waste gas system performance during operation in their various modes.
3. To demonstrate operation of the recycle holdup tank vent eductor.
4. To demonstrate operation and performance of the gas decay tank drain pump and the drain system components.

B. Prerequisites

1. Required component testing, instrument calibration, and system flushing/cleaning are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. The auxiliary gas system is available to supply nitrogen to the waste gas system and oxygen to the catalytic hydrogen recombiners.
4. The ACCW system is available to supply cooling water to the waste gas compressors and catalytic hydrogen recombiners.
5. The reactor makeup water system is available to provide water to the waste gas compressors, catalytic hydrogen recombiners, and waste gas decay tank drain header.
6. The instrument air system is available to provide instrument air to waste gas system instruments and control valves.
7. A temporary source of hydrogen is available for checking the operation of the catalytic hydrogen recombiner.

C. Test Method

1. System component control circuits are verified, including response to normal control, interlock, and alarm signals.
2. The waste gas compressors and catalytic hydrogen recombiners are operated in their various modes, and their performance characteristics are verified.
3. Recycle holdup tank venting is performed, and the satisfactory operation of the recycle holdup tank vent eductor is verified.
4. The gas decay tank drain pump and system performance characteristics are verified during operation to remove water from a gas decay tank.

D. Acceptance Criteria

1. The waste gas compressors, catalytic hydrogen recombiners, and waste gas system components respond to normal control, interlock, and alarm signals. (See NSSS Startup Manual, SU-4.1.6.)

2. The waste gas compressors, catalytic hydrogen recombiners, and waste gas system are operable and their controls operate as described in section 11.3.2.

3. The recycle holdup tank vent eductor operates properly to perform recycle holdup tank venting. (See NSSS Startup Manual, SU-4.1.6.)

4. The gas decay tank drain pump and drain system performance characteristics are within design specifications. (See table 11.3.2-7.)

14.2.8.1.49 Liquid Waste Processing System Preoperational Test (See table 14.2.1-1 for Unit 2)

A. Objectives

1. To verify operation of the control circuitry of the liquid waste processing pumps, including the waste monitor, waste evaporator concentrates holdup tank, laundry and hot shower tank, chemical drain, spent resin sluice, and RCDT.

2. To demonstrate the operation and verify the operating characteristics of the liquid waste processing system, including pumps, valves, and tanks.

B. Prerequisites

Required component testing and instrument calibration are complete.

C. Test Method

1. The control circuitry and operation of system pumps and valves, including response to safeguards signals and high radiation isolation signals, will be verified.

2. The system will be operated and performance characteristics will be verified.

D. Acceptance Criteria

1. The performance characteristics of the system pumps are as designed. (See table 11.2.1-2.)

2. The pump and valve controls and interlocks are as designed. (See NSSS Startup Manual, SU-4.1.4.)

14.2.8.1.50 Waste Evaporator Preoperational Test (To be performed prior to utilization of the system)

A. Objectives

1. To demonstrate the operability of the waste evaporator and its associated pumps, valves, tanks, and control circuits.

2. To verify the capability of the waste evaporator to produce the required distillate output.

B. Prerequisites

1. Required component testing and instrument calibration are complete.

2. The ACCW system is available to supply cooling water.
3. The auxiliary steam system is available to supply steam to the waste evaporator when required.
4. Water is available with sufficient boron concentration to produce a batch of 12-percent boric acid in the evaporator when required.[2]

C. Test Method

1. The control circuitry and operation of system pumps and valves will be verified.
2. The system will be operated using demineralized water, and performance characteristics will be measured.
3. The system will be operated using borated water, and the evaporator output will be verified.

D. Acceptance Criteria

1. System pumps, valves, and interlocks operate as designed. (See NSSS Startup Manual, SU-4.1.5.)
2. The system operating characteristics are within the design limitations. (See table 11.2.1-2.)

14.2.8.1.51 Backflushable Filter System Preoperational Test

A. Objective

To demonstrate operation of the backflushable filter system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

1. Verify manual and automatic (Unit 1 only) system controls.
2. Verify system flowrates.
3. Verify alarms, indicating instruments, and status lights are functional.

D. Acceptance Criterion

The backflushable filter system operates as described in section 11.4.

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[2] Initial operation is planned for 4-percent boric acid and will be preoperationally tested accordingly. Later, operation up to 12-percent boric acid will be permitted following preoperational testing to 12-percent boric acid.

14.2.8.1.52 (Deleted)

14.2.8.1.53 Boron Recycle System Preoperational Test

A. Objectives

1. To demonstrate the operability of the boron recycle system, including the recycle evaporator and its associated pumps, valves, tanks, and control circuits.
2. To verify the capability of the recycle evaporator to produce the required distillate output.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. The ACCW system is available to supply cooling water.
3. The auxiliary steam system is available to supply steam to the system when required.
4. Water is available with sufficient boron concentration to produce a batch of 4-percent boric acid in the evaporator when required.

C. Test Method

1. The control circuitry and operation of system pumps and valves will be verified.
2. The system will be operated using demineralized water, and performance characteristics will be measured.
3. The recycle evaporator will be operated using borated water, and the evaporator output will be verified.

D. Acceptance Criteria

1. System valves, pumps, and interlocks operate as designed. (See NSSS Startup Manual, SU-4.1.2.)
2. The system operating characteristics are within design limitations. (See table 9.3.4-3.)

14.2.8.1.54 Containment, Auxiliary, Control, and Fuel Handling Buildings Drains System Preoperational Test (See table 14.2.1-1 for Unit 2.)

A. Objectives

1. To demonstrate operation of the containment, auxiliary, control, and fuel handling buildings drains system.
2. To demonstrate the ability of the containment cooler condensate measuring system to detect an increase in the amount of water collected by the system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.
5. Verify system supply/inlet flow paths.

C. Test Method

1. Verify manual and automatic system controls.
2. Verify system discharge flow paths.
3. Verify alarms, indicating instruments, and status lights are functional.
4. A known amount of water is added to the containment coolers condensate measuring system, and its detection ability is verified.

D. Acceptance Criterion

1. The containment, auxiliary, control, and fuel handling buildings drains system operates as described in subsection 9.3.3.
2. The detection ability of the containment coolers condensate measuring system is verified.

14.2.8.1.55 Diesel Generator Fuel Oil System Preoperational Test

A. Objectives

1. To demonstrate the capability of the system to provide an adequate fuel supply to the diesel generator fuel oil day tanks.
2. To demonstrate operability of associated instrumentation and controls.

B. Prerequisites

1. Required component testing, instrument calibration, and system flushing/cleaning are complete.
2. Required electrical power supplies and control circuits are operational.

C. Test Method

1. Fuel oil is transferred from the fuel oil storage tank to the fuel oil day tanks by means of the transfer pumps. Appropriate flow parameters are recorded.
2. Simulation of day tank levels is utilized to verify the transfer pump automatic operation.

D. Acceptance Criterion

The performance of the diesel generator fuel oil system is in accordance with the design criteria and subsection 9.5.4.

14.2.8.1.56 Service Air System Preoperational Test

A. Objectives

1. To demonstrate operation of air compressors, air dryers, and isolation valves.
2. To verify that the system meets design air quality requirements.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing has been completed.
2. Permanently installed instrumentation is properly calibrated and operable.
3. Turbine plant closed cooling water system is available.
4. Appropriate ac and dc power sources are available.

C. Test Method

1. Simulate temperature and pressure signals to verify alarms and operation of isolation valves.
2. Operate compressors to verify operation.
3. Simulate safety signal to verify operation of the containment isolation valves.
4. Take batch samples of the service air for analysis.

D. Acceptance Criterion

Compressors, air dryers, and isolation valves perform as described in subsections 6.2.4 and 9.3.1.

14.2.8.1.57 Instrument Air System Preoperational Test

A. Objectives

1. To demonstrate operation of air dryers and isolation valves.
2. To verify that the system meets design air quality requirements.
3. To demonstrate that equipment designed to be supplied with instrument air is not supplied with service air.
4. To demonstrate that operation of components requiring large quantities of air does not cause excessive instrument air system pressure transients.

B. Prerequisites

1. Required construction acceptance testing is complete.
2. Required permanently installed instrumentation is properly calibrated and operable.
3. Service air system is operable.
4. Appropriate ac and dc power sources are available.

C. Test Method

1. Simulate pressure signals to verify alarms and operation of isolation valves.
2. Operate air dryers to verify operation.
3. Simulate safety signals to verify operation of the containment isolation valves.
4. Take batch samples of the instrument air for analysis.
5. Verify that, with service air shut off, there are no crossties from the service air system that introduce air into the instrument air system.
6. Simultaneously operate large users of instrument air and monitor the instrument air system pressures.

D. Acceptance Criterion

Air dryers and isolation valves perform as described in subsections 6.2.4 and 9.3.1.

14.2.8.1.58 Fire Protection System Preoperational Test

A. Objective

To demonstrate operation of the fire protection system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

1. Demonstrate operation of the fire detection system.
2. Demonstrate operation of the fire protection water system:
   a. Demonstrate the head and flow characteristics of the diesel engine-driven fire water pumps, the electric motor-driven fire water pumps, and the operation of all auxiliaries.
   b. Verify control logic.
   c. Demonstrate flow paths of the fire protection system.
3. Demonstrate proper operation of the fire protection halon system.
4. Verify installation of portable fire protection apparatus.

D. Acceptance Criterion

The fire protection system operates as described in subsection 9.5.1.

14.2.8.1.59 Fuel Handling and Vessel Servicing Preoperational Test

A. Objectives

1. To demonstrate operation of the spent fuel handling tool, new fuel assembly handling tool, control drive shaft unlatching tool, and irradiation sample handling tool.
2. To demonstrate operation of the reactor vessel stud tensioner, and the polar crane, including the control circuits and interlocks.
3. To verify the crane has completed static testing at 125 percent and operational testing at 100 percent of rated load.

B. Prerequisites

1. The required component testing and instrument calibration are complete.
2. The fuel transfer system is available to support the testing of the fuel handling equipment.
3. The dummy subassembly and an irradiation sample holder are available for testing the fuel handling tool.

C. Test Method

1. Verify operation of the fuel handling equipment using the fuel transfer system and a dummy fuel assembly.
2. Verify fuel handling tool operations.
3. Verify control circuitry for the polar crane.
4. Verify the polar crane static and dynamic load testing has been completed.

D. Acceptance Criteria

1. The fuel handling tools are able to perform according to design. (See paragraph 9.1.4.2.4 and 9.1.4.3.14.)
2. The polar crane and its associated interlocks and control circuits perform as specified in its equipment technical manual.
3. The polar crane has completed static testing at 125 percent and operational testing at 100 percent of rated load (see ANSI B30.2, OSHA P 1910.)

14.2.8.1.60 Fuel Building Hoists and Elevator Preoperational Test

A. Objectives

1. To demonstrate operation of the spent fuel cask bridge crane, fuel handling machine, and new fuel elevator, including their control circuits and interlocks.
2. To verify the spent fuel cask bridge crane has completed static testing at 125 percent and operational testing at 100 percent of rated load.
3. To verify the fuel handling machine has completed static testing at 125 percent and operational testing using the dummy fuel assembly.

B. Prerequisite

Required component testing and instrument calibrations are complete.

C. Test Method

1. Verify the operability of the control circuits and associated interlocks of the cranes and elevator.
2. Verify the spent fuel cask bridge static and dynamic loading testing has been completed.
3. Verify that the fuel handling machine static and dynamic load testing has been completed.

D. Acceptance Criteria

1. The fuel building hoists and new fuel elevator perform all functions and operations as specified in their equipment technical manuals.
2. The spent fuel cask bridge crane has completed static testing at 125 percent and operational testing at 100 percent of rated load. (ANSI B30.2, OSHA P 1910.)
3. The fuel handling machine can lift 125 percent of its rated load and can transfer the dummy fuel assembly between the transfer cart and the fuel racks. (FSAR paragraphs 9.1.4.3.1.D and 9.1.4.3.1.F.)

14.2.8.1.61 Fuel Transfer System Preoperational Test

A. Objectives

1. To demonstrate operation of the fuel transfer system control circuits and associated interlocks.
2. To verify the ability of the refueling machine, new fuel elevator, fuel transfer car, spent fuel bridge crane, spent fuel cask handling crane, and associated fuel handling tools to transfer a dummy fuel assembly.
3. To perform refueling machine static load testing at 125 percent of rated load and operational load testing using the dummy fuel assembly.

B. Prerequisites

1. The required component testing and instrument calibration are complete.
2. There is a means available to lubricate the bearings of the fuel transfer equipment.
3. A dummy fuel assembly is available.

C. Test Method

1. The refueling machine will be load tested at 125 percent of rated load.
2. A dummy fuel assembly will be transferred from the new fuel pit to the refueling machine in the containment and back to the spent fuel pool to verify the operation of the fuel transfer equipment.

D. Acceptance Criteria

1. The fuel transfer system interlocks and interlock bypasses perform as specified in the equipment technical manual.
2. The fuel transfer equipment is able to transfer a dummy fuel assembly in and out of containment in accordance with the equipment technical manual.
3. The refueling machine can lift 125 percent of its rated load and can transfer the dummy fuel assembly between the transfer cart and the reactor vessel (paragraph 9.1.4.3.1.F (safety)).

14.2.8.1.62 Refueling Machine Preoperational Test

A. Objectives

1. To demonstrate the operation of refueling machine, its control circuits, and interlocks.
2. To demonstrate the handling of thimble plugs and rod cluster control assembly (RCCA) by the refueling machine.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. A dummy fuel assembly is available.
3. The lower internals are installed in the reactor vessel.

C. Test Method

1. Operability of the refueling machine and control circuits of the bridge, trolley, hoist, and gripper interlocks will be verified.
2. Handling of thimble plugs and RCCAs will be demonstrated.

D. Acceptance Criteria

1. Control circuits and interlocks associated with the refueling machine allow the equipment to perform all necessary functions in accordance with the equipment technical manual.
2. The refueling machine is capable of proper handling of thimble plugs and RCCAs.

14.2.8.1.63 Refueling Machine Indexing Preoperational Test

A. Objectives

1. To verify the indexing of the refueling machine and establish bridge rail reference points for future operations.
2. To demonstrate the ability to transfer the dummy fuel assembly to the reactor vessel and determine that the lower core plate pins align with the bottom nozzle of the fuel assembly.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. A dummy fuel assembly is available.
3. The lower internals are installed in the reactor vessel.

C. Test Method

1. Correct indexing will be verified using the dummy fuel assembly at the following core locations:

   a. At each location across the core in an east-west direction.
   b. At each location across the core in a north-south direction.
   c. At each location around the outer core periphery.

2. The dummy fuel assembly will be lowered into position in the core, and its alignment will be verified.

D. Acceptance Criteria

1. The refueling machine core indexes to all core locations.
2. The lower core plate pins align with the bottom nozzle of the dummy fuel assembly when it is transferred to a core location.

14.2.8.1.64 Diesel Generator Preoperational Test

A. Objectives

1. Demonstrate the operability of each diesel breaker and associated interlocks.
2. Demonstrate the operation of air start, lube oil, and jacket cooling water systems.
3. Demonstrate diesel reliability by performing 35 consecutive starts with no failures of each diesel.
4. Demonstrate the ability of the diesel generators to synchronize with the offsite power system.
5. Determine the fuel oil consumption of each diesel while operating under continuous rating load conditions.
6. Verify that all automatic diesel generator trips except engine overspeed, low lube oil pressure, high jacket water temperature, and generator differential are automatically bypassed upon loss of voltage on the emergency bus concurrent with a safety injection signal.
7. Verify that, with the diesel generator operating in the test mode connected to its bus, a simulated safety injection signal overrides the test mode by returning the diesel generator to standby operation.
8. Verify that the following diesel generator lockout features prevent the diesel generator starting:

   a. Turning gear engaged.
   b. Emergency stop.

9. Demonstrate full load carrying capability for 24 h.

B. Prerequisites

1. Required construction acceptance tests are complete.
2. Required electrical power supplies and control circuits are operational.
3. The NSCW system is available.
4. The diesel generator fuel oil system is available.
5. Adequate ventilation for the diesel generator building is available.

C. Test Method

1. The control logic of the diesel breakers, diesel start circuit, and support pumps and valves will be verified.
2. The operability of the diesel air start, lube oil, and jacket cooling water system will be verified.
3. There will be a demonstration of 35 consecutive starts of each diesel.
4. The diesel generators will be started, voltage and frequency control demonstrated, phase rotation verified, and the diesels synchronized to offsite power and loads.
5. During the testing, fuel oil consumption will be monitored with the diesels operating at the continuous load rating.
6. With a simulated loss-of-offsite-power signal, the proper diesel trips and bypasses will be verified.
7. With the diesel connected to its bus, a simulated safety injection signal will cause it to return to standby operation.
8. The diesel generator lockout features will be verified.
9. Demonstrate full load carrying capability for 24 h, of which 22 h are at a load equivalent to the continuous rating of the diesel generator and 2 h at a load equivalent to the 2-h rating of the diesel generator.

D. Acceptance Criteria

1. The controls, interlocks, and operation of the diesel generator breakers and support systems are as designed.
2. Each diesel must have 35 consecutive starts within the required time without a failure.
3. Each diesel generator is capable of being synchronized with offsite power.
4. Upon receipt of safeguards signals, the diesels operate as designed.
5. The diesel generator lockout features operate in accordance with design.
6. The diesel fuel oil consumption does not exceed the design requirements.
7. Each diesel satisfactorily completes the full-load test.

14.2.8.1.65 Auxiliary Building Flood Retaining Rooms Alarm System Preoperational Test

A. Objective

To demonstrate operation of auxiliary building flood retaining rooms alarm.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

Verify alarms, indicating instruments, and status lights.

D. Acceptance Criterion

The auxiliary building flood retaining room alarm system operates as described in subsection 9.3.3.

14.2.8.1.66 Main and Unit Auxiliary Transformers Preoperational Test

A. Objectives

1. To demonstrate the operation of protective relaying, alarms, and control devices associated with the main and unit auxiliary transformers.
2. To demonstrate the energization of the unit auxiliary transformers.

B. Prerequisites

1. Required construction acceptance testing has been completed.
2. Related ac and dc power sources are available.

C. Test Method

1. Simulate fault conditions to verify alarms and operation of protective relaying circuits.
2. The unit auxiliary transformers are energized. Voltage and phase rotation are verified and recorded.

D. Acceptance Criterion

Performance of the main and unit auxiliary transformers is in accordance with the design criteria and subsection 8.3.1.

14.2.8.1.67 Reserve Auxiliary Transformers Preoperational Test

A. Objectives

1. To demonstrate the operation of protective relaying, alarms, and control devices associated with the startup transformers.
2. To demonstrate the energization of the startup transformers.

B. Prerequisites

1. Required construction acceptance testing has been completed.
2. Related ac and dc power sources are available.

C. Test Method

1. Simulate fault conditions to verify alarms and operation of protective relaying circuits.
2. The transformers are energized. Voltage and phase rotation are verified and recorded.

D. Acceptance Criterion

The performance of the startup transformers is in accordance with the design criteria and subsection 8.3.1.

14.2.8.1.68 Non-Class 1E ac Distribution Preoperational Test

A. Objectives

1. To demonstrate the operability of bus supply breaker control logic and interlocks.
2. To demonstrate the automatic fast transfer from the unit auxiliary source or startup source is within design specifications.
3. To demonstrate the operation of system instrumentation and controls.
4. To demonstrate the energization of all buses and distribution panels.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. Required electrical power supplies and control circuits are operational.

C. Test Method

1. Operability of bus supply breakers is demonstrated with the bus deenergized.
2. Simulation of automatic fast transfer initiation signals is utilized to demonstrate automatic fast transfer scheme operability as applicable.
3. Buses are energized from their alternate supplies. Voltage, phase rotation, and phase angles are measured and recorded.

D. Acceptance Criterion

The 13.8-kV, 4.16-kV, 480-V, and instrument 120-V-ac systems (non-class 1E) perform as described in subsection 8.3.1.

14.2.8.1.69 125-V dc (non-class 1E) Preoperational Test

A. Objectives

1. To demonstrate the capacity of each battery.
2. To demonstrate the capability of the batteries to supply the required loads for the required period of time.

B. Prerequisites

1. Required construction acceptance testing has been completed.
2. Required electrical power supplies and control circuits are operational.
3. Battery room ventilation is available.
4. Required resistive load devices are available.

C. Test Method

1. To determine capacity, each battery will be discharged, using the design-specified rate, to its minimum design voltage limit.
2. Each battery will be discharged using the required discharge rates and time periods.
3. Each individual battery cell will be measured following the discharge test.

D. Acceptance Criteria

1. Battery capacity will be verified to be 90 percent of the manufacturer's rating.
2. Each battery can carry the required load for the required time period.
3. Each battery cell is within the minimum design voltage limit at the end of the service test.

14.2.8.1.70 4.16-kV (class 1E) System Preoperational Test

A. Objectives

1. To demonstrate the operation of normal and alternate incoming supply breakers of 4.16-kV (class 1E) buses.
2. To demonstrate that the buses can be energized by the normal and alternate sources via these breakers.
3. To verify operation of interlocks, protective relaying, and ESF sequencer trip signals.

B. Prerequisites

1. Required construction acceptance testing has been completed.
2. Instruments required to perform the preoperational test shall be calibrated and operable prior to their use in that test.
3. Appropriate ac and dc power sources available.

C. Test Method

1. Operate breaker from control points and verify operation of switches, indicating lights, interlocks, and alarms.
2. Simulate abnormal conditions and verify responses and alarms.
3. The 4.16-kV (class 1E) buses are energized via their normal and alternate sources; bus voltage and phase rotation are recorded.

D. Acceptance Criterion

The 4.16-kV (class 1E) system performs as described in subsection 8.3.1.

14.2.8.1.71 Class 1E Standby Power Supply (4.16-kV Diesel Generator Sequencer) Preoperational Test

A. Objectives

1. Verify the control logic and operation of bus undervoltage and degraded grid undervoltage relays.
2. Verify the control logic of the load shed and load sequencer.

B. Prerequisites

1. Required construction acceptance tests are complete.
2. Required electrical power supplies are operational.
3. The required instrumentation calibration is complete.

C. Test Method

1. The sequencer control logic will be verified by simulating a loss of offsite power and by simulating a safeguard initiation and recording the response of the sequencer cabinets.
2. The logic for the bus undervoltage and degraded grid relays will be verified.

D. Acceptance Criteria

1. The sequencer, the bus undervoltage relays, and the degraded grid relays operate in accordance with design.
2. The loading intervals for the sequencer are within the design limits.

14.2.8.1.72 480-V (class 1E) Switchgear Preoperational Test

A. Objective

To perform the initial operation of the 480-V class 1E load centers and to demonstrate that they can be energized and operated.

B. Prerequisites

1. Required construction acceptance testing has been completed.
2. Required permanently installed instrumentation is properly calibrated and operable.

3. Required electrical ac power supplies and dc control circuits are operational and available.

C. Test Method

1. The manual controls, annunciation, and instrumentation for the 480-V class 1E load centers and their 4-kV feeder breakers are checked for correct operation.

2. The 480-V class 1E load centers are energized. Voltage and phase rotation are verified and recorded.

D. Acceptance Criteria

1. The manual controls, automatic sequencer controls, annunciation, and instrumentation for the 480-V class 1E load centers and their 4-kV feeder breakers operate correctly.

2. The voltage and phase rotation of each 480-V class 1E load center is within design specifications.

3. All 480-V class 1E load centers must perform as described in subsection 8.3.1.

14.2.8.1.73 480-V (class 1E) Motor Control Center Preoperational Test

A. Objectives

1. To demonstrate the operation of associated controls, interlocks, alarms, and solid-state trip devices.

2. To demonstrate energization of the applicable motor control centers (MCCs).

B. Prerequisites

1. Required construction acceptance testing has been completed.

2. Required ac and dc power sources are available.

C. Test Method

1. Operability of all MCC supply breakers is demonstrated.

2. Fault conditions that verify alarms and operation of solid state trip devices and protective relaying are simulated.

3. The MCCs are energized, with voltage and phase rotation recorded and verified.

D. Acceptance Criterion

The performance of the 480-V MCC (class 1E) system is in accordance with the design criteria and subsection 8.3.1.

14.2.8.1.74 125-V dc class 1E Preoperational Test

A. Objectives

1. To demonstrate the battery capacity.

2. To demonstrate the capability of the batteries to supply the required loads for the required period of time.

3. To demonstrate the ability of the battery chargers to carry the load.

4. To demonstrate the ability of the battery chargers to recharge the battery in the required time period, while carrying the largest combined continuous steady-state load expected on the dc system.

B. Prerequisites

1. Required construction acceptance testing has been completed.

2. Required electrical power supplies and control circuits are operational.

3. Required battery room ventilation is available.

4. Required resistive load devices are available.

C. Test Method

1. To determine capacity, each battery will be discharged, using the 2.75-h rate, to its minimum design voltage limit.

2. The battery will be charged to its fully charged condition, while the battery charger is simultaneously carrying the largest combined continuous steady-state load expected on that dc system.

3. Each battery will be discharged using the required discharge rates and time periods.

4. The battery charger will be subjected to its rated load for the required 8-h period.

5. Each individual battery cell will be measured following the discharge test.

D. Acceptance Criteria

1. Battery capacity will be verified to be 90 percent of the manufacturer's rating.

2. The battery charger can charge the battery to a fully charged condition following a design basis discharge within 12 h while carrying the largest combined continuous steady-state load expected on that dc system.

3. Each battery can carry the required load for 2 3/4 h without reaching its minimum discharged state.

4. The battery charger can carry nameplate-rated load as follows:

a. Systems A and B - 400 A for 8 h.
b. System C - 300 A for 8 h.
c. System D - 200 A for 8 h.

14.2.8.1.75 Vital 120-V-ac class 1E Preoperational Test

A. Objectives

1. To demonstrate that the 120-V (class 1E)-ac distribution panels can be fed from their normal source inverters and from their backup source transformers by manual transfer.

2. To verify operation of system instrumentation and controls, including breaker protective interlocks.

B. Prerequisites

1. Required construction acceptance testing is complete.
2. Required component testing and instrumentation calibration are complete.
3. Appropriate ac and dc power sources are available.

C. Test Method

1. Normal source inverters are energized from 125-V-dc switchgear, and inverter input voltages are recorded.
2. The 120-V (class 1E)-ac distribution panels are energized from normal source inverters, and distribution panel voltages are recorded.
3. Normal source inverters are energized from a 480-V MCC (where available), and inverter input voltages are recorded.
4. The 120-V (class 1E)-ac distribution panels are energized from normal source inverters, and ac distribution panel voltages are recorded.
5. The 120-V (class 1E)-ac distribution panels are energized from their backup source transformer by manual transfer, and distribution panel voltages are recorded.
6. The system breakers are operated, and breaker interlocks are verified.

D. Acceptance Criterion

All 120-V (class 1E)-ac distribution panels perform as described in subsection 8.3.1, and their operation is in conformance with the Technical Specification limits.

14.2.8.1.76 Essential Lighting System Preoperational Test

A. Objectives

1. To demonstrate the capability of the essential lighting system to provide adequate illumination levels.
2. To verify operation of system instrumentation and controls.

B. Prerequisites

1. Required construction acceptance testing is complete.
2. Required electrical power supplies and control circuits are operable.
3. The essential lighting system has been energized.

C. Test Method

System function, correct electrical feeds, and correct fixture location will be verified per plan drawings.

D. Acceptance Criterion

The essential lighting system functions as described in subsection 9.5.3 provides illumination to areas of the plant designated by the VEGP design manual.

14.2.8.1.77 Emergency Lighting System Preoperational Test

A. Objectives

1. To demonstrate the capability of the emergency lighting system to provide adequate illumination levels.
2. To verify operation of system instrumentation and controls.

B. Prerequisites

1. Required electrical power supplies and control circuits are operable.
2. The emergency lighting system has been energized.

C. Test Method

1. The ability of the emergency lighting system is verified by simulating a loss of the normal and essential lighting and observing that the emergency system automatically activates.
2. Illumination levels and operation times will be verified.

D. Acceptance Criterion

The performance of the emergency lighting system tested is in accordance with the design criteria and subsections 8.3.1 and 9.5.3.

14.2.8.1.78 Offsite Communication System Preoperational Test

A. Objective

To demonstrate operation of the offsite communication system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Support systems are available.

C. Test Method

1. Verify operation of the plant telephone branch exchange (PABX) system to provide adequate communication between onsite stations.
2. Verify operation of the PABX system in relation to the following offsite communication systems:
    a. Hotline to Georgia Power Company (GPC) general office production department.
    b. Southern Bell telephone system lines.
    c. Two telephone lines from GPC's Atlanta private branch exchange.

D. Acceptance Criterion

The offsite communication system operates as described in subsection 9.5.2.

14.2.8.1.79 Inplant Communication System Preoperational Test

A. Objective

To demonstrate the adequacy of the inplant communication system to provide reliable communications between plant areas and to verify the operability of the emergency alarm system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support system is available.
5. Plant equipment that contributes to the ambient noise level is in operation if possible.

C. Test Methods

1. Verify the telephone/page system operability.
2. Verify the sound-powered system operability.
3. Verify the tone generator operability.

D. Acceptance Criterion

The inplant communication system operates as described in subsection 9.5.2.

14.2.8.1.80 Heat Tracing System Preoperational Test

A. Objective

To demonstrate operation of the heat tracing system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

1. Verify the operability of the heat tracing circuits.

2. Verify alarms, indicating instruments, and status lights are functional.

D. Acceptance Criterion

The heat tracing system automatically controls the associated heat tracing circuits, in accordance with system operation requirements as described in subsection 9.3.4.

14.2.8.1.81 (Material deleted)

14.2.8.1.82 Plant Annunciator System Preoperational Test

A. Objectives

1. To demonstrate operation of the plant annunciator system.
2. To verify that the annunciator will transfer to the alternate power source upon loss of primary power source.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete to the extent required to perform the preoperational test.
2. Component testing and instrumentation calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Methods

1. Self-test features of the system shall be tested and verified to be operational.
2. Verify appropriate annunciator response to each retransmitting contact when it is actuated and verify the isolation device panels are operational.
3. Transfer of power sources shall be tested and verified operational.

D. Acceptance Criterion

The plant annunciator system operates as described in chapter 18.

14.2.8.1.83 Emergency Response Facility Post-Accident Monitoring and Sampling Systems Preoperational Test

A. Objective

To demonstrate operation of the emergency response facility post-accident monitoring and sampling systems.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Method

1. Verify instrument response to simulated external inputs.
2. Verify sampling system operation.

D. Acceptance Criteria

1. Post-accident sampling is described in subsection 9.3.2. The post-accident sampling system (PASS) was deleted in Technical Specification Amendments 123/101.
2. The post-accident monitoring system operates as described in subsection 7.5.3.
3. Verify the turbine plant sampling system operates as described in subsection 9.3.2.
4. Verify the nuclear sampling system operates as described in subsection 9.3.2.

14.2.8.1.84 Reactor Protection System and Engineered Safety Features Actuation System (ESFAS) Logic Preoperational Test

A. Objectives

1. To demonstrate operation of the ESFAS and reactor protection system and their ability to initiate appropriate reactor trip and safety actuation signals on receipt of simulated input signals.
2. To demonstrate operation of the reactor protection system block and permissive interlocks.
3. To demonstrate the functioning of the reactor protection system instrumentation inputs and BOP protection system instrumentation inputs to the SSPS including interlock and alarm functions.
4. To demonstrate operation of the reactor trip breakers and reactor trip bypass breakers, including breaker interlock, alarm, and tripping functions.
5. To demonstrate that the undervoltage coil and the shunt trip coil will function independently to trip the reactor trip breakers and bypass breakers following initiation of a manual reactor trip.
6. To demonstrate that reactor trip functions occur at design setpoints using simulated signals inserted at the NSSS and BOP process and protection rack inputs.

B. Prerequisites

1. Required component testing and instrument calibrations are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. Required slave output relay field wiring has been electrically isolated to prevent operation of components controlled by the relays except where specifically removed in the preoperational test procedure.
4. Plant system and components to be operated during testing are aligned prior to actuation of the particular system or component.

C. Test Method

1. Appropriate input signals are simulated and proper operation of the reactor protection logic matrix is verified by observing operation of reactor trip breakers and/or undervoltage coils.
2. Appropriate input signals are simulated and proper operation of the ESFAS logic matrix is verified by observing operation of the master relays and slave output relays.
3. Operation of the reactor protection system block and permissive interlocks is verified by observing proper response on receipt of appropriate input signals.
4. Operation of the reactor protection and the BOP protection panel inputs to the reaction protection system including setpoints, interlock, and alarm functions, are verified.
5. Operation of the reactor trip breakers and reactor trip bypass breakers and their associated trip, interlock, and alarm functions are verified.

D. Acceptance Criteria

1. The ESFAS and reactor protection systems operate as designed to initiate the appropriate reactor trip and safety actuation signals. (See NSSS Startup Manual, SU-2.7.2 and SU-2.7.3.)
2. Reactor protection system block and permissive interlocks operate in accordance with system design. (See drawings 1X6AA02-225, 1X6AA02-226, 1X6AA02-227, 1X6AA02-228, 1X6AA02-229, 1X6AA02-230, 1X6AA02-231, 1X6AA02-232, 1X6AA02-233, 1X6AA02-234, 1X6AA02-235, 1X6AA02-236, 1X6AA02-237, 1X6AA02-238, 1X6AA02-239, 1X6AA02-240, 1X6AA02-494, 1X6AA02-495, 1X6AA02-496, and 1X6AA02-519, table 7.2.1-2, and table 7.3.1-3.)
3. The reactor protection instrumentation inputs and BOP protection instrumentation inputs into the SSPS setpoint, interlock, and alarm functions, operate. (See paragraph 7.2.2.3.)
4. The reactor protection system logic trains respond during testing. (See paragraph 7.2.2.3).
5. The reactor trip breakers and reactor trip bypass breakers operate as designed to provide normal trip, interlock, and alarm functions. (See paragraph 7.2.2.3.)

14.2.8.1.85 Safeguards Test Cabinet (STC) Testing Capability

A. Objectives

1. To demonstrate that the ESFAS slave relays field wiring is terminated and that the ESFAS components actuate to the correct position.
2. To demonstrate that ESFAS slave relays can be operated from the safeguards test cabinet.

B. Prerequisites

1. Required component testing and instrument calibrations are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. Required slave output relay field wiring has been electrically isolated to prevent operation of components controlled by the relays except where specifically removed in the preoperational test procedure.
4. Plant system and components to be operated during testing are aligned prior to actuation of the particular system or component.
5. The reactor protection system logic trains and slave relays have been operationally checked with the reactor protection system test panels.

C. Test Method

1. Each ESFAS slave relay is actuated by the STC and relay operation is verified by contact operation, component actuation, or electrical test.
2. The ESFAS slave relays are actuated by the STC or simulated trip signals, and correct (proper positioning) component actuation and field wiring are verified by actual response or electrical checks (electrical checks are verifying a component breaker will change states, as required, when in the test position).

D. Acceptance Criteria

1. The ESFAS slave relay function is designed to actuate the ESFAS components. (See paragraph 7.3.1.1.1.1.)
2. The STC can actuate each slave relay individually.

14.2.8.1.86 Nuclear Instrumentation System Preoperational Test

A. Objective

To demonstrate the operability of the nuclear instrumentation source, intermediate, and power range channels, including their ability to supply signals for operating appropriate alarm and trip signals and indicating reactor power levels.

B. Prerequisites

1. Nuclear instrumentation system installation, calibration, and alignments are complete.
2. Required electrical power supplies are energized and operational.

3. The nuclear instrumentation system has been energized for at least 4 h.

C. Test Method

1. The ability of the source, intermediate, and power range nuclear instrumentation circuitry to respond to test signals is verified.
2. The source, intermediate, and power range instrumentation trip functions and alarm setpoints are verified.
3. Operation of the source range nuclear instrumentation audible count rate circuitry is verified.
4. Operation of the power range nuclear instrumentation comparator and rate circuitry is verified.

D. Acceptance Criteria

1. The source, intermediate, and power range nuclear instrumentation circuitry responds properly to test signals. (See NSSS Startup Manual, SU-2.9.1.)
2. The source, intermediate, and power range trip functions and alarm setpoints are within design specifications. (See NSSS Startup Manual, SU-2.9.1.)
3. The source range nuclear instrumentation audible count rate circuitry and power range nuclear instrumentation comparator and rate circuitry operate within design requirements. (See NSSS Startup Manual, SU-2.9.1.)

14.2.8.1.87 Process and Effluent Radiological Monitoring System Preoperational Test

A. Objective

To demonstrate operation of the process and effluent radiological monitoring system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.
4. Suitable check sources are available.

C. Test Method

1. The operation of each monitor will be verified.
2. The control logic, annunciation, and power failure alarms of each monitor will be verified.

D. Acceptance Criterion

The process and effluent radiological monitoring system operates as described in section 11.5.

14.2.8.1.88 Incore Instrumentation System Preoperational Test

A. Objectives

1. To demonstrate operation of the incore movable detector system control equipment, drive assemblies, and transfer devices, including proper rotation, position indication, and limit switch actuation.
2. To demonstrate operation of the gas purge and leak detection systems associated with the incore movable detectors.

B. Prerequisites

1. The incore movable detector system installation is complete except for installation of the retractable guide thimbles.
2. Required component installation checks and instrument calibrations are complete.
3. Required electrical power supplies and control circuits are energized and operational.
4. A dummy cable is available for test performance.

C. Test Method

1. The functioning of the incore movable detector system control equipment drive assemblies and transfer devices is verified during operation in the various modes using a dummy cable.
2. Position indication and limit switch actuations are verified during test performance.
3. Operation of the gas purge and leak detection systems is verified by observing purge gas flow and observing drain system response to simulated high-level signal.

D. Acceptance Criteria

1. The drive assemblies insert and withdraw the detectors in the proper paths of the ten-path rotary transfer devices and operate in the high and low speeds in the Manual mode. Incore Instrumentation Manual Volume 1 1/2X6AB08-59.
2. The drive assemblies operate in the automatic mode for the Normal, Calibrate, Emergency, Common Group and Storage paths. Incore Instrumentation Manual Volume 1 1/2X6AB08-59.
3. The safety, withdraw, top, and bottom limit switches operate properly. Incore Instrumentation Manual Volume 1 1/2X6AB08-59.
4. The leak detection system is capable of detecting a leak. Incore Instrumentation Manual Volume 1 1/2X6AB08-59.

5. System limit switch setpoints are acceptable and in accordance with the manufacturer's instructions. (See NSSS Startup Manual, SU-2.9.3.)

14.2.8.1.89 Reactor Control, Rod Control, and Digital Rod Position Indication Preoperational Test

A. Objectives

1. To demonstrate the functioning of the rod control system and component cabinets.
2. To demonstrate operation of the rod control system in response to interlock signals.
3. To perform initial energization and calibration and demonstrate the functioning of the digital rod position indication system.

B. Prerequisites

1. Required component testing and instrument calibrations are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. The reactor vessel head is in place with all CRDM cables connected.
4. The integrated head and CRDM cooling system is in its normal operational alignment.

C. Test Method

1. The rod control system is initially operated, and functioning of the system and component cabinets is verified.
2. Rod control interlock actuation signals are simulated and response of the rod control system is verified.
3. The digital rod position indication system is initially energized and calibrated, and functioning of the system is verified through use of system-installed light-emitting diodes, annunciators, and a test fixture.

D. Acceptance Criteria

1. The rod control system and component cabinets operate as designed. (See NSSS Startup Manual, SU-2.5.2.)
2. The rod control system operates as designed in response to interlock signals (See table 7.7.1-1)
3. The digital rod position indication system operates as designed to provide rod position indication and alarms. (See section 7.7.1.3.2.)

14.2.8.1.90 CRDM Motor Generator Set Preoperational Test

A. Objectives

1. To demonstrate operation of the CRDM motor generator sets and system components including proper generator phasing for parallel operation.

2. To demonstrate operation of the motor generator set control system during motor generator starting, running, and parallel operation including verification of interlock and alarm functions.

B. Prerequisites

1. Required component testing and instrument calibrations are complete.
2. Required electrical power supplies are energized and operational.
3. A three-phase load bank is available for motor generator set testing under loaded conditions.

C. Test Method

1. CRDM motor generator set and system component control circuits including interlock and alarm functions are verified.
2. Generator phasing for parallel generator operation is verified.
3. Operation of the CRDM motor generator sets and control system is verified during starting, running, and parallel operation.

D. Acceptance Criteria

1. Generator phasing is proper for parallel motor generator set operation. (See Equipment Technical Manual.)
2. The CRDM motor generator sets and control system operates in accordance with design requirements during starting, running, and parallel operation. (See Equipment Technical Manual.)
3. The motor generator and control system alarms and interlocks function as designed. (See Equipment Technical Manual.)

14.2.8.1.91 CRDM Initial Timing Preoperational Test

A. Objectives

1. To demonstrate CRDM current command timing, rod speed, and bank overlap unit operation.
2. To demonstrate that the integrated head and CRDM cooling system maintain CRDM temperature within acceptable limits.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. For the CRDM cooling checkout, the plant is at or near normal operating temperature, and pressure and hot functional testing is in progress.
4. The integrated head and CRDM cooling system is in its normal operational alignment.

C. Test Method

1. CRDM withdrawal and insertion are initiated; and proper current command sequence, timing, and rod speed signal voltages are verified.

2. Proper operation of the bank overlap unit to control rod bank sequence and movement is verified.

3. Adequacy of the integrated head and CRDM cooling system for maintaining CRDM temperature is verified by measuring CRDM coil resistances and calculating the coil temperatures.

D. Acceptance Criteria

1. The CRDMs operate as verified by current traces showing current command sequence and timing and measurement of rod speed signal voltages. (See NSSS Startup Manual, SU-2.5.1.)

2. The bank overlap unit operates to control rod movement and sequence. (See NSSS Startup Manual, SU-2.5.2.)

3. The integrated head and CRDM cooling system maintains CRDM coil temperatures within design limits. (See Equipment Technical Specification.)

14.2.8.1.92 Seismic Monitoring System Preoperational Test

A. Objective

To demonstrate operation of the seismic monitoring system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.

2. Instrumentation calibration is complete.

3. Test instrumentation is available and calibrated.

C. Test Method

1. Verify operability of internal calibration devices by recording calibration records on all applicable sensors.

2. Verify system response to a simulated seismic event by actuating the appropriate trigger units, recording accelerograph outputs, and playing back the records for analysis.

3. Verify all alarms and indicators.

4. Verify installation and operation of the peak recording accelerographs.

D. Acceptance Criterion

The seismic monitoring system operates as described in subsection 3.7.4.

14.2.8.1.93 Nuclear Sampling System Preoperational Test

A. Objectives

1. To demonstrate the capability of the nuclear sampling system, liquid and gaseous, to collect representative samples from the RCS and auxiliary system process streams.
2. To verify the operation of system valves and control circuitry, including the response to safety signals.
3. To initially energize, calibrate, and verify the operation of the gross-failed fuel detector system.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. ACCW is available for cooling the sample streams.
3. The systems to be sampled are at their normal pressure and temperature.
4. The CVCS is available to receive discharge from the nuclear sampling station.

C. Test Method

1. Samples are taken from each of the sample points; flows are adjusted; and flow, pressure, and temperature data are recorded.
2. The gross-failed fuel detector system is energized, and its operability is verified.
3. Control circuit and operability of the system valves are verified.

D. Acceptance Criteria

1. The sample system flows, pressures, and temperatures are within specifications such that normal samples can be taken and analyzed. (See Design Basis DC-1211, DC-1212.)
2. The system valve controls and interlocks function as designed. (See table 6.2.4-1.)

14.2.8.1.94 Miscellaneous Leak Detection System Preoperational Test

A. Objective

To demonstrate operation of the miscellaneous leak detection system.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration are complete.
3. Test instrumentation is available and calibrated.

4. Support systems are available.

C. Test Method

Verify alarms, indicating instruments, and status lights are functional.

D. Acceptance Criterion

The miscellaneous leak detection system operates as described in subsection 9.3.3.

14.2.8.1.95 Metal Impact Monitoring System Preoperational Test

A. Objectives

1. To perform initial calibration of the metal impact monitoring system.
2. To demonstrate operation of the metal impact monitoring system and establish the alarm signal level for use during preoperational testing.

B. Prerequisites

1. Metal impact monitoring system accelerometers have been mechanically fastened to the RCS.
2. Required electrical power supplies and control circuits are energized and operational.
3. For system operation demonstration portions of the test, the RCS hot functional test is in progress.

C. Test Method

1. The metal impact monitoring system initial calibration is performed in accordance with the manufacturer's instructions, including verification of accelerometer calibration.
2. During RCS hot functional testing, the metal impact monitoring system is set up for initial operation, and system operation is verified. The alert signal level for use during hot functional testing will be established.

D. Acceptance Criteria

1. Metal impact monitoring system calibration is acceptable and in accordance with the manufacturer's instructions. (See Equipment Technical Manual.)
2. System operates for RCS monitoring during hot functional testing.

14.2.8.1.96 Integrated Control Logic Safety Injection Preoperational Test

A. Objectives

1. To demonstrate that all safety-related equipment starts or repositions properly when the appropriate safeguard signals are actuated and that response is obtained when the actuation signals are reset.

2. To demonstrate operation of the offsite power distribution system and startup transformers when the safety-related components are actuated by safeguard signals.

3. To demonstrate the validity of the assumptions used and the analytical results of the class 1E electrical system voltage analyses (performed during HFT for Unit 2).

4. To demonstrate that the diesel generators start and accelerate to rated speed and voltage within the required time interval.

B. Prerequisites

1. Required preoperational testing, component testing, and instrument calibrations are complete.

2. Required electrical power supplies and control circuits are energized and operational.

3. The reactor vessel head has been removed, and the vessel and refueling cavity have been prepared to receive a large quantity of water.

4. The RWST contains an adequate supply of demineralized water for the performance of this test.

5. The condensate storage tanks contain an adequate supply of demineralized water for the performance of this test.

6. Plant systems and components to be operated during testing are aligned prior to actuation of the safeguard signals.

C. Test Method

1. Appropriate safeguard signals are initiated and equipment starting and repositioning are verified.

2. The time required for the diesel generators to reach rated speed and voltage following safeguards actuation is determined.

3. Operation of the offsite power distribution system and startup transformers are verified during the safeguard sequence.

4. Class 1E electrical system power and voltage drop data are taken during the safeguards sequence for Unit 1 and during HFT for Unit 2 and compared with analytical results of the voltage analyses.

5. Safety-related component response is verified when the appropriate safeguard signals are reset.

D. Acceptance Criteria

1. Safety-related equipment starts or repositions when appropriate safeguard signals are actuated, and equipment response is obtained when the actuation signals are reset. (See drawings 1X3D-AA-K02A, 2X3D-AA-K02A, 1X3D-AA-K02B, and 2X3D-AA-K02B.)

2. The offsite power distribution system and startup transformers operate during and following safeguard sequence loading.

3. The diesel generators start and accelerate to rated speed and voltage within the design specification time interval. (See Design Basis DC-1821.)

4. Voltage drop test results and the analytically derived voltage values provide good correlation to support the validity of the voltage analyses analytical results (performed during HFT for Unit 2). (See Design Basis DC-1821.)

14.2.8.1.97 Integrated Safeguards and Blackout Sequence Preoperational Test

A. Objectives

1. To demonstrate that the diesel generators start, offsite power is isolated, loads are shed, and the emergency bus loads are properly sequenced on the diesel generators; to verify voltage and frequency are maintained during sequencing; and to verify that proper train separation of safety-related components is maintained upon initiation of a safeguards actuation signal coincident with a simulated loss of offsite power.
2. To demonstrate that, upon interruption of the onsite source, the emergency bus loads are shed and that, upon restoration of the onsite source, subsequent loading is through the load sequencer.
3. To demonstrate that, upon a simulated loss of offsite power or upon a simulated loss of offsite power in conjunction with a safeguards actuation signal, the emergency buses are deenergized and loads are shed from the emergency buses.
4. To demonstrate that, upon a simulated loss of offsite power, the diesel generators start and emergency bus loads are sequenced on the diesel generators.
5. To demonstrate the functional capability of the diesel generators at full-load temperature conditions by simulating a loss of offsite power immediately following the full-load capability test.
6. To demonstrate that the continuous operation rating of the diesel generator is not exceeded, upon initiation of a simulated loss of offsite power or safeguards actuation coincident with a simulated loss of offsite power.
7. To demonstrate the capability of the diesel generators to reject a load equal to or greater than the largest single load associated with that diesel generator while maintaining speed and voltage, and the capability to reject load equal to its continuous rating without overspeed tripping of the diesel.
8. To demonstrate the capability of the diesel generators to synchronize with and transfer its loads to the offsite power source while the generator is loaded with its emergency bus loads.

B. Prerequisites

1. Required preoperational testing, component testing, and instrument calibrations are complete.
2. Required electrical power supplies and control circuits are energized and operational.

3. The reactor vessel head has been removed, and the vessel and refueling cavity have been prepared to receive a large quantity of water.
4. The RWST contains an adequate supply of demineralized water for the performance of this test.
5. The condensate storage tank contains an adequate supply of demineralized water for the performance of this test.
6. Plant systems and components to be operated during testing are aligned prior to actuation of the safeguard signals.

C. Test Method

1. Each train diesel generator and its emergency bus are functionally tested, with the opposite train ac and dc buses deenergized, by initiating a safeguards actuation signal coincident with a simulated loss of offsite power; load sequencing and train separation of safety-related components are verified; and the absence of voltage on other train ac and dc buses will be verified.
2. With the diesel generator supplying the emergency bus loads and offsite power not available, a loss of the diesel generator is simulated, and load shedding and subsequent loading by the load sequencer are verified.
3. Emergency bus deenergization and load shedding are verified on a simulated loss of offsite power and on a safeguards actuation signal in conjunction with a simulated loss of offsite power.
4. A loss of offsite power is simulated and diesel generator starting and emergency bus load sequencing are verified.
5. Immediately following the diesel generator full-load capability test, a loss of offsite power is simulated, and diesel generator functioning is verified.
6. Following initiation of a simulated loss of offsite power and a safeguards actuation coincident with a simulated loss of offsite power, diesel generator loading is checked to ensure that the diesel generator continuous operation rating is not exceeded.
7. A load equal to or greater than the largest single load associated with each diesel generator and a load equal to the diesel generator continuous rating are instantaneously removed from each diesel generator, and diesel generator load rejection capability is verified.
8. With each diesel generator loaded with the emergency bus loads, the capability to synchronize each diesel generator with the offsite source and to transfer load is verified.

D. Acceptance Criteria

1. Each train diesel generator functions as designed following initiation of a safeguards actuation signal coincident with a simulated loss of offsite power, emergency bus loads are sequenced on the bus, and train independence is verified. (See Design Basis DC-1821.)
2. On a simulated loss of the diesel generator with offsite power not available, the loads are shed from the emergency buses and subsequent loading is in accordance with design requirements. (See paragraph 8.3.1.1.3.)

3. The emergency buses are deenergized and loads are shed from the emergency buses upon a simulated loss of offsite power coincident with a safeguards actuation signal. (See paragraph 8.3.1.1.3.)
4. The diesel generators function as designed on a simulated loss of offsite power including a simultaneous start of the redundant units, and emergency bus loads are sequenced on the bus. (See Design Basis DC-1821.)
5. With the diesel generators at full-load temperature conditions, the diesel generators will function as designed following a simulated loss of offsite power. (See paragraph 8.3.1.1.3.)
6. The diesel generator loading is within design specifications following a simulated loss of offsite power or safeguards actuation coincident with a simulated loss of offsite power. (See paragraph 8.3.1.1.3.)
7. The diesel generator functions following rejection of a load equal to or greater than its largest associated load and rejection of load equal to its continuous rating. (See paragraph 8.3.1.1.3.)
8. With the diesel generators loaded with emergency bus loads, the diesel generators function properly during synchronizing and transferring loads to the offsite power source. (See paragraph 8.3.1.1.3.)

14.2.8.1.98 ESFAS Master and Slave Relay Preoperational Test

This test incorporated into 14.2.8.1.84, 14.2.8.1.85, and 14.2.8.1.106.

14.2.8.1.99 Containment Local Leakrate Preoperational Test

A. Objective

To determine the leakage rate of the containment penetrations and isolation valves.

B. Prerequisites

1. Containment isolation valves are closed by normal activation methods.
2. Associated piping is drained, and vent paths for leakage are established as required. (See table 6.2.6-1.)

C. Test Method

The containment penetrations and isolation valves are leak tested by performing type B and type C tests, in accordance with 10 CFR 50, Appendix J.

D. Acceptance Criterion

The combined leakage from containment penetrations and isolation valves is a 0.6 L where L is a leakage of 0.20 percent by weight of the containment air per 24 h at a pressure of not less than P (design accident pressure). (See subsection 6.2.6).

14.2.8.1.100 Reactor Containment Structural Integrity Test

A. Objective

To demonstrate the structural integrity of the reactor containment building.

B. Prerequisites

1. Containment penetrations are installed, and penetration leak tests are complete.
2. Containment penetrations, including equipment latches and personnel airlocks, are closed.

C. Test Method

The containment is pressurized to the test value and deflection measurements, and concrete crack inspections are made to determine that the actual structural response is within the limits predicted by the design analyses.

D. Acceptance Criterion

The containment structural response is within the limits predicted by design analyses (FSAR 3.8.1.7).

14.2.8.1.101 High-Efficiency Particulate Air Filters Preoperational Test

A. Objective

To demonstrate operation of the high-efficiency particulate air (HEPA) filters.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing is complete.
2. The ventilation systems containing HEPA filters and charcoal absorbers have been air balanced and are operational and available to support this test.
3. Component testing and instrument calibration is complete.
4. Test instrumentation is available and calibrated.
5. Support systems are available.

C. Test Method

1. HEPA filters and charcoal absorbers will be tested in place.
2. Verify alarms, indicating instruments, and status lights.
3. Verify design airflow.

D. Acceptance Criterion

The HEPA filters operate as described in subsection 6.5.1 and section 9.4.

14.2.8.1.102 Emergency Core Cooling System Sump Preoperational Test

A. Objectives

1. To demonstrate the operating characteristics of the RHR and containment spray pumps during the recirculation phase of safety injection.
2. To demonstrate the piping pressure drop from the containment sump to the suction of the RHR and containment spray pumps while in the recirculation mode.
3. To verify that vortex control and pressure drop across the sump screens were evaluated with a laboratory model study.

B. Prerequisites

1. Required preoperational testing, instrument calibration, and system flushing/cleaning are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. The RWST contains an adequate supply of demineralized water for performance of the test.
4. A temporary piping jumper connects each of the RHR and containment spray suction pipes at the containment sump.
5. Temporary piping is connected to the containment spray pump discharge lines to allow removal of the containment spray water from containment.

C. Test Method

1. Each RHR pump is started, taking a suction on the RWST through the temporary piping jumper to simulate the recirculation mode and pump operating characteristics; and piping pressure drops are recorded while discharging to the reactor vessel.
2. Each containment spray pump is started, taking a suction on the RWST through the temporary piping jumper to simulate the recirculation mode and pump operating characteristics; and piping pressure drops are recorded while discharging out the temporary discharge pipe.

D. Acceptance Criteria

1. The RHR and containment spray pumps operate as designed in the simulated recirculation mode. (See 6.2.2.2.2.3.2 and 6.3.2.2.4.)
2. The suction piping pressure drop for each pump is within the design limits. (See 6.2.2.2.2.3.2 and 6.3.2.2.4.)

14.2.8.1.103 Thermal Expansion Testing

A. Objectives

1. To verify that essential nuclear steam supply system and BOP components, piping, support, and restraint deflections are unobstructed and within design specifications.

2. To verify that thermal movements for safety-related snubbers, as delineated in the Technical Specifications and whose system operation temperature exceeds 250°F, are within design specifications.

B. Prerequisites

1. This test is conducted simultaneously with hot functional testing.
2. Supports, restraints, and hangers are installed, and reference points and predicted movements are established.
3. Temporary instrumentation is installed as required to monitor the expansion of the components under test.
4. A preservice examination has been performed on snubbers as defined in paragraph 14.2.8.1.103.A.2.

C. Test Method

1. During the reactor coolant system heatup and cooldown, deflection data are recorded.
2. Snubber thermal movements are verified by recording positions during initial system heatup and cooldown.
3. Snubber swing clearance is verified at specified heatup and cooldown intervals.

D. Acceptance Criteria

1. There shall be no evidence of blocking of the thermal expansion of any piping or components, other than by installed supports, restraints, and hangers (paragraph 3.9.B.2.1).
2. Spring hanger movements must remain within the hot and cold setpoints, and snubbers must not become fully retracted or extended.
3. Piping and components must return to their approximate baseline cold position.

14.2.8.1.104 Power Conversion and Emergency Core Cooling System Dynamics Test

A. Objective

To demonstrate during specified transients that the system's monitored parts respond in accordance with design calculations.

B. Prerequisites

1. Reference points for measurement of the systems are established, and required temporary instrumentation is installed and calibrated.
2. Hot functional testing is in progress for those portions of testing requiring elevated temperatures and pressure.
3. All subject systems are available for the specified dynamic operations.

C. Test Method

Deflection measurements are recorded during various plant transients.

D. Acceptance Criteria

1. The movements due to flow-induced loads shall not exceed design limits (subsection 3.9.2).

2. Flow-induced movements and loads will not cause malfunctions of plant equipment or instrumentation.

14.2.8.1.105 Remote Shutdown Preoperational Test

A. Objectives

1. To demonstrate the capability to cool down the plant from the hot standby condition to the cold shutdown condition using controls and instrumentation located outside the control room.

2. To demonstrate the capability to control plant parameters during a simulated loss of ac power using manual control and the steam-driven auxiliary feedwater pump.

B. Prerequisites

1. The controls and instrumentation associated with the remote shutdown panel are available.

2. Hot functional testing is in progress with the RCS temperature above that at which the RHR system is in operation.

C. Test Method

1. Transfer control from the control room to the remote shutdown panel.

2. Check the functioning of instrumentation, controls, alarms, and interlocks.

3. Remotely cool the plant down to the point of establishing RHR.

4. Remotely establish RHR, and reduce reactor coolant temperature approximately 50°F.

5. Demonstrate manual control of plant parameters using the atmospheric relief valves and the steam-driven auxiliary feedwater pump.

6. Transfer control back to the control room.

D. Acceptance Criteria

1. Transfer of control from the control room to the remote shutdown panel is achieved in accordance with design requirements. (See table 7.4.2-1.)

2. The potential for cooling the plant down from hot standby to cold shutdown has been demonstrated. (See Design Basis DC-1624.)

14.2.8.1.106 Reactor Trip System and ESFAS Response Time Test

A. Objective

1. To demonstrate that the reactor trip system and ESFAS response times are less than the maximum allowable response times as specified in the accident analysis.

2. To demonstrate that the containment isolation valves reposition in less than or equal to the required operating time.

B. Prerequisites

1. Required reactor protection system alignments, calibration, and testing are complete.
2. Required electrical power supplies and control circuits are energized and operational.
3. Special test circuitry test equipment and test instrumentation required for test performance are available.

C. Test Method

1. The response times of each primary sensor associated with the reactor trip system and the ESFAS will be obtained by testing with an appropriate test device.
2. Reactor trip system and ESFAS actuation signals are applied at the process sensor output, and the time interval is measured until loss of stationary gripper coil voltage and until the ESF equipment is capable of performing its safety function.
3. Total reactor trip system and ESFAS response time is computed by summing the results of the previous test methods.
4. Individual components and valve response times are accomplished in the system preoperational test procedures.

D. Acceptance Criterion

1. The total reactor trip system and total ESFAS response times do not exceed the allowable response times as specified.
2. Containment isolation valves reposition in less than or equal to the required time. (See table 6.2.4-1.)

14.2.8.1.107 Extraction Steam Test

A. Objectives

1. To verify proper operation and control of each extraction line isolation and drain valve.
2. To verify proper operation of each nonreturn valve.

B. Prerequisites

The required portions of the following prerequisites are completed as necessary to support the preoperational test:

1. Construction acceptance testing is complete.
2. Component testing and instrument calibration is complete.
3. Test instrumentation is available and calibrated.
4. Support systems are available.

C. Test Methods

1. Extraction line isolation and drain valves will be tested to verify the valves function in accordance with the control logic.
2. Each nonreturn valve will be observed to verify that the valve shuts following a turbine trip signal.

D. Acceptance Criteria

1. The extraction line isolation valves close and drain valves open following a turbine trip signal.
2. Each nonreturn valve closes following a turbine trip.

14.2.8.1.108 Condensate and Feedwater Chemical Injections System Test

A. Objective

To verify the operability of the condensate and feedwater chemical injection system.

B. Prerequisites

1. Required construction acceptance testing is complete.
2. Required system flushing/cleaning is complete.
3. Required electrical power supplies and control circuits are operational.
4. The condensate and feedwater systems are available.

C. Test Method

1. The operating parameters of the positive displacement chemical feed pumps and the wet layup pumps will be measured.
2. The mixing capability of the mixing pump will be verified.

D. Acceptance Criterion

1. The isolation valves operate as described in 10.4.7.2.2.5.
2. The automatic functions required to add chemicals performed as described in 10.4.7.5.

14.2.8.1.109 Proteus Computer Preoperational Test

A. Objectives

1. To verify computer hardware is operational.
2. To verify all digital inputs are conditioned correctly.
3. To verify computer software functions correctly.

B. Prerequisites

1. Required electrical power supplies are operational.
2. Test instrumentation is available and calibrated.

C. Test Method

1. Diagnostic programs are run on each section of the hardware.
2. Test signals are injected into the computer to simulate all digital inputs.
3. Software tests are run to verify operability of software.

D. Acceptance Criteria

1. The diagnostic programs or other functional tests are performed on computer hardware without error.
2. The computer conditions all digital inputs in agreement with the Proteus computer input/output list.
3. Applicable software routines run without error when subject to the applications program test procedures.

14.2.8.1.110 Equipment Building HVAC and Piping Penetration Preoperational Test

This test has been deleted.

14.2.8.1.111 Steam Generator Blowdown Processing System

A. Objective

To demonstrate that the steam generator blowdown processing system accepts water from each steam generator blowdown line, processes the blowdown as required, and delivers the processed water to the condensate system and the waste water retention basin.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing has been completed.
2. Component testing and instrument calibration has been completed.
3. Test instrumentation is available and has been calibrated.
4. Support systems are available.

C. Test Method

1. Verify manual and automatic system controls.
2. Verify flowrates, pressures, and temperatures.
3. Verify processed water quality.
4. Verify indications (flow, pressure, and temperature), status lights, and alarms.
5. Verify system isolation using simulated signals.
6. Verify flowpaths.

D. Acceptance Criterion

The steam generator processing system operates as described in subsection 10.4.8.

14.2.8.1.112 Main Turbine System Valves Preoperational Test

A. Objective

To demonstrate the functional performance of the main turbine system stop valves, including the following:

1. Operability of the main stop valves, control valves, intermediate stop valves, and intercept valves, including the actuation times from a turbine trip signal. Testing is to be performed in accordance with vendor requirements.

B. Prerequisites

The required portions of the following prerequisites are completed, as necessary, to support the preoperational test:

1. Construction acceptance testing completed.
2. Component testing and instrument calibration completed.
3. Test instrumentation available and calibrated.
4. Support systems available.

C. Test Method

1. The main stop valves, control valves, intermediate stop valves, and intercept valves will be tested in accordance with vendor requirements.
2. The various interfaces from control and trip signals will be simulated as required during testing.

D. Acceptance Criterion

The main turbine system stop valves operate as described in Section 10.2.

14.2.8.1.113 125-V-dc Class 1E Minimum Load Voltage Verification

A. Objectives

1. To measure the voltage drops, at nominal battery voltage, to 125-V-dc class 1E inverters and power-operated valves.
2. To determine the voltage which would be available at the 125-V-dc class 1E inverters and power-operated valves if the batteries were discharged to the minimum voltage limit.
3. To verify that the voltage available to 125-V-dc class 1E inverters and power-operated valves exceed the design minimum.

B. Prerequisites

1. Required construction acceptance testing is complete.

2. The 125-V-dc class 1E inverters and power-operated valves are operable.

3. Required load test devices are available.

C. Test Method

1. Each 125-V-dc class 1E inverter will be loaded to its design capacity and the voltage drop from the battery to the inverter input measured.

2. Each 125-V-dc class 1E power-operated valve will be operated and the voltage drop from the battery to the motor or solenoid measured.

3. The minimum available voltage at each 125-V-dc class 1E inverter and power-operated valve will be determined from the measured voltage drops and the battery minimum voltage limit.

D. Acceptance Criteria

1. The minimum available input voltage for the 125-V-dc class 1E inverters equals or exceeds 105 V-dc (paragraph 8.3.2.1.1).

2. The minimum available input voltage for the 125-V-dc class 1E power-operated valves equals or exceeds 100 V-dc or the equipment minimum-rated operating voltage, whichever is lower (paragraph 8.3.2.1.1).

14.2.8.1.114 Steady State Vibration Monitoring of Safety Related and High-Energy Piping

A. Objective

Demonstrate that steady state vibrations of safety- related and high-energy piping are within acceptable limits.

B. Prerequisites

1. The subject systems are available for operation at the specified conditions.

2. Portable instrumentation is available and calibrated.

C. Test Method

1. Specified safety-related and high-energy piping runs are screened qualitatively for perceptible vibration by visual inspection.

2. All piping observed to be vibrating is monitored with portable instrumentation. Results are assessed versus quantitative screening criteria or, if necessary, using standard stress evaluation techniques.

D. Acceptance Criterion

Steady state vibrations of safety-related and high- energy piping are within the allowable stress limits defined in 3.9.B.2.1-B.2.

14.2.8.1.115 AMSAC (ATWS Mitigating System Actuation Circuitry System)

A. Objectives

1. Demonstrate the capability of the AMSAC processing system to perform the required logic and timing functions.
2. Demonstrate that the AMSAC system produces the required output signals in response to specific input signals.

B. Prerequisites

7300 Process Protection and Control System is calibrated and available to the extent necessary to provide the required input signals to the AMSAC system.

C. Test Method

1. Internal AMSAC testing will be accomplished to verify input, output, setpoint, timer, and logic functions.
2. Overall tests will be accomplished to verify correct AMSAC output trip, and actuation signals are generated in response to specified input signal levels.

D. Acceptance Criteria

1. The AMSAC processing system performed the required logic and timing functions.
2. The AMSAC system produced the required output signals in response to specified input signals.

14.2.8.2 Startup Test Procedures

The following are the test abstracts for each startup test listed in table 14.2.1-2.

14.2.8.2.1 RCS Final Leak Test

A. Objective

To determine the amount of identified and unidentified leakage from the RCS and verify that the leakage is within allowable limits.

B. Prerequisites

1. Core loading is complete.
2. The RCS is at operating temperature and pressure.

C. Test Method

The identified and unidentified RCS leakage rates are determined by monitoring the system water inventory over a specified period of time.

D. Acceptance Criterion

Identified and unidentified RCS leakage is within Technical Specification limits.

14.2.8.2.2 Pressurizer Heater and Spray Capability and Continuous Spray Flow Verification Test (See Table 14.2.1-2 for Unit 2)

A. Objectives

1. To establish the optimum continuous spray flowrate.
2. To determine the effectiveness of the pressurizer heaters and normal control spray.
3. To demonstrate the depressurization rate by turning off pressurizer heaters and by using auxiliary spray.

B. Prerequisites

1. The RCS is at no-load operating temperature and pressure.
2. Reactor coolant pumps are operating.
3. Pressurizer heaters are operable.

C. Test Method

1. While maintaining constant pressurizer level, spray bypass valves are adjusted until a minimum flow is achieved which maintains the temperature difference between the spray line and the pressurizer within acceptable limits.
2. With the pressurizer spray valves closed, pressurizer heaters are energized, and the time to raise the pressurizer pressure a specified amount is recorded.
3. With the pressurizer heaters deenergized, both spray valves are fully opened, and the time to lower the pressurizer pressure a specified amount is recorded.
4. All except one reactor coolant pump are turned off, and the pressurizer heaters are turned off. The depressurization rate is then observed with its effect on margin to saturation temperature.
5. The pressurizer heaters are reestablished and auxiliary spray is initiated; charging and steam flow are then varied to observe the effects on RCS depressurization.

D. Acceptance Criteria

1. The spray bypass valves are throttled so that the minimum flow necessary to keep the spray line warm is achieved.
2. The pressurizer pressure response to the opening of the pressurizer spray valves and to the actuation of all pressurizer heaters is within the limits described in Section 5.4.10.

14.2.8.2.3 RCS Flow Measurement Test

A. Objectives

1. To obtain the necessary data to interrelate reactor coolant elbow tap differential pressure as a measurement of RCS flowrate, confirming

reactor coolant flow is equal to or greater than the design flow listed in table 5.1.2-1.

2. To perform a calorimetric flow measurement at less than 75-percent power, confirming RCS flow is equal to or greater than the design flow listed in table 5.1.2-1.

B. Prerequisites

1. Required instrument calibration is complete.
2. Required support systems are operational.
3. The reactor core is installed, and the plant is at normal operating temperature and pressure prior to initial criticality.

C. Test Method

1. Elbow tap differential pressure is taken prior to criticality and RCS flow is calculated.
2. A calorimetric is performed at the 50-percent power plateau and RCS flow is calculated.

D. Acceptance Criterion

Reactor coolant flowrate in each loop is equal to or greater than the design flow listed in table 5.1.2-1.

14.2.8.2.4 RTD Bypass Valve Flow Measurement Test

A. Objectives

1. To determine the flowrate necessary to achieve the design objectives for reactor coolant transport time in each resistance temperature detector (RTD) bypass loop.
2. To measure the flowrate in each RTD bypass loop and verify that the reactor coolant transport times in these loops (from the bypass loop connection on the main coolant loop to the last RTD) are acceptable.
3. To establish the low flow alarm setpoint for the total bypass loop flow for each reactor coolant loop.

B. Prerequisites

1. The installed pipe length measurements are made with the plant cold before insulation is installed.
2. The RCS is at operating temperature and pressure.

C. Test Method

1. The flowrate necessary to achieve the design reactor coolant transport time for each hot and cold leg bypass loop is calculated utilizing the piping length of each leg.
2. The hot and cold leg RTD bypass loop flow data are recorded at operating temperature and pressure.

D. Acceptance Criteria

1. Using actual piping lengths, the minimum flowrates were calculated to meet the transport time specified in the Westinghouse NSSS Startup Manual.
2. Actual flowrates in all RTD bypass manifolds exceed the minimum required flowrate required to achieve a 1.0.
3. Each RTD loop bypass low flow alarm has been verified to activate at the flow specified in the Westinghouse NSSS Startup Manual.

14.2.8.2.5 RCS Flow Coastdown Test

A. Objectives

1. To measure the rate at which reactor coolant flow changes, subsequent to tripping all reactor coolant pumps.
2. To measure various delay times associated with the loss of flow accident.

B. Prerequisites

1. Required component testing and instrument calibration are complete.
2. Required electrical power supplies and control circuits are operational.
3. The reactor core is installed, and the plant is at normal operating temperature and pressure with all reactor coolant pumps running.

C. Test Method

Flow coastdown stabilization and loss of coolant delay-time data are recorded while tripping all reactor coolant pumps.

D. Acceptance Criteria

1. The rate of change of reactor coolant flow for all reactor coolant pumps tripped is less than the rate of change assumed in section 15.0 within design specifications.
2. The RCS low-flow delay times are less than or equal to those assumed in the safety analysis for loss of flow in section 15.0.

14.2.8.2.6 Reactor Protection Test

A. Objectives

1. To verify that initial trip setpoint adjustments have been made prior to initial unit startup and to specify which trip setpoint adjustments will require readjustment during startup.
2. To obtain a record of all trip setpoints.

B. Prerequisites

1. Reactor trip instrumentation has been aligned and calibrated with setpoints adjusted to values given in the Technical Specifications or the unit test documents.

2. Reactor trip instrumentation has been energized for a time sufficient to achieve stability.

C. Test Method

1. Trip setpoints are reviewed and documented prior to criticality.
2. During startup and test operations, specific setpoints noted for readjustment on the data sheets are readjusted and final setpoint values recorded.

D. Acceptance Criteria

1. Initial reactor trip setpoints are verified to be within design criteria and in conformance with or more conservative than values in the Technical Specifications.
2. Setpoints readjusted during startup and testing are noted, and a final record of all setpoints is obtained.

14.2.8.2.7 Core Loading Instrumentation and Neutron Source Requirements Test

A. Objectives

1. To verify alignment, calibration, and neutron response of the temporary core loading instrumentation prior to the start of fuel loading.
2. To verify the neutron response of the nuclear instrumentation system (NIS) source range channels prior to the start of fuel loading.
3. To verify the neutron response of the temporary and NIS source range instrumentation prior to resumption of fuel loading following any delay of 8 h or more.

B. Prerequisites

1. Temporary core loading instrumentation package is available.
2. Operational alignment of the NIS that is to be performed prior to core loading is complete.
3. Plant and temporary instrumentation has been calibrated, and calibration data are available.

C. Test Method

1. A portable neutron source (1 to 5 Ci) and preshipment equipment checkout data are used to verify proper alignment, calibration, and neutron response of the temporary core loading instrumentation and the NIS source range instrumentation.
2. A portable neutron source (1 to 5 Ci) or movement of a source-bearing fuel element is used to produce the desired change in neutron level to verify the neutron response of the temporary core loading instrumentation and the NIS source range instrumentation prior to resumption of fuel loading, following any delay of 8 h or more.
3. A statistical evaluation of ten observations for each channel may be performed to verify operability of the equipment as an alternative to Test Method No. 2.

D. Acceptance Criterion

Neutron instrumentation is operational and calibrated and indicates the appropriate change in count rate as the neutron level is varied. (NIS source range instrumentation is calibrated in accordance with paragraph 14.2.8.2.10.)

14.2.8.2.8 Thermal Power Measurement and Statepoint Data Collection Test

A. Objectives

1. To determine reactor power by performing a heat balance at 30-, 50-, 75-, 90-, and 100-percent power.
2. To identify instrumentation for statepoint data collection.

B. Prerequisites

1. The following equipment has been installed and checked out and is operational: sensor for measuring steam generator feedwater temperature, differential pressure measuring devices for determining feedwater flow to each steam generator, and pressure gauges to measure steam pressure at steam generator outlets.
2. The following control systems are in automatic: pressurizer pressure and level and steam generator level.

C. Test Method

1. Collect data and calculate thermal power.
2. Obtain statepoint data, compute the average for each parameter measured, convert to the appropriate units, and summarize the data for each RCS loop.
3. Calibrate nuclear instruments to the power determined by the heat balance.

D. Acceptance Criterion

This test is for the collection of data.

14.2.8.2.9 Incore Detector Test

A. Objectives

1. To set up and demonstrate the operation of the incore instrumentation system.
2. To verify its adequacy for incore flux mapping.

B. Prerequisites

1. Reactor vessel upper internals are installed, and the reactor vessel head is installed with the studs tensioned.
2. The initial fuel loading has been completed.
3. Rotation and limit switch operation of the incore detector system has been verified.

4. Testing is performed at hot shutdown.

C. Test Method

1. A dummy cable will be inserted into each thimble to demonstrate system operation.
2. Detectors will be inserted into the thimbles to demonstrate system operation.

D. Acceptance Criterion

The system operates per its technical manual and is capable of taking a flux map.

14.2.8.2.10 Operational Alignment of Nuclear Instrumentation Test

A. Objectives

To establish and determine voltage settings, trip settings, operational settings, alarm settings, and overlap of channels on source range, intermediate range, and power range instrumentation from prior to initial criticality to at or near full reactor power.

B. Prerequisite

The NIS has been aligned in accordance with the Westinghouse Nuclear Instrumentation System Manual and the Precautions, Limitations, and Setpoints for Westinghouse NSSS.

C. Test Method

1. Functions will be calibrated, tested, and verified, utilizing permanently installed controls and adjustment mechanisms.
2. Operational modes of the source range, intermediate range, and power range channels will be set for their proper functions, in accordance with the test instructions.

D. Acceptance Criteria

1. The NIS demonstrates the ability to achieve the operational adjustments according to the Westinghouse Technical Manual.
2. The NIS demonstrates an overlap of indication between the source and intermediate ranges and the intermediate and power range instrumentation.

14.2.8.2.11 (Material deleted)

14.2.8.2.12 Rod Control System Test

A. Objective

To demonstrate and document that the rod control system performs the required control and indication functions just prior to initial criticality.

B. Prerequisites

1. RCS is at no-load operating temperature and pressure.
2. The NIS source range channels are aligned and operable.

C. Test Method

1. With the reactor at no-load temperature and pressure, just prior to initial criticality, the operation of the rod control system in various modes is verified.
2. The operation of status lights, alarms, and indicators is verified.

D. Acceptance Criteria

1. The rod control system will withdraw and insert each rod bank.
2. The rod position and indication system tracks each rod bank as it is being moved.
3. The control banks overlap system starts and stops rod movement at the designated bank positions.

14.2.8.2.13 CRDM Operational Test

A. Objectives

1. To demonstrate the operation of the CRDM under both cold and hot plant conditions.
2. To provide verification of slave cycler timing.

B. Prerequisites

1. The RCS is filled and vented at cold shutdown.
2. All rods are fully inserted.
3. Nuclear instrumentation channels are available.
4. A fast-speed oscillograph (Visicorder or equivalent) to monitor test parameters is available.

C. Test Method

1. With the reactor core installed and the reactor in the cold shutdown condition, confirm that the slave cycler devices supply operating signals to the CRDM stepping magnet coils.
2. Verify operation of all CRDMs under both cold and hot standby conditions. The CRDM magnet coil currents are recorded.

D. Acceptance Criterion

The CRDMs conform to the requirements for proper mechanism operation and timing as described in the magnetic CRDM instruction manual.

14.2.8.2.14 Rod Drop Time Measurement Test

A. Objectives

1. To determine the rod drop time of each RCCA under full-flow conditions, with the reactor at normal operating temperature and pressure.
2. To verify the operability of the control rod deceleration device.

B. Prerequisites

1. Initial core loading has been completed.
2. Source range channels are in operation.
3. All rods are fully inserted.

C. Test Method

1. Withdraw each RCCA, interrupt the electrical power to the associated CRDM, measure and record the rod drop time, and verify control rod deceleration.
2. Perform a minimum of three additional drops for each control rod whose drop time falls outside the two-sigma limit, as determined from the drop times obtained for each test condition.

D. Acceptance Criteria

1. The rod drop times are acceptable in accordance with plant Technical Specifications.
2. The control rod is slowed by the control rod deceleration device during rod drop testing.

14.2.8.2.15 Rod Position Indication Test

A. Objective

To verify that the rod position indication system satisfactorily performs required indication and alarm functions for each individual rod and that each rod operates satisfactorily over its entire range of travel.

B. Prerequisites

1. The RCS is at no-load operating temperature and pressure.
2. At least one reactor coolant pump in service with reactor coolant boron concentration not less than specified in the Technical Specifications for refueling shutdown.

C. Test Method

Rod banks are individually withdrawn from and reinserted into the core, according to the test procedure, while recording analog output voltage, control room position readout, and the group step counters.

D. Acceptance Criterion

The rod position indication system performs the required indication and alarm functions, as described in section 7.7.1.3, and each rod operates over its entire range of travel.

14.2.8.2.16 Operational Alignment of Process Temperature Instrumentation Test

A. Objective

To align ΔT and T_{avg} process instrumentation under isothermal conditions prior to criticality and at power.

B. Prerequisites

1. All reactor coolant pumps are operating.
2. The RCS average temperature must be at the hot no-load average temperature ±2°F.

C. Test Method

1. Align ΔT and T_{avg} according to test instructions at isothermal conditions prior to criticality and at approximately 75-percent power. Extrapolate the 75-percent data to determine ΔT and T_{avg} values for the 100-percent plateau.
2. At or near 100-percent power, check the alignment of the ΔT and T_{avg} channels for agreement with the results of the thermal power measurement.

D. Acceptance Criteria

1. The T_{avg} should be within 0.5°F of the value calculated from the T_{hot} and T_{cold} converter outputs.
2. The T_{hot} and T_{cold} measured values are within 1.2°F of the spare RTDs for each channel.
3. The measured ΔT for each channel is within 1 percent of the calorimetric power at power and 0.5 percent at isothermal conditions.

14.2.8.2.17 Startup Adjustments of RCS Test

A. Objective

To obtain the optimum plant efficiency.

B. Prerequisites

1. The RCS is at no-load operating temperature and pressure.
2. The RCS temperature is being controlled by the dump valves.

C. Test Method

1. Obtain system temperature and steam pressure data at steady-state conditions for zero power and at hold points during power escalations.

2. Evaluation of this data will provide the basis for adjustments to the RCS.

D. Acceptance Criteria

1. Steam generator pressure is within ±10 psi of design at 100-percent power.

2. The full load T_{avg} shall not exceed 588.5°F.

14.2.8.2.18 RCS Sampling for Core Loading Test

A. Objective

To verify correct and uniform boron concentration, prior to core loading in the RCS and directly connected auxiliary systems.

B. Prerequisites

1. Boric acid storage tanks, transfer pumps, and associated piping and equipment are available for use.

2. The RCS is filled with reactor grade water which has been borated to the concentration specified in the initial core loading procedure.

C. Test Method

1. The RCS will be filled and on recirculation to maintain a uniform boron concentration.

2. Four samples will be obtained at approximately four equidistant depths from the surface to the bottom of the reactor vessel. Also, a sample will be taken from the operating RHR loop.

3. These samples will be analyzed to verify that there is a uniform boron concentration in the RCS.

D. Acceptance Criteria

1. The samples obtained from the designated sample points shall have a minimum boron concentration of 2000 ppm.

2. The reactor vessel samples are within 50 ppm of each other.

14.2.8.2.19 Metal Impact Monitoring System Test

A. Objectives

1. To perform final calibration of the metal impact monitoring system.

2. The alarm signal level will be established for startup testing and operations.

B. Prerequisites

1. Required electrical power supplies and control circuits are energized and operational.

2. Metal impact monitoring system accelerometers have been installed at the proper locations.

C. Test Method

1. The metal impact monitoring system calibration is performed in accordance with the manufacturer's instructions including verification of accelerometer calibration.

2. After fuel loading, the RCS metal impact monitoring system operation is verified. The alert signal level is established.

D. Acceptance Criteria

1. A metal impact monitoring system calibration is acceptable and in accordance with the manufacturer's instructions.

2. The system operates for RCS monitoring during startup testing and power operations.

14.2.8.2.20 Initial Fuel Loading Test Sequence

A. Objective

1. To establish the conditions under which the initial fuel loading is to be accomplished.

2. To accomplish initial fuel loading in a safe and orderly manner.

B. Prerequisites

1. Testing prior to initial fuel loading is completed sufficiently to demonstrate the operability of required systems and components.

2. Temporary and permanent source range channels are operable.

3. At least one path for boron addition to the RCS is available.

4. Uniform boron concentration in the RCS is maintained by recirculation with at least one RHR pump and is sufficient to ensure required boron concentration in accordance with the Technical Specifications during fuel loading.

5. Containment integrity is established in accordance with applicable Technical Specifications.

C. Test Method

1. Fuel is inserted into the reactor vessel in accordance with the prespecified loading sequence.

2. Neutron count rate is monitored on temporary and permanent source range detectors.

3. Core reactivity is monitored through plots of inverse neutron count rate ratio.

D. Acceptance Criterion

The core is assembled in accordance with the prespecified configuration.

14.2.8.2.21 Inverse Count Rate Ratio Monitoring for Fuel Loading Test

A. Objective

To describe the frequency, core conditions, and methods for obtaining nuclear monitoring data during initial core loading.

B. Prerequisites

1. Temporary and plant source range nuclear instrumentation has been operational for a minimum of 60 min to allow the instruments to attain stable operating conditions.
2. The plant is prepared for initial core loading.

C. Test Method

1. Data from the temporary and permanent nuclear monitoring channels are used to assess the core multiplication factor during core loading operations.
2. The inverse count rate ratio is plotted and evaluated to prevent any unexpected deviation from the required core multiplication factor.

D. Acceptance Criteria

Acceptance criteria are not applicable to this test.

14.2.8.2.22 Determination of Core Power Range for Physics Testing

A. Objectives

1. To determine the reactor power level at which effects from fuel heating is detectable.
2. To establish the range of neutron flux in which zero power reactivity measurements are to be performed.

B. Prerequisites

1. The reactor is critical and stable in the intermediate range.
2. Control rods are sufficiently deep in the core to allow positive reactivity insertion by rod withdrawal.
3. Reactor coolant temperature is established at a value that minimizes the moderator temperature coefficient reactivity feedback.

C. Test Method

1. Withdraw control rod bank and allow the neutron flux level to increase until nuclear heating effects are indicated by the reactivity computer.
2. Record the reactivity computer picoammeter flux level and, if possible, the corresponding intermediate range channel currents at which nuclear heating occurs, to obtain zero power testing range.

D. Acceptance Criterion

This procedure is provided to collect data and to determine the power range at which zero power testing will be conducted.

14.2.8.2.23 Precritical Test Sequence

A. Objectives

To specify the sequence of events which constitute the precritical test program.

B. Prerequisite

Plant system conditions are established as required by the individual test instructions within this sequence.

C. Test Method

This instruction will establish the sequence for required testing after core loading, until the plant has completed all precritical testing and reached the hot shutdown condition.

D. Acceptance Criteria

Acceptance criteria are contained in the various individual tests conducted during this time.

14.2.8.2.24 Dynamic Automatic Steam Dump Control Test

A. Objective

To verify automatic operation of the T_{avg} steam dump control system, demonstrate controller setpoint adequacy, and obtain final settings from steam pressure control of the condenser dump valves.

B. Prerequisites

1. The steam dump control system has been aligned and calibrated to initial settings.
2. The plant is at no-load temperature and pressure.
3. The condenser vacuum has been established.
4. The reactor is critical.

C. Test Method

1. Reactor power is increased to less than 10 percent by rod withdrawal and steam dump to condenser to demonstrate setpoint adequacy.
2. Pressure controller setpoint is increased prior to switching to T_{avg} control, which rapidly modulates open condenser dump valves.
3. Simulate turbine operating conditions with reactor at power, then simulate a turbine trip resulting in the rapid opening of the steam dump valves.

D. Acceptance Criteria

1. The plant trip controller responds to maintain a stable T_{avg}. After steady state power is achieved, no divergent oscillations in temperature occur.
2. The loss of load controller responds properly to maintain a specified stable T_{avg}. After steady state power is achieved, no divergent oscillations in temperature occur.

3. The steam header pressure controller responds to maintain a stable pressure at normal no-load pressure.

14.2.8.2.25 Automatic Steam Generator Level Control Test

A. Objective

To verify the stability of the automatic steam generator level control system following simulated transients at low-power conditions and the operation of the variable speed feature of the feedwater pumps.

B. Prerequisites

1. The reactor is critical, and at less than 10-percent power.
2. The steam generator level control system has been checked and calibrated.
3. Steam generator alarm setpoints have been set for each generator.

C. Test Method

1. Induce simulated steam generator level transients to verify steam generator level control response.
2. Verify the variable speed features of the main feedwater pumps by manipulation of controllers and test input signals.

D. Acceptance Criteria

1. Steam generator level overshoot (undershoot) is less than 4 percent following a level change; the level will track to within 2 percent of the level setpoint within three reset time constants following a control transfer, level, or setpoint change.
2. Feedwater pump discharge pressure oscillations are less than 3 percent within three reset time constants following a steamflow change.
3. The main feedwater regulating valves open and stabilize in response to various steam flow conditions.

14.2.8.2.26 Power Coefficient Determination Test

A. Objective

To verify nuclear design predictions of the doppler-only power coefficient.

B. Prerequisites

1. Reactor power level is at the specified testing plateau (30-, 50-, 75-, and greater than 90-percent power). Reactor coolant temperature and pressure, and rod cluster control (RCC) bank configurations are at normal operating conditions and are stable.
2. All subsystems which affect overall plant transient response, except the rod control system, should, whenever possible, be in automatic control.

C. Test Method

1. Plant conditions are stabilized at selected power plateaus (30-, 50-, 75-, and greater than 90-percent). The plant load is varied, using the turbine-generator controller, in a manner approximating step changes of about 2- to 4-percent power.

2. The doppler-only power coefficient verification factor is determined from correlative measurements of reactor coolant system temperature and core thermal power output.

D. Acceptance Criterion

The doppler-only power coefficient verification factor agrees with the value given in the nuclear fuel design report.

14.2.8.2.27 Load Swing Test

A. Objective

To verify nuclear plant transient response, including automatic control system performance, when step-load changes are introduced to the turbine-generator at 30-, 75-, and 100-percent power levels.

B. Prerequisite

The plant is operating in a steady state condition at the desired power level.

C. Test Method

The turbine-generator output is changed as rapidly as possible to achieve a step load increase or decrease. Selected plant parameters are monitored and recorded during the load transients.

D. Acceptance Criterion

The primary and secondary control systems, with no manual intervention, maintain reactor power, RCS temperature, pressurizer pressure and level, steam generator levels and pressures within acceptable ranges during steady-state and transient operation.[3]

14.2.8.2.28 Process and Effluent Radiation Monitoring System Test

A. Objectives

1. For monitors which:

   a. Will be used for establishing conformance with the safety limits or limiting conditions for operation that will be included in the facility technical specifications, or,
   b. Are classified as engineered safety features or will be relied on to support or ensure operation of the engineered safety features within design limits, or,

[3] Control system response is reviewed and adjustments to the control systems are made, if necessary, prior to proceeding to the next power plateau.

c. Are assumed to function or for which credit is taken in the accident analysis of the facility, and,
d. Will be used to process, store, control, or limit the release of radioactive materials.

2. The objectives are:

a. To verify the calibration of the process and effluent radiation monitor against an acceptable standard.
b. To establish baseline activity and background levels.
c. To demonstrate that process and effluent radiation monitoring systems are responding correctly by performing independent laboratory or other analyses.

B. Prerequisites

1. The plant is stable at the desired power level.

2. The sampling systems for the process and effluent systems are operable.

C. Test Method

1. Calibrations are performed with the use of radioactive sources to verify proper operation of the monitors and detectors.

2. Samples are collected, analyzed with laboratory instruments, and compared with the process and effluent monitor to verify proper operation.

3. Background levels are established at low power (less than 5-percent power); background levels and baseline activity levels determined by sampling are established at 100-percent power to monitor the buildup of activity.

D. Acceptance Criteria

1. Process and effluent radiation monitors are calibrated against radioactive standards.

2. Baseline activities are established.

3. With the exception of ARE-0014, laboratory analyses agree satisfactorily, to the extent practical given sensitivity and energy response, with the process and effluent radiation monitors.

14.2.8.2.29 Axial Flux Difference Instrumentation Calibration Test

A. Objective

To derive calibration factors for overpower and overtemperature ΔT setpoints based on incore flux data, calorimetric data, and excore nuclear instrumentation detector currents.

B. Prerequisites

1. The NIS power range isolation amplifiers have been aligned.

2. The core average axial offset from the incore data is available from flux maps.

3. Average core thermal power from calometrics, performed during the flux mapping, is available.

C. Test Method

Collect data, as required by test instruction, at 75-percent power, perform $F\Delta I$ calculations to the calibration factors, and extrapolate results for use at the 100-percent power plateau.

D. Acceptance Criterion

Calibration factors agree with Technical Specifications.

14.2.8.2.30 Flux Map Test

A. Objective

To determine the reactor core power distributions for various control rod configurations at 30-, 50-, 75-, and 100-percent power.

B. Prerequisites

1. Incore instrumentation and process computer are operable.
2. Reactor is critical, and power level is established as necessary.

C. Test Method

Use data collected from incore detectors to generate a flux map.

D. Acceptance Criterion

The core flux distribution indicated by the flux maps are acceptable and in accordance with plant Technical Specifications.

14.2.8.2.31 Automatic Reactor Control System Test

A. Objectives

To demonstrate the capability of the reactor control system to respond to input signals.

B. Prerequisites

1. The reactor is at equilibrium at the power level specified by the startup test program reference document.
2. Setpoints and controls for the pressurizer, steam generator, steam dump, and feedwater pump have been checked and set to proper values.

C. Test Method

T_{avg} is varied from the T_{ref} setpoint to verify the transient recovery capabilities of the automatic RCS.

D. Acceptance Criterion

T_{avg} should return to within ± 1.5°F of the T_{ref} setpoint.

14.2.8.2.32 (Material Deleted)

14.2.8.2.33 (Material Deleted)

14.2.8.2.34 Pseudo Rod Ejection Test (See Table 14.2.1-2 for Unit 2)

A. Objective

To verify that the power distribution resulting from a simulated, ejected RCCA is within acceptable limits.

B. Prerequisites

1. Excore and incore nuclear instrumentation and incore thermocouples are operable and have been calibrated.
2. The test is performed with the reactor critical at 0-percent power.

C. Test Method

1. Simulate rod ejection accident at the required power levels by withdrawing an RCCA.
2. Record data and calculate power distribution.

D. Acceptance Criterion

The power distribution resulting from the withdrawn RCCA is consistent with the power distribution assumed in the plant safety analysis.

14.2.8.2.35 RCCA Bank Worth Measurement at Zero Power Test (See Table 14.2.1-2 for Unit 2)

A. Objectives

1. To determine the differential and integral reactivity worth of RCCA banks.
2. To determine that the highest worth rod is less than or equal to the value used in the Nuclear Fuel Design Report.

B. Prerequisites

1. The reactor is critical at zero power, with no-load temperature and pressure stable.
2. The reactivity computer is installed and calibrated.

C. Test Method

RCCA bank worths are determined by constant addition and/or dilution of boron in the RCS, causing rod movement to compensate for the boron concentration changes. This rod movement will cause step changes in reactivity which are used to compute the bank worths.

D. Acceptance Criteria

1. The RCCA bank worths are in accordance with the Westinghouse Startup Program Reference Document.

2. The worth of the highest worth rod is less than or equal to the value used in the Nuclear Fuel Design Report.

14.2.8.2.36 Control Systems Checkout Test

This material is included in the Load Swing Test, 14.2.8.2.27; the System Generator Level Control Test, 14.2.8.2.25; and the Automatic Steam Dump Control Test, 14.2.8.2.24.

14.2.8.2.37 Boron Endpoint Determination Test

A. Objective

To determine the critical reactor coolant system boron concentration appropriate to an endpoint configuration (RCC configuration).

B. Prerequisites

1. The reactor is critical at zero power, with no-load temperature and pressure stable.

2. The reactivity computer has been installed.

C. Test Method

1. The control rods are positioned close to the desired endpoint configuration, with the RCS boron concentration stabilized.

2. The rods are quickly moved to the desired endpoint configuration without boron concentration adjustment.

3. The change in reactivity is measured, and this is converted to an equivalent change in boron concentration which will yield the endpoint for the given rod configuration.

D. Acceptance Criterion

The results of the boron endpoint calculations agree with the acceptance criteria of the Westinghouse Startup Program Reference Document.

14.2.8.2.38 Inverse Count Rate Ratio Monitoring for Approach to Initial Criticality Test

A. Objective

To describe methods for obtaining and evaluating data used as an indication of core reactivity during the approach to initial criticality.

B. Prerequisites

1. Both source range and intermediate range nuclear channels alarms, trip functions, and indicating devices have been checked out and calibrated.

2. Both source range and intermediate range nuclear channels have been energized a minimum of 60 min to ensure stable operation.

C. Test Method

1. Obtain baseline count rates prior to rod withdrawal and boron dilution. After each increment of rod withdrawal and periodically during boron dilution, count rates are obtained and inverse count rate ratio is evaluated.

2. Core reactivity is monitored during the approach to criticality.

D. Acceptance Criteria

Acceptance criteria are not applicable to this test.

14.2.8.2.39 Initial Criticality Test

A. Objective

To achieve initial criticality in a controlled manner.

B. Prerequisite

Plant system conditions are established as required in the plant Technical Specifications.

C. Test Method

1. At preselected points during rod withdrawal and/or boron dilution, data are taken and the inverse count rate ratio is plotted to enable an extrapolation to be carried out to predict the expected critical point.

2. Initial criticality may be achieved by boron dilution or, if desired, by rod withdrawal.

D. Acceptance Criterion

The reactor is critical with the flux level established at approximately 1×10^{-8} amps on the intermediate range nuclear instrumentation.

14.2.8.2.40 Isothermal Temperature Coefficient Measurement Test

A. Objective

To determine isothermal temperature coefficient and then derive the moderator temperature coefficient from the isothermal data.

B. Prerequisites

1. The reactor is critical at zero power, with no-load temperature and pressure stable.

2. The reactivity computer has been installed.

C. Test Method

The isothermal temperature coefficient is determined by heating/cooling the RCS at a constant rate and plotting temperature versus reactivity. The moderator temperature may be derived from isothermal data.

D. Acceptance Criterion

The average of the measured values of the isothermal and the derived moderator temperature coefficient agrees with acceptance criteria given in the Westinghouse Startup Program Reference Document.

14.2.8.2.41 Initial Criticality and Low-Power Test Sequence

A. Objective

To define the sequence of tests and operations, beginning with initial criticality, which constitutes the low-power testing program.

B. Prerequisite

Plant system conditions are established as required by the individual test instruction within this sequence.

C. Test Method

Individual test instruction will establish the plant conditions required for initial criticality and during the low-power testing program following initial criticality.

D. Acceptance Criterion

Relevant acceptance criteria are provided in each of the test procedures.

14.2.8.2.42 Biological Shield Survey Test (See Table 14.2.1-2 for Unit 2)

A. Objectives

1. To document the radiation levels in accessible locations of the plant outside of the biological shield while at power.
2. To obtain baseline radiation levels for comparison with future measurements of level buildup with operation.

B. Prerequisites

1. Radiation survey instruments are calibrated.
2. Background radiation levels are measured in designated locations prior to initial criticality.
3. The plant is stable at the applicable power level.

C. Test Method

Gamma and neutron radiation dose rates at designated locations will be measured at 30, 50, 75, and 100 percent of reactor power.

D. Acceptance Criterion

Radiation levels have been demonstrated acceptable for full-power operation and are within the limits described in section 12.3.

14.2.8.2.43 Dynamic Response Test

A. Objectives

1. To verify during power range testing that stress analysis of essential nuclear steam supply system (NSSS) and BOP components under transient conditions is in accordance with design.
2. Points will be tested to resolve discrepancies from hot functional testing, to test modifications made since hot functional testing, and to test systems not tested during hot functional testing, such as main feedwater.

B. Prerequisites

1. Temporary instrumentation is installed, as required, to monitor the deflections of components under test.
2. Points are monitored and baseline data are established.

C. Test Method

Deflection measurements are recorded during various plant transients.

D. Acceptance Criteria

1. The movements due to flow-induced loads shall not exceed the stress analysis of the monitored points.
2. Flow-induced movements and loads will not cause malfunctions of plant equipment or instrumentation.

14.2.8.2.44 Steady-State Core Performance Test

This material has been incorporated into 14.2.8.2.8 and 14.2.8.2.30.

14.2.8.2.45 Remote Shutdown Test

A. Objectives

1. To demonstrate the capability of performing a safe plant shutdown.
2. To maintain the plant in a hot standby condition from outside the control room.

B. Prerequisites

1. Reactor power is greater than 20 percent.
2. The equipment and instrumentation associated with the remote shutdown panel are available.
3. Plant systems are in the normal operating mode, with the turbine-generator in operation.

C. Test Method

1. Transfer control from the control room to the remote shutdown panel.
2. Check the instrumentation, control interlocks, and alarms.

3. Perform a safe shutdown from the remote shutdown panel.
4. Demonstrate the capability to achieve and maintain the plant in a hot standby condition from the remote shutdown panel for a minimum of 30 min.

D. Acceptance Criteria

1. Transfer of control from the control room to the remote shutdown panel is achieved.
2. Remote shutdown instrumentation, controls, and interlocks function.
3. The ability to perform a safe shutdown and to achieve and maintain hot standby conditions from the remote shutdown panel has been demonstrated.

14.2.8.2.46 Loss of Offsite Power at Greater Than 10-Percent Power Test

A. Objective

To demonstrate plant response following a plant trip with no offsite power available.

B. Prerequisites

1. The reactor will be operated at approximately 10-to 20-percent power with the reserve auxiliary transformers supplying station electrical load.
2. The test will occur prior to ascending from the nominal 20-percent power plateau.

C. Test Method

1. Manually start equipment, as required, for protection of nonsafety-related secondary equipment.
2. Isolate the reserve auxiliary transformers.
3. Verify reactor trip.
4. Verify automatic starting and automatic loading of diesel generators.
5. Verify that natural circulation conditions have been established.
6. Verify that the RCS can be maintained in a stable shutdown condition utilizing the power-operated atmospheric relief valves to remove decay heat.
7. Certain nonsafety-related, secondary plant equipment may be immediately restarted.

D. Acceptance Criteria

1. The automatic transfer from the reserve auxiliary transformers to the onsite standby power supply is demonstrated.
2. The primary plant can be maintained in a stable condition on natural circulation using the battery and diesel power supplies for 30 minutes.

14.2.8.2.47 Natural Circulation Test (See Table 14.2.1-2 for Unit 2)

A. Objectives

1. To demonstrate the capability to remove decay heat by natural circulation.

2. To demonstrate that primary parameters can be controlled within acceptable limits.

B. Prerequisites

1. Reactor coolant pumps are operating.

2. Primary system is at normal operating temperature and pressure.

C. Test Method

1. The test will be initiated by tripping off all reactor coolant pumps.

2. Natural circulation will be verified by observing the response of the hot leg and cold leg temperature instrumentation in each loop for natural circulation stabilization period and the ability to maintain the cooling mode.

3. Core exit thermocouples will be monitored to assess core flow distribution.

D. Acceptance Criteria

1. Natural circulation must occur in the primary system, and primary temperatures and pressures are within the design limits of subsection 5.1.2.

2. Delta T across the reactor core is less than the full-power delta T.

3. Decay heat removal capability is demonstrated by maintaining natural circulation conditions for a minimum of one hour.

14.2.8.2.48 Thermal Expansion Test

A. Objective

To demonstrate that essential NSSS and BOP components can expand without obstruction and that the expansion is in accordance with design. Also, during cooldown the components return to their approximate baseline cold position. Testing will be conducted to resolve discrepancies from hot functional testing and to test modifications made since hot functional testing was completed. Systems not tested during hot functional will be tested (e.g., main feedwater).

B. Prerequisite

Temporary instrumentation is installed, as required, to monitor the deflections for the components under test.

C. Test Method

For the components being tested the following will apply:

1. During plant heatup and cooldown, deflection data are recorded.

2. Snubber movements are verified by recording hot and cold positions.

D. Acceptance Criteria

For the components being tested, the following will apply:

1. There shall be no evidence of blocking of the thermal expansion of any piping or component, other than by installed supports, restraints, and hangers (paragraph 3.9.B.2.1).
2. Spring hanger movements must remain within the hot and cold setpoints, and snubbers must not become fully retracted or extended.
3. Piping and components must return to their approximate baseline cold position.

14.2.8.2.49 Primary and Secondary Chemistry Test

A. Objective

To verify proper water quality in the RCS and secondary coolant system.

B. Prerequisite

The plant is at steady state condition at approximately 0-, 30-, 50-, 75-, and 100-percent power.

C. Test Method

Samples will be taken and analyzed to determine the chemical and radiochemical concentrations.

D. Acceptance Criterion

The chemical and radiochemical control systems are capable of maintaining the water chemistry within limits specified by the manufacturer and listed in the Technical Specifications.

14.2.8.2.50 Power Ascension Test Sequence

A. Objective

To define the sequence of operations, beginning at approximately 5-percent power, which constitutes the power ascension testing program.

B. Prerequisite

Plant system conditions are established, as required, by the individual test instruction within this sequence.

C. Test Method

The sequence of operations and tests are presented along with detailed instructions, specific plant conditions, and test procedures.

D. Acceptance Criteria

Relevant acceptance criteria are provided in each of the test procedures.

14.2.8.2.51 (Material Deleted)

14.2.8.2.52 Large Load Reduction Test

A. Objectives

1. To demonstrate satisfactory plant transient response to various specified large-load reductions.
2. To monitor the RCS during these transients.
3. To optimize the RCS setpoints, if necessary.

B. Prerequisite

The plant is operating in a steady-state condition at the desired power level.

C. Test Method

1. The turbine-generator output is manually reduced to achieve an approximate 50-percent load reduction.
2. Selected plant parameters are monitored and recorded during the load transients.
3. If necessary, RCS setpoints are adjusted until optimal response is obtained.

D. Acceptance Criterion

The control systems, with no manual intervention, maintain reactor power, RCS temperature, pressurizer level and pressure, and steam generator levels and pressures without causing a reactor or turbine trip, or lift primary or secondary safety valves during, or following, the transient operation.

14.2.8.2.53 Plant Trip from 100-Percent Power Test

A. Objectives

1. To verify the ability of the plant automatic control systems to sustain a trip from 100-percent power and to bring the plant to stable conditions following the transient.
2. To determine the overall response time of the hot leg RTD.
3. To optimize the control systems setpoints, if necessary.

B. Prerequisite

The plant is operating in a steady-state condition at full power.

C. Test Method

1. The turbine-generator is tripped by opening the generator main breakers.
2. Selected plant parameters are monitored and recorded.
3. If necessary, the control systems setpoints are adjusted to obtain optimal response.

D. Acceptance Criteria

1. Following a 100-percent load rejection, primary and secondary control systems and operator actions can stabilize RCS temperature, pressurizer pressure and level, and steam generator levels to no-load operating temperature and pressure.

2. The steam dump control system operates to prevent opening of primary and secondary safety valves.

14.2.8.2.54 (This test deleted)

14.2.8.2.55 Plant Performance

A. Objectives

1. To monitor secondary plant systems under loaded conditions during power ascension testing.

2. To make adjustments to plant secondary systems, as necessary, to improve plant efficiency.

B. Prerequisites

1. The plant is at the specified power level.

2. The applicable secondary systems are in service.

C. Test Method

1. This procedure provides for the monitoring and collection of data to ensure that the plant is operated in the most efficient manner.

2. This procedure provides for minor adjustments to be made to secondary systems to obtain greater efficiency.

D. Acceptance Criteria

No acceptance criteria are required.

14.2.8.2.56 Nuclear Steam Supply System Acceptance Test

A. Objectives

1. To demonstrate the reliability of the NSSS by maintaining the plant at rated output (+0 percent, -5 percent) for 100 h without a load reduction resulting from an NSSS malfunction.

2. To measure the NSSS output and compare with its warranted rating.

B. Prerequisite

The plant is in normal operating conditions at full power (+0 percent, -5 percent).

C. Test Method

The plant will be maintained at steady-state conditions with NSSS thermal output between 95 percent and 100 percent. Selected plant parameters are monitored and recorded.

D. Acceptance Criteria

1. The reliability of the NSSS has been demonstrated by operating the plant at 95 to 100 percent of the rated thermal output for 100 h without a load reduction resulting from an NSSS malfunction.

2. The NSSS is capable of developing the warranted output.

14.2.8.2.57 At-Power Intercomparison of Reactor Protection System Inputs and Plant Computer Outputs Test

A. Objective

To demonstrate that the inputs to the reactor protection system and the plant computer are in satisfactory agreement with one another.

B. Prerequisites

1. The plant is at steady-state conditions at the applicable power level (30, 50, 75, and 100 percent).

2. Permanently installed instrumentation is calibrated and operable.

C. Test Method

1. Take simultaneous readings of the inputs to the reactor protection system, as read from the main control board or other accessible locations, and the output of the plant computer.

2. Evaluate the readings taken.

D. Acceptance Criterion

The inputs of the reactor protection system and outputs of the plant computer agree within the combined accuracy of the individual instruments.

14.2.8.2.58 Ventilation Capability Test

A. Objective

To verify that various HVAC systems for the containment and areas housing ESF continue to maintain design temperatures.

B. Prerequisite

The plant is operating at or near the desired power (0, 50, and 100 percent).

C. Test Method

1. Record temperature readings in specified areas while operating with normal ventilation lineups.

2. Record temperature readings in specified areas while operating the designed minimum number of HVAC components consistent with existing plant conditions.
3. Record surface concrete temperatures adjacent to the high temperature piping penetrations and at selected locations on the concrete shielding (at 100-percent power only).
4. Record ESF room cooler temperatures and compute the heat transfer capability (UA).

D. Acceptance Criteria

1. Temperature conditions are maintained in the containment, and ESF areas in accordance with table 9.4.1-2.
2. Concrete surface temperatures are maintained in accordance with paragraph 9.4.6.1.
3. Verify that the UA for each ESF room cooler has the required capacity.(4)

14.2.8.2.59 Gross Failed Fuel Detector Test (See Table 14.2.1-2 for Unit 2)

A. Objective

1. Calibrate the gross-failed fuel detector.
2. To establish baseline activity levels.

B. Prerequisites

1. Required electrical power supplies and control circuits are energized and operational.
2. A neutron source is available.

C. Test Method

1. Using a neutron source, the detector is calibrated and alarms checked.
2. At specified power levels (25 percent, 100 percent) baseline activity levels are recorded.

D. Acceptance Criteria

1. The gross-failed fuel detector is calibrated in accordance with the Westinghouse technical manual.
2. Base line activity levels are established.

[4] Should a cooling coil fall below rated capacity, water and/or air flow will be raised through the coil to required capacity.

14.2.8.2.60 Ultimate Heat Sink Heat Rejection Capability Test (See Table 14.2.1-2 for Unit 2)

A. Objectives

1. To determine the heat rejection capabilities of the NSCW cooling towers while operating under partial heat load conditions.
2. To evaluate the heat rejection capabilities of the NSCW cooling towers determined under test conditions and demonstrate that they meet the design requirements.

B. Prerequisites

1. The NSCW systems are operable.
2. The plant is operated in a manner to provide the NSCW system with a heat load.

C. Test Method

Heat rejection capability of NSCW cooling tower cells will be measured in accordance with CTI Bulletin ATC-105.

D. Acceptance Criteria

The heat rejection capability of the NSCW cooling towers, based on test conditions, is in accordance with design requirements.

TABLE 14.2.1-1 (SHEET 1 OF 7)

PREOPERATIONAL TEST PROCEDURES

Title	Test Abstract FSAR Paragraph
Main steam system	14.2.8.1.1[(1)]
Steam dump	14.2.8.1.2[(2)]
Condensate system	14.2.8.1.3[(3)]
Main feedwater system	14.2.8.1.4[(1)]
Motor-driven auxiliary feedwater (AFW) system	14.2.8.1.5
Turbine-driven AFW system	14.2.8.1.6
Reactor coolant system (RCS)	14.2.8.1.7
Pressure relief tank	14.2.8.1.8
RCS hydrostatic	14.2.8.1.9
RCS leak rate	14.2.8.1.10
Pressurizer pressure and level control	14.2.8.1.11
Reactor coolant pump initial operation	14.2.8.1.12
RCS hot functional	14.2.8.1.13[(10)]
Reactor internals and RCS vibration	14.2.8.1.14
Resistance temperature detectors/ thermocouple cross-calibration	14.2.8.1.15
Resistance temperature detector bypass flow measurements	14.2.8.1.16
Residual heat removal system	14.2.8.1.17
Chemical and volume control system	14.2.8.1.18
Letdown system	14.2.8.1.19
Boron thermal regeneration system	14.2.8.1.20[(13)]

TABLE 14.2.1-1 (SHEET 2 OF 7)

Title	Test Abstract FSAR Paragraph
Boric acid blender	14.2.8.1.21(5)
Safety injection system	14.2.8.1.22
Safety injection check valve	14.2.8.1.23
Safety injection accumulator	14.2.8.1.24
Boron Injection Tank and recirculation test (test deleted)	14.2.8.1.25
Containment spray system	14.2.8.1.26
Reactor makeup water storage tank and degasifier system	14.2.8.1.27
Refueling water storage tank	14.2.8.1.28
Condenser air ejection system	14.2.8.1.29(3)
Circulating water system	14.2.8.1.30(3)
Spent fuel pool cooling and cleanup system	14.2.8.1.31
Auxiliary component cooling water	14.2.8.1.32
Nuclear service cooling water	14.2.8.1.33
Component cooling water	14.2.8.1.34
AFW pumphouse heating, ventilation, and air-conditioning (HVAC)	14.2.8.1.35
Fuel handling building HVAC system	14.2.8.1.36(4)
Essential chilled water system	14.2.8.1.38
Control building HVAC	14.2.8.1.39(5)
Auxiliary building HVAC	14.2.8.1.40
Diesel generator building HVAC	14.2.8.1.41

TABLE 14.2.1-1 (SHEET 3 OF 7)

Title	Test Abstract FSAR Paragraph
Containment heat removal system	14.2.8.1.42
Containment integrated leak rate test	14.2.8.1.43(7)
Control rod drive mechanism (CRDM) cavity and vessel support cooling system	14.2.8.1.44
Post-loss-of-coolant accident purge exhaust	14.2.8.1.45
Hydrogen monitor and removal system	14.2.8.1.46
Containment air purification cleanup system	14.2.8.1.47
Gaseous waste processing system	14.2.8.1.48(5)
Liquid waste processing system	14.2.8.1.49(5)(14)
Waste evaporator	14.2.8.1.50(11)
Backflushable filter system	14.2.8.1.51
Boron recycle system	14.2.8.1.53(4)
Containment, auxiliary, control, and fuel handling buildings drains system	14.2.8.1.54(1)(4)(15)
Diesel generator fuel oil system	14.2.8.1.55
Service air system	14.2.8.1.56(2)
Instrument air system	14.2.8.1.57
Fire protection system	14.2.8.1.58(5)
Fuel handling and vessel servicing	14.2.8.1.59(5)
Fuel building hoists and elevator	14.2.8.1.60(4)(5)
Fuel transfer system	14.2.8.1.61
Refueling machine	14.2.8.1.62

TABLE 14.2.1-1 (SHEET 4 OF 7)

Title	Test Abstract FSAR Paragraph
Refueling machine indexing	14.2.8.1.63
Diesel generator	14.2.8.1.64
Auxiliary building flood retaining rooms alarm system	14.2.8.1.65
Main and unit auxiliary transformers	14.2.8.1.66(3)
Reserve auxiliary transformers	14.2.8.1.67(3)
Non-Class 1E ac distribution	14.2.8.1.68(3)
125 V dc (non-Class 1E)	14.2.8.1.69(3)
4.16 kV (Class 1E)	14.2.8.1.70
Class 1E standby power supply (4.16-kV diesel generator sequencer)	14.2.8.1.71
480 V (Class 1E) switchgear	14.2.8.1.72
480 V (Class 1E) motor control center	14.2.8.1.73
125 V dc Class 1E	14.2.8.1.74
Vital 120-V-ac Class 1E	14.2.8.1.75
Essential lighting system	14.2.8.1.76
Emergency lighting system	14.2.8.1.77
Offsite communication system	14.2.8.1.78(3)
Inplant communication system	14.2.8.1.79(3)
Heat tracing system	14.2.8.1.80(3)
(Material deleted)	14.2.8.1.81
Plant annunciator system	14.2.8.1.82
Emergency response facility and post-accident monitoring and sampling systems	14.2.8.1.83(1)(5)

TABLE 14.2.1-1 (SHEET 5 OF 7)

Title	Test Abstract FSAR Paragraph
Reactor protection system and engineered safety features actuation system (ESFAS) logic	14.2.8.1.84
Safeguards test cabinet testing capability	14.2.8.1.85
Nuclear instrumentation system	14.2.8.1.86
Process and effluent radiological monitoring system	14.2.8.1.87
Incore instrumentation system	14.2.8.1.88
Reactor control, rod control, and digital rod position indication	14.2.8.1.89
Control rod drive mechanism (CRDM) motor generator set	14.2.8.1.90
CRDM initial timing	14.2.8.1.91
Seismic monitoring system	14.2.8.1.92(4)
Nuclear sampling system	14.2.8.1.93
Miscellaneous leak detection system	14.2.8.1.94(2)
Metal impact monitoring system	14.2.8.1.95(6)
Integrated control logic safety injection	14.2.8.1.96(10)
Integrated safeguards and blackout sequence	14.2.8.1.97
ESFAS master and slave relay (test deleted)	14.2.8.1.98
Containment local leak rate	14.2.8.1.99
Reactor containment structural integrity test	14.2.8.1.100
High-efficiency particulate air filters	14.2.8.1.101(1)
Emergency core cooling system (ECCS) sump	14.2.8.1.102

TABLE 14.2.1-1 (SHEET 6 OF 7)

Title	Test Abstract FSAR Paragraph
Thermal expansion	14.2.8.1.103
Power conversion and ECCS dynamics	14.2.8.1.104
Remote shutdown	14.2.8.1.105
Reactor trip system and ESFAS response time	14.2.8.1.106
Extraction steam	14.2.8.1.107(3)
Condensate and feedwater chemical injection system	14.2.8.1.108(2)
Proteus computer	14.2.8.1.109(9)
Equipment building HVAC and piping penetration (test deleted)	14.2.8.1.110
Steam generator blowdown processing system	14.2.8.1.111
Main turbine system preoperational test	14.2.8.1.112
125-V dc, Class 1E minimum load voltage verification	14.2.8.1.113
Steady State Vibration Monitoring of Safety Related and High Energy Piping	14.2.8.1.114
AMSAC (ATWS Mitigating System Actuation Circuitry) System	14.2.8.1.115(12)

The notes below identify how the test abstracts are being implemented on Unit 2. These notes also identify where acceptance tests will be used on Unit 2 instead of a preoperational test as identified in chapter 14 and other chapters of the FSAR.

(1) This test abstract will be performed on Unit 2 via preoperational test(s) on the safety-related portions and acceptance tests(s) on the nonsafety related portions.

(2) This test abstract will be performed as an acceptance test on Unit 2. Any safety-related item(s) in this test abstract will be performed in another preoperational test.

TABLE 14.2.1-1 (SHEET 7 OF 7)

(3) This test abstract will be performed as an acceptance test on Unit 2.

(4) The system(s) covered by this test abstract are common to Units 1 & 2 and were tested during the Unit 1 preoperational test program. Interfaces, if any, with Unit 2 will be covered in the Unit 2 preoperational or acceptance test of the system that has the interface.

(5) The system(s) covered by this test abstract are common to Units 1 & 2. The portions of the system(s) tested during Unit 1 testing will not be repeated during the Unit 2 preoperational test.

(6) Parts of this test abstract will be performed as part of 14.2.8.1.13, RCS Hot Functional Preoperational Test Procedure, and will not have a unique preoperational test as on Unit 1. The initial calibration of the metal impact monitor, items A.1., C.1, and D.1 of 14.2.8.1.95, in preparation for hot functional testing, will be accomplished via vendor checkout.

(7) This test abstract may be performed as a surveillance procedure under the controls of the preoperational test program.

(8) The NPSH test and ultimate heat sink rejection capability test required by this test abstract do not have to be performed on Unit 2.

(9) The portions of this test abstract concerning the Internal Application Programs will not be performed on Unit 2 for those programs that were successfully debugged and unchanged between Units. (i.e., image tape of Unit 1 master tape used on Unit 2).

(10) The testing described in items A3, C4, and D4 of 14.2.8.1.96 will be performed during conduct of 14.2.8.1.13 HFT for Unit 2.

(11) The waste evaporator test will be performed prior to equipment use, but is not scheduled prior to Unit 2 OL.

(12) Since this design modification was not incorporated into Unit 1 prior to fuel load, this test abstract is applicable to Unit 2 only.

(13) On Unit 2, the Units 1 and 2 common chiller will be used to verify system performance during HFT. The Unit 2 chiller performance will be verified by use of an external heat source to simulate HFT condition.

(14) The portion of the preoperational test procedure which demonstrates the capability to transfer resin will not be performed on Unit 2 since the capability to transfer resin on similar equipment was demonstrated on Unit 1.

(15) Flow-path verification will not be accomplished for flow discharge paths that result in discharging to a common system that is contaminated and could result in increasing contamination in Unit 2.

TABLE 14.2.1-2 (SHEET 1 OF 4)

STARTUP TEST PROCEDURES

Title	Test Abstract FSAR Paragraph
RCS final leak test	14.2.8.2.1
Pressurizer heater and spray capability and continuous spray flow verification test	14.2.8.2.2(2)
RCS flow measurement	14.2.8.2.3
Resistance temperature detector (RTD) bypass valve flow measurement	14.2.8.2.4
RCS flow coastdown	14.2.8.2.5
Reactor protection	14.2.8.2.6
Core loading instrumentation and neutron source requirements	14.2.8.2.7
Thermal power measurement and statepoint data collection	14.2.8.2.8
Incore detector test	14.2.8.2.9
Operational alignment of nuclear instrumentation	14.2.8.2.10
Material deleted	14.2.8.2.11
Rod control system	14.2.8.2.12
CRDM operational test	14.2.8.2.13
Rod drop time measurement	14.2.8.2.14
Rod position indication	14.2.8.2.15
Operational alignment of process temperature instrumentation	14.2.8.2.16

TABLE 14.2.1-2 (SHEET 2 OF 4)

Title	Test Abstract FSAR Paragraph
Startup adjustments of RCS	14.2.8.2.17
RCS sampling for core loading	14.2.8.2.18
Metal impact monitoring system	14.2.8.2.19
Initial fuel loading	14.2.8.2.20
Inverse count rate ratio monitoring for fuel loading	14.2.8.2.21
Determination of core power range for physics testing	14.2.8.2.22
Precritical test sequence	14.2.8.2.23
Dynamic automatic steam dump control	14.2.8.2.24
Automatic steam generator level control	14.2.8.2.25
Power coefficient determination	14.2.8.2.26
Load swing	14.2.8.2.27
Process and effluent radiation monitoring	14.2.8.2.28
Axial flux difference instrumentation calibration	14.2.8.2.29
Flux map	14.2.8.2.30
Automatic reactor control system	14.2.8.2.31
(Material deleted)	14.2.8.2.32
(Material deleted)	14.2.8.2.33
Pseudo rod ejection	14.2.8.2.34(3)
RCCA and bank worth measurement at zero power	14.2.8.2.35(4)

TABLE 14.2.1-2 (SHEET 3 OF 4)

Title	Test Abstract FSAR Paragraph
Control system checkout (test deleted)	14.2.8.2.36
Boron endpoint determination	14.2.8.2.37
Inverse count rate ratio monitoring for the approach to initial criticality	14.2.8.2.38
Initial criticality	14.2.8.2.39
Isothermal temperature coefficient measurement	14.2.8.2.40
Initial criticality and low-power test sequence	14.2.8.2.41
Biological shield survey	14.2.8.2.42(5)
Dynamic response	14.2.8.2.43
Steady-state core performance test (test deleted)	14.2.8.2.44
Remote shutdown	14.2.8.2.45
Loss of offsite power at greater than 10-percent power	14.2.8.2.46
Natural circulation	14.2.8.2.47(3)
Thermal expansion	14.2.8.2.48
Primary and secondary chemistry	14.2.8.2.49
Power ascension test sequence	14.2.8.2.50
Material deleted	14.2.8.2.51
Large load reduction	14.2.8.2.52
Plant trip from 100-percent power	14.2.8.2.53
Steam generator moisture carryover	14.2.8.2.54

TABLE 14.2.1-2 (SHEET 4 OF 4)

Title	Test Abstract FSAR Paragraph
Plant performance	14.2.8.2.55
Nuclear steam supply system acceptance test	14.2.8.2.56
At-power intercomparison of reactor protection system inputs and plant computer output test	14.2.8.2.57
Ventilation capability test	14.2.8.2.58(7)
Gross failed fuel detector test	14.2.8.2.59(6)
Ultimate heat sink heat rejection capability test	14.2.8.2.60(1)

(1) The NPSHS test and ultimate heat sink rejection capability test required by this test abstract do not have to be performed on Unit 2

(2) Units 1 and 2 are essentially identical plants and it is not necessary to repeat all of this test abstract on Unit 2 when no significant new data or test results would be obtained. The portions (A.3, C.4, and C.5) relating to natural circulation will not be performed on Unit 2.

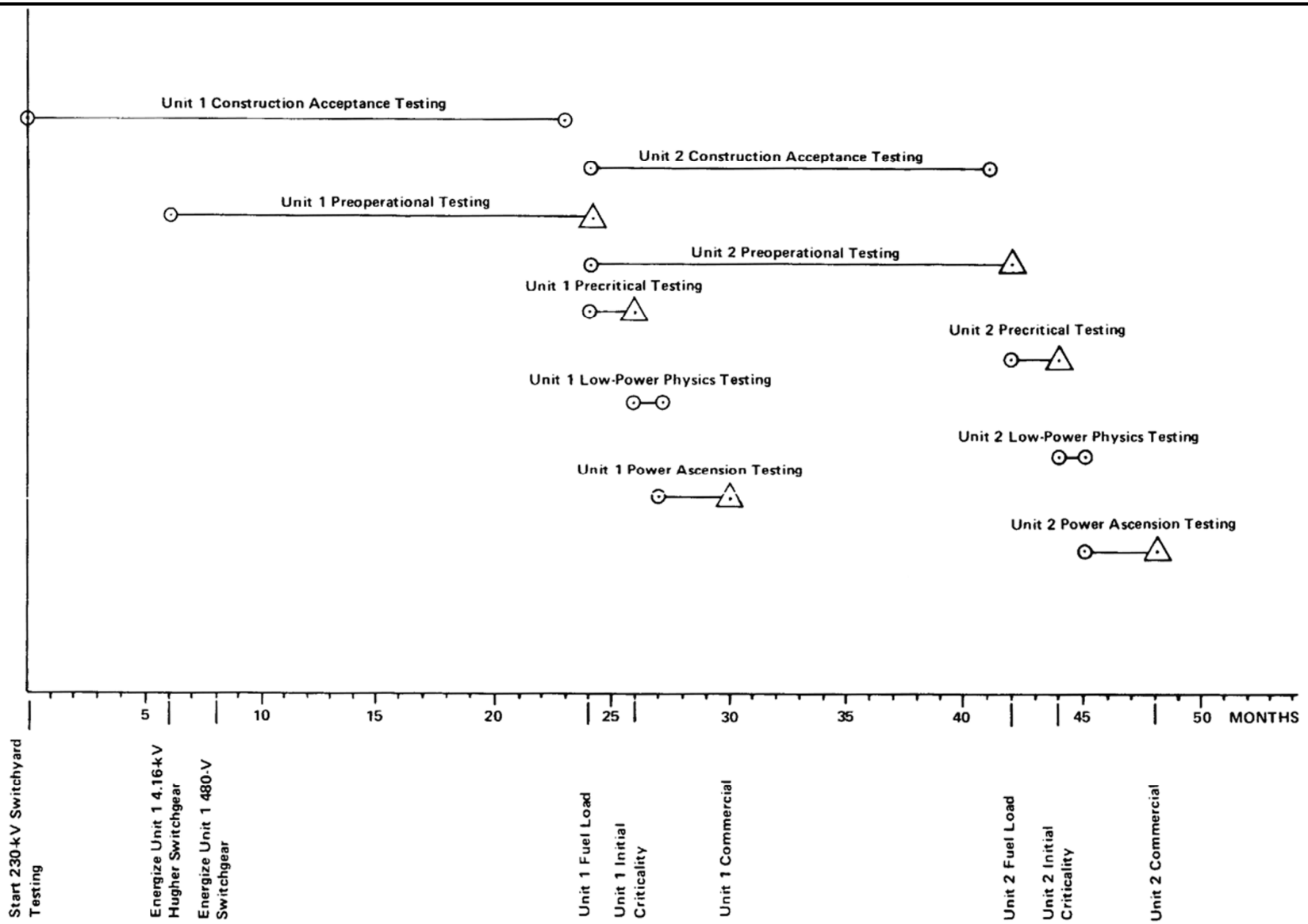
(3) Units 1 and 2 are essentially identical plants and the extremely close agreement between the predicted data and the Unit 1 test results demonstrates that the Unit 2 behavior can be accurately predicted. Performing this test abstract will not result in obtaining any significant new data or test results. This test abstract will not be performed on Unit 2.

(4) Units 1 and 2 are essentially identical plants and the extremely close agreement between the predicted data and the Unit 1 test results demonstrates that the Unit 2 behavior can be accurately predicted. Performing this test abstract will not result in obtaining any significant new data or test results. Acceptance criterion D.2 concerning pseudo-ejected rod will not be performed on Unit 2.

(5) Radiation surveys are required at the 50 % and 100 % power only. Radiation surveys at the 30 % and 75 % power plateau will not be performed on Unit 2.

(6) Calibration with a neutron source will not be performed for Unit 2. This is accomplished per preoperational test abstract 14.2.8.1.93. A calibration check will be performed instead.

(7) Test method C.4 and acceptance criteria D.3 will not be performed on Unit 2. The design similarity does not require that the UA be reverified.



REV 14 10/07

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VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

INITIAL TEST
PROGRAM MILESTONE SCHEDULE

FIGURE 14.2.7–1